

OCTOBER 28, 2025

STATE OF VERMONT AGENCY OF HUMAN SERVICES

HOSPITAL TRANSFORMATION ANALYTICAL SUPPORT



Prepared For:

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This proposal is valid for 180 days

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October 28, 2025

Vermont Agency of Human Services

Office of Health Care Reform

280 State Drive

Waterbury, VT 05671



Attn: Matthew Rockwell, Administrative Services Director

Subject: Proposal Submission – *Hospital Transformation Analytical Support Project*

Dear Mr. Rockwell:

On behalf of **Laurel Health Advisors, LLC**, in partnership with **Charm Economics** and **Lewis & Ellis, Inc.**, we are pleased to submit our proposal to support the **State of Vermont's Hospital Transformation Analytical Support Project**. Our team brings the technical expertise, policy insight, and analytical precision to assist the **Agency of Human Services (AHS)**, the **Green Mountain Care Board (GMCB)**, and the **Office of Health Care Reform (OHCR)** in advancing the State's vision for a sustainable, high-performing, and equitable healthcare system.

Vermont's commitment—through **Acts 167 (2024)** and **68 (2023)** and its collaboration with the **CMS Rural Health Transformation Program**—represents a bold and forward-looking approach to health system reform. The Laurel Health Team is uniquely positioned to help the State realize this vision through actionable, data-driven insights that link hospital sustainability with affordability, quality, and access. We understand Vermont's health policy environment deeply, having previously supported the **Vermont Department of Health** and **Agency of Human Services** in developing a **statewide data governance framework** and advancing the operationalization of the **Health Information Exchange (HIE)**.

Our multidisciplinary team combines three nationally recognized organizations:

- **Laurel Health Advisors**, founded in 2020, brings national leadership in **Medicaid policy, data governance, and health system transformation**, led by **Dr. Yael Harris**, a former senior HHS official, and **Jason Goldwater, PMP, LSSBB**, a nationally respected expert in health IT and analytics.
- **Charm Economics**, led by **Dr. Adam Block**, contributes deep expertise in **economic modeling, rate setting, and health system forecasting**.
- **Lewis & Ellis (L&E)** provides industry-leading **actuarial and financial modeling expertise** for state Medicaid agencies and CMS, ensuring that all fiscal and reimbursement analyses are methodologically sound and compliant with federal standards.

Together, the Laurel Health Team will deliver a rigorous, transparent, and comprehensive analytic framework that enables Vermont to model, evaluate, and implement transformation strategies that enhance hospital viability, strengthen access, and contain system-wide costs.

We are confident that our demonstrated experience with **state-level transformation initiatives** in **Washington, Connecticut, Oklahoma, Oregon, and New Jersey**, and our proven success in integrating **data analytics, financial modeling, and policy alignment** make us the ideal partner for this work. Our approach is grounded in collaboration, transparency, and measurable results, ensuring that Vermont's transformation decisions are supported by evidence, stakeholder engagement, and the highest analytic standards.

We appreciate the opportunity to submit this proposal and look forward to partnering with the State of Vermont to advance its hospital transformation goals. Please feel free to contact me at **jgoldwater@lh-advisors.net** or **301-529-4245** should you require additional information or clarification.

Sincerely,

A handwritten signature in black ink that reads "Jason Goldwater". The script is fluid and cursive, with the first letter of each word being capitalized and larger than the others.

Jason C. Goldwater
President and Chief Operating Officer

Table of Contents

Introduction.....	1
Company Information	2
Understanding of the Problem	3
Relevant Experience of the Laurel Health Team.....	4
Technical Approach	5
Task 2.11: Hospital and Regional Access Modeling.....	5
Data Integration and Model Design	5
Scenario Modeling and Predictive Analytics	6
Benchmarking and Comparative Analysis.....	7
Alignment with Vermont’s Needs.....	8
Task 2.1.2: Modeling the Impacts of Proposed Reforms on Cost, Quality, Access, and Sustainability	8
Integrated Analytic Framework.....	8
Modeling Methodology	9
Scenario Simulation and Impact Evaluation	10
Alignment with Vermont’s Needs.....	10
Task 2.1.3: Alignment with Vermont Legislation and Federal Initiatives	10
Legislative and Policy Crosswalk	11
Policy and Programmatic Modeling.....	11
Use of Tools and Data	12
Alignment with Vermont’s Needs.....	13
Task 2.2.1: Cost Savings – Enabling Data-Driven Reforms that Lower Statewide Healthcare Costs and Improve Affordability for Vermonters.....	13
Modeling Methods and Tools	14
Analytical Tools and Visualization	15
Pathway from Analysis to Business Value	15
Alignment with Vermont’s Goals.....	16
Task 2.2.2: Customer Service Improvement – Enhancing Hospital and Regional Service Quality, Efficiency, and Access through Evidence-Based Modeling and Benchmarking	16
Integrated Performance and Quality Framework.....	17
Modeling Methodology and Analytic Tools	17
Translating Analysis into Business Value and Customer Impact.....	18
Alignment with Vermont’s Goals.....	19

Task 2.2.3: Risk Reduction – Reducing Operational, Financial, and Quality-of-Care Risks through Predictive Analytics and Transformation Planning	19
Comprehensive Risk Modeling Framework	19
Modeling Methodology and Analytical Tools	20
Application of Results to Vermont’s Transformation Planning	21
Tools for Risk Reduction and Continuous Monitoring	21
Alignment with Vermont’s Goals	22
Task 2.2.4: Compliance – Ensuring Transformation Initiatives Meet State and Federal Standards and Align with CMS and Vermont Legislative Requirements	22
Compliance Integration Framework	23
Technical Approach and Tools	23
Deliverables and Analytical Tools	25
Pathway from Analysis to Business Value	25
Alignment with Vermont’s Goals	26
Task 2.2.5: Transparency and Accountability – Delivering Evidence-Based Reports and Visualizations to Support Open, Data-Driven Decision-Making.....	26
Transparency and Accountability Framework	26
Technical Approach and Analytical Tools	27
Deliverables	28
Pathway from Analysis to Business Value	29
Alignment with Vermont’s Goals	29
<i>Past Performance</i>	30
Experience with Vermont	30
Other Past Performance	31
<i>Personnel</i>	33
<i>Management Plan</i>	38
Organizational Capacity	39
Roles and Responsibilities	40
Risk Management	43
Communication Management	45
Quality Assurance	48
Timeline	51
<i>References</i>	54



Introduction

The **Laurel Health Team**, a partnership among **Laurel Health Advisors**, **Charm Economics**, and **Lewis & Ellis (L&E)**, is honored to submit this proposal to support the **State of Vermont** in advancing its **Hospital Transformation Analytical Support Project**. Vermont stands at a pivotal moment in its ongoing effort to create a healthcare system that is sustainable, equitable, and accountable. The State's vision, as outlined in **Acts 167 (2024)** and **68 (2023)** and through its collaboration with the **Centers for Medicare & Medicaid Services (CMS) Rural Health Transformation Program**, reflects a commitment to ensuring that every Vermonter has access to high-quality, affordable care while maintaining the financial viability of hospitals and healthcare providers statewide. Achieving that vision requires rigorous analytics, economic modeling, and data-driven insights that can inform sound policy and strategic decision-making.

The Laurel Health Team brings the comprehensive expertise and capacity necessary to meet Vermont's needs. Together, our organizations have supported numerous state and federal health agencies in developing transformative health system models that integrate **data governance, financial sustainability, hospital access analysis, actuarial modeling, and policy alignment**. Laurel Health Advisors provides nationally recognized leadership in healthcare data analytics, interoperability, and Medicaid transformation; Charm Economics contributes advanced economic and quantitative modeling expertise; and Lewis & Ellis brings deep actuarial and rate-setting experience across state Medicaid programs, CMS, and hospital rate regulation.

Our approach combines **technical precision, policy understanding, and collaborative engagement**. We will deliver analytic tools and modeling frameworks that help Vermont quantify the effects of proposed reforms on cost, access, quality, and sustainability—ensuring that the

Why Laurel Health Advisors?

- **Proven Experience in Statewide Transformation:** Laurel Health Advisors has successfully led complex health system transformation and Medicaid reform initiatives for states including Washington, Connecticut, Oklahoma, Oregon, and New Jersey, providing actionable models that balance fiscal sustainability, access, and quality.
- **Deep Understanding of Vermont's Health System:** Laurel has worked directly with the Vermont Department of Health and Agency of Human Services on data governance, health information exchange (HIE) operations, and hospital transformation planning.
- **Comprehensive, Interdisciplinary Expertise:** Together with Charm Economics and Lewis & Ellis, the team unites nationally recognized experts in data analytics, actuarial modeling, health economics, and hospital rate regulation, ensuring each deliverable integrates technical rigor with practical policy insight.
- **National Leadership in Health Policy and Analytics:** Laurel Health Advisors' leadership team—Dr. Yael Harris and Jason Goldwater (PMP, LSSBB) bring decades of federal and state experience, having supported CMS, HRSA, and ONC on projects advancing value-based care, telehealth, and data interoperability.
- **Commitment to Transparency, Quality, and Results:** Laurel's project management and quality assurance framework emphasizes data integrity, stakeholder engagement, and measurable outcomes, ensuring Vermont receives clear, evidence-based analyses to guide hospital sustainability and transformation decisions.

State can make informed, evidence-based decisions. The Laurel Health Team is committed to working hand in hand with the **Agency of Human Services (AHS)**, the **Green Mountain Care Board (GMCB)**, and the **Office of Health Care Reform (OHCR)** to deliver actionable results that strengthen the healthcare system, enhance transparency, and build long-term capacity for data-driven governance.

With a strong record of success supporting Medicaid and hospital transformation initiatives in states such as **Washington, Connecticut, Oklahoma, New Jersey, and Oregon**, and deep familiarity with Vermont's policy landscape, the Laurel Health Team stands ready to deliver the technical excellence, strategic insight, and collaborative spirit needed to advance Vermont's hospital and regional transformation goals.

Company Information

Laurel Health Advisors, LLC is a **limited liability company (LLC)** founded in **2020** and headquartered in **Washington, D.C.** The firm operates as an SBA-certified **woman-owned small business** specializing in **health policy, data analytics, and health system transformation**. Laurel Health Advisors provides strategic, analytic, and technical consulting services to federal and state agencies, healthcare organizations, and community-based entities working to improve healthcare quality, access, and affordability.

Since its founding, Laurel Health Advisors has grown into a **multidisciplinary firm with more than 10 core staff and subject-matter experts** supported by an extended network of specialized consultants and data scientists. The firm's professional team includes **health policy analysts, actuaries, economists, epidemiologists, data engineers, and project managers** who collaborate seamlessly across disciplines to deliver evidence-based insights and solutions.

Laurel Health Advisors maintains the **technical infrastructure and operational resources** necessary to manage complex, data-driven projects of statewide and national scope. The firm operates secure, cloud-based analytic environments compliant with **HIPAA and federal information security standards**, as well as robust tools for **data integration, predictive modeling, and geospatial analysis** (Python, R, SQL, ArcGIS, Tableau, and Power BI). Laurel's **project management systems**, built around **PMI and Lean Six Sigma methodologies**, ensure disciplined oversight, risk control, and quality assurance across all engagements.

The company's **organizational resources** are augmented by strong partnerships with **Charm Economics** and **Lewis & Ellis (L&E)**, enabling Laurel to offer clients a complete range of expertise spanning **healthcare finance, actuarial modeling, policy analysis, and advanced analytics**. This collaborative structure allows Laurel Health Advisors to deliver the analytical depth, flexibility, and scalability required to meet Vermont's needs for rigorous, transparent, and actionable hospital transformation support.

Understanding of the Problem

The State of Vermont, through the Agency of Human Services' Office of Health Care Reform, is seeking analytical support to advance hospital and regional health system transformation across the state. This need arises from Vermont's ongoing challenge of balancing cost containment, sustainability of hospital operations, access to care, and quality improvement—particularly in a small, rural state where hospitals serve as both healthcare providers and critical economic anchors. The State recognizes that a data-driven understanding of the interdependencies between cost, access, quality, and workforce capacity must inform hospital transformation.

Vermont's hospitals face structural and operational pressures from declining inpatient volumes, workforce shortages, rising costs, and uneven access across regions. Demographic shifts, aging populations, and limited economies of scale in rural areas compound these challenges. In this context, transformation planning requires rapid-cycle modeling and predictive analytics that can integrate claims, clinical, cost report, and administrative data to simulate the impact of reform proposals at the facility, regional, and statewide levels.

The project is situated within Vermont's broader reform initiatives, Act 167, Act 68, and CMS's Rural Health Transformation Program, and requires analytic capabilities that can produce evidence-based, transparent, and benchmarked outputs. These insights must enable the State and the Green Mountain Care Board (GMCB) to evaluate how potential service line changes, shared services, and regionalization efforts affect financial viability, patient access, quality, and overall system performance. The ultimate aim is to support reform decisions that reduce total system cost while improving affordability, access, and outcomes for Vermonters.

Specific Issues of Greatest Concern to Vermont

1. **Financial Sustainability and Affordability** – Many Vermont hospitals operate with narrow margins or deficits. The State must model how proposed reforms—such as service line consolidation or regional partnerships—affect hospital solvency, workforce needs, and long-term financial sustainability, while lowering costs for payers, employers, and consumers.
2. **Access and Regionalization** – Rural and regional disparities in access to services persist. Vermont seeks to understand the trade-offs between regionalization for efficiency and maintaining reasonable geographic and seasonal access, including drive-time modeling, cross-border utilization (NY, NH, MA), and analyses of traffic and patient flow.
3. **Workforce and Capacity Optimization** – Workforce shortages and uneven bed utilization require analysis of staffing implications, acuity management, and capacity modeling to ensure that transformation proposals preserve or improve access without overburdening specific hospitals or regions.
4. **Quality, Safety, and Outcomes** – Vermont aims to ensure that any transformation aligns with national quality benchmarks (e.g., CMS 5-Star, Leapfrog, PSI 90) and evidence-



based service line thresholds, ensuring reforms enhance rather than erode patient safety and care quality.

5. Data Integration and Transparency – The State’s need for a unified, secure analytic dataset that combines state-de-identified data with vendor-provided multi-sector data (claims, cost reports, EHR extracts, etc.) is central to achieving reliable, transparent, and replicable modeling results that can be shared with stakeholders and the public.
6. Alignment with Federal and State Policy – Vermont’s transformation must align with federal initiatives, such as the CMS Rural Health Transformation Program, and with state legislation (Acts 167 and 68), which require compliance, standardization, and compatibility with existing reporting and monitoring systems.

Relevant Experience of the Laurel Health Team

The Laurel Health Team, comprising Laurel Health Advisors, Charm Economics, and Lewis & Ellis, offers the combined expertise necessary to meet Vermont’s analytical, technical, and policy goals for hospital transformation.

- Laurel Health Advisors brings extensive experience in state and federal healthcare reform, data integration, and analytics for system transformation. The firm has led projects for multiple states and federal agencies, including the Centers for Medicare & Medicaid Services (CMS) and state Medicaid programs, developing evidence-based models for value-based care, telehealth integration, and hospital system performance. Laurel’s expertise in data ingestion, predictive modeling, and quality and access evaluation will enable Vermont to produce the rapid-cycle analytics and benchmarking this contract requires.
- Charm Economics specializes in healthcare market analysis and financial modeling, with particular strengths in hospital cost structures, reimbursement modeling, and rate-setting analysis. Their capabilities in Medicare cost report analytics and economic scenario modeling are directly aligned with Vermont’s need to simulate the financial and operational impacts of regional transformation initiatives.
- Lewis & Ellis, an actuarial and health economics firm with deep experience in hospital rate modeling, insurance premium analysis, and healthcare cost containment, provides the technical depth to assess payer mix implications, rate sustainability, and the effects of reform scenarios on affordability and insurance markets. Their expertise complements Laurel and Charm’s policy and systems analysis, ensuring comprehensive modeling across financial, operational, and clinical dimensions.

Together, this team integrates policy insight, economic and actuarial modeling, and technical analytics—a combination that uniquely positions them to deliver the data-driven insights, transparency, and rigor required for Vermont’s hospital transformation initiative. Their collective approach emphasizes collaboration with state and regional stakeholders, alignment with CMS

and legislative mandates, and the delivery of clear, evidence-based recommendations to guide the State toward a sustainable, high-performing healthcare system.

Technical Approach

We will employ a comprehensive, data-driven, and collaborative technical approach to support Vermont's hospital and regional transformation planning. Our methodology integrates multi-sector data sources (claims, clinical, administrative, and cost reports) with state-provided de-identified data extracts to generate actionable analytic insights that directly inform policy and operational decision-making. Grounded in national best practices and tailored to Vermont's legislative and regulatory landscape (Act 167, Act 68, and CMS's Rural Health Transformation Program), our approach emphasizes rapid-cycle analytics, predictive modeling, and evidence-based benchmarking to evaluate the financial, operational, and quality impacts of proposed reforms. Through transparent modeling, robust stakeholder engagement, and alignment with AHS and the Green Mountain Care Board (GMCB), the team will deliver clear, defensible analyses that help the State balance cost containment, quality, and access, ensuring that transformation initiatives strengthen Vermont's hospital system and improve health outcomes statewide.

Task 2.11: Hospital and Regional Access Modeling

We will implement a rigorous, evidence-based, and transparent analytic framework to perform hospital and regional access modeling for the State of Vermont. This task will assess Vermont's current and projected patterns of hospital utilization, capacity, and access, with particular attention to geographic distribution, seasonal variations, and cross-border care with neighboring states (New Hampshire, New York, and Massachusetts). The work will provide Vermont with actionable insights into how proposed transformation initiatives may affect patient flow, accessibility, and equity across the health system.

Data Integration and Model Design

Our approach begins with integrating **multi-source data assets**, including State de-identified claims data, hospital discharge and cost report data, and Medicare and Medicaid utilization data. These datasets will be harmonized using our established data ingestion and normalization protocols, which employ standardized taxonomies (such as NPI, DRG, CPT, and ICD-10) and uniform provider, patient, and service-line identifiers. Using secure data exchange through State-approved SFTP, the team will construct an **analytic base file (ABF)** capable of supporting

longitudinal and comparative analyses across Vermont's hospital network. An overview of our process is shown in Exhibit 1.

The modeling process will use **geospatial and statistical techniques** to evaluate access and utilization patterns. Drive-time and patient flow analyses will be performed using **geographic information system (GIS) tools** such as QGIS, which will integrate road network, topographic, and weather-adjusted variables to model access under different conditions. Seasonal drive-time analyses will account for travel impediments typical of Vermont's winter months, using meteorological data overlays to provide realistic access estimates. These models will be validated against historical utilization data and patient origin studies to ensure fidelity to observed patterns.

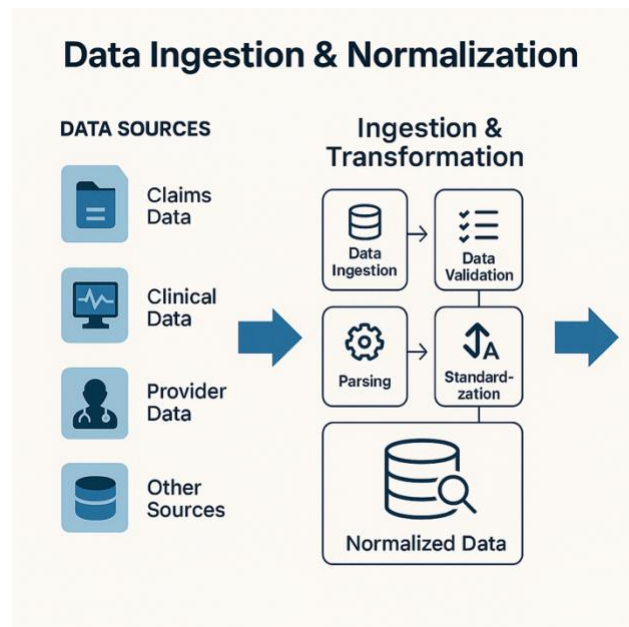


Exhibit 1: Laurel Health's Data Ingestion and Normalization Process

Scenario Modeling and Predictive Analytics

Once the baseline access model is established, the team will conduct **scenario-based simulations** to assess the impact of proposed or hypothetical hospital service line changes, consolidations, or regional partnerships. These simulations will evaluate how reforms may alter travel times, service availability, patient referral flows, and hospital throughput. Predictive analytics will be applied using machine learning models, such as gradient boosting and logistic regression, to project utilization and access impacts under alternative configurations. These models will generate metrics including population-weighted travel time, bed occupancy ratios, and network adequacy scores, which will be benchmarked against national standards such as CMS network adequacy guidelines and HRSA Health Professional Shortage Area criteria. An overview of potential predictive modeling approaches is shown in Exhibit 2.

Exhibit 2: Potential Predictive Models

Predictive Model	Independent Variable(s)	Dependent Variable(s)	Results of Analysis	Benefit to Vermont
Financial Sustainability Model	Hospital revenue mix, cost-to-charge ratio, patient volume, staffing ratios	Operating margin, days cash on hand	Identifies hospitals at risk of financial distress and forecasts the impact of proposed reforms on fiscal stability	Enables the State to target technical assistance and rate interventions to preserve access and system sustainability



Predictive Model	Independent Variable(s)	Dependent Variable(s)	Results of Analysis	Benefit to Vermont
Access and Drive-Time Model	Geographic distance, travel time, service line location, seasonal weather data	Population-weighted travel time, access index	Quantifies geographic disparities in care access and predicts how service changes affect regional availability	Supports regional planning and equitable service distribution in rural and underserved areas
Quality and Outcome Model	Staffing levels, readmission rates, acuity, care coordination measures	Patient safety indicators, readmissions, mortality rates	Predicts how structural or workforce changes influence care quality and outcomes	Helps Vermont ensure reforms maintain or improve quality while containing costs
Workforce Capacity Model	Workforce supply, vacancy rates, wage levels, patient volume	Staffing adequacy, service coverage, burnout risk index	Projects future workforce shortages by region and specialty	Guides workforce planning and informs funding for training and retention programs
Cost Savings and ROI Model	Service line integration, administrative costs, utilization rates	Total cost of care per capita, ROI	Estimates financial savings from shared services, telehealth, and process improvements	Demonstrates fiscal value of reforms and supports policy justification for affordability initiatives
Risk Prediction Model	Payer mix, service volume, cost structure, quality metrics	Probability of financial or operational risk	Identifies hospitals and regions at elevated risk under various reform scenarios	Reduces likelihood of hospital closures and system disruption through proactive mitigation
Compliance and Alignment Model	Act 167/68 mandates, CMS Rural Health Transformation KPIs	Compliance score, audit readiness index	Evaluates alignment of proposed reforms with legislative and federal program requirements	Ensures Vermont reforms meet all statutory and funding obligations

Benchmarking and Comparative Analysis

We will benchmark Vermont's hospital and regional access patterns against **peer states** with similar geographic and demographic characteristics, such as Maine and New Hampshire. Our team will identify outliers in access, travel time, and capacity utilization to guide recommendations for resource allocation and regional service planning. Comparative analyses will also examine cross-border care utilization, quantifying the proportion of Vermont residents

receiving care in adjacent states and the resulting financial and access implications for Vermont's hospital system.

Alignment with Vermont's Needs

This methodology aligns precisely with Vermont's requirements under Act 167 and Act 68 for transparent, data-guided decision-making. The State seeks to understand how service line transformations, regional partnerships, and shared services arrangements may influence access to hospital-based and community-oriented care. By combining **quantitative access modeling** with **financial and operational data**, the Laurel Health Team will provide Vermont with a multidimensional perspective on the trade-offs between efficiency and accessibility. The modeling results will directly inform policy deliberations by the Agency of Human Services (AHS) and the Green Mountain Care Board (GMCB), supporting equitable, evidence-based decisions that sustain hospital viability and improve care access for all Vermonters.

The final outputs will include a comprehensive set of **data visualizations, scenario models, and policy briefs** that communicate findings in clear, reproducible formats suitable for both technical and public review. This technical rigor, combined with the team's demonstrated experience in similar state-level transformation initiatives, ensures that Vermont will receive a robust, defensible, and actionable hospital and regional access model to guide its transformation planning

Task 2.1.2: Modeling the Impacts of Proposed Reforms on Cost, Quality, Access, and Sustainability

We will apply a comprehensive and empirically grounded analytic framework to assess the projected impacts of Vermont's proposed health system reforms. The objective of this task is to develop **integrated, dynamic models** that evaluate how hospital and regional transformation initiatives influence **costs, quality, access, and financial sustainability** across Vermont's healthcare delivery system. The approach is designed to provide the Agency of Human Services (AHS) and the Green Mountain Care Board (GMCB) with transparent, evidence-based tools to simulate the outcomes of proposed changes and to guide strategic decisions that align with Act 167, Act 68, and the CMS Rural Health Transformation Program.

Integrated Analytic Framework

The modeling framework will combine **financial, operational, and population health data** from State-provided and vendor-supplied sources to generate multidimensional simulations of Vermont's healthcare landscape. We will lead data integration and model design, leveraging its secure data ingestion architecture and normalization procedures to align diverse datasets, including:

- Vermont hospital discharge data and Medicare Cost Reports;
- Medicaid claims and all-payer claims data; and
- State workforce and facility capacity data

These data sources will be combined within a unified analytic environment using **Python, R, and SQL-based data pipelines**, hosted within a secure analytic sandbox compliant with State privacy and security protocols.

Modeling Methodology

We will employ three complementary model types to quantify the effects of proposed reforms:

1. System Dynamics Modeling (SDM):

This model will capture the interdependencies among cost, access, workforce, and utilization variables across Vermont's healthcare ecosystem. Using software such as **Vensim** and **Stella Architect**, SDM will simulate how changes in one area (e.g., consolidating service lines or implementing shared administrative functions) propagate through the system over time. For example, a regionalization proposal that reduces inpatient capacity in one area will be evaluated for its cascading effects on emergency department volume, transfer rates, and travel times, as well as its financial impact on affected hospitals.

2. Econometric and Cost Modeling:

We will apply **multivariate regression and cost-elasticity models** to project financial outcomes for specific reform scenarios. These models will analyze relationships between operational inputs (labor, supplies, administrative overhead) and outputs (revenue, margin, per-unit costs) using tools such as **Stata, R, and Python's StatsModels library**. The models will be calibrated using Vermont-specific data, allowing estimation of marginal and total cost impacts under various configurations, such as shared service agreements, outpatient migration, or value-based payment adjustments.

Lewis & Ellis will integrate these results with **hospital-level pro forma models**, derived from Medicare Cost Reports, to assess sustainability under multiple revenue and volume assumptions. This will include sensitivity analyses for payer mix changes, Medicaid rate adjustments, and shifts in commercial reimbursement.

3. Predictive Quality and Access Models:

We will deploy **supervised machine learning algorithms**, including **random forests** and **gradient boosting**, to predict the effect of proposed reforms on quality indicators and access metrics. Model outputs will include predicted changes in:

- Potentially avoidable hospitalizations (PAH) and emergency visits;
- CMS quality indicators such as PSI-90 composites, readmission rates, and mortality measures; and
- Access measures, including population-weighted travel time and average time to specialty or primary care.

Model training will leverage Vermont's historical hospital and claims data as well as national benchmark datasets. Validation will be performed using cross-validation and backtesting against observed outcomes from comparable reform initiatives in other states.



Scenario Simulation and Impact Evaluation

Each reform proposal will be evaluated through a structured set of **scenario simulations** that compare baseline conditions to modeled outcomes under alternative configurations. Examples of scenarios include:

- The consolidation of surgical or maternity services between regional hospitals;
- The creation of shared administrative or IT functions across hospital systems.
- Expansion of advanced primary care and telehealth services; and
- The implementation of hospital-at-home or acute care substitution models.

For each scenario, the models will produce quantitative estimates of changes in:

- **Cost** (total system costs, per capita expenditures, and payer-specific impacts);
- **Access** (drive times, service availability, workforce distribution);
- **Quality** (national benchmark scores, adverse event rates, readmission rates); and
- **Sustainability** (operating margin, liquidity ratios, and long-term financial viability).

These outcomes will be visualized through **interactive dashboards and geospatial maps**, developed using **Tableau** and **ArcGIS**, to enable AHS and GMCB to interpret results at both the hospital and regional levels.

Alignment with Vermont's Needs

This modeling framework is precisely aligned with Vermont's requirements to evaluate proposed transformation initiatives in a rigorous and transparent manner. The approach integrates **financial, clinical, and operational modeling** to provide a comprehensive assessment of the trade-offs inherent in reform decisions. By combining system dynamics with econometric and predictive analytics, the models will deliver clear insights into **how reforms affect costs, service accessibility, and quality outcomes**—ensuring that Vermont can pursue hospital transformation strategies that are sustainable, equitable, and evidence-based.

All results will be accompanied by methodological documentation, reproducible code, and detailed interpretation suitable for presentation to AHS leadership, the Green Mountain Care Board, and other stakeholders, supporting Vermont's commitment to data-driven and transparent healthcare reform.

Task 2.1.3: Alignment with Vermont Legislation and Federal Initiatives

We will ensure that all analytical work under this engagement is rigorously aligned with Vermont's legislative framework (Act 167 and Act 68) and with relevant federal initiatives, including the **Centers for Medicare & Medicaid Services (CMS) Rural Health Transformation Program**. This task focuses on developing an integrated alignment and compliance model that links Vermont's hospital and regional transformation efforts to the State's policy goals for **affordability, access, quality, and system sustainability**, while ensuring

consistency with federal funding and performance requirements. The team will combine **policy analytics, legislative crosswalks, and quantitative compliance modeling** to ensure every reform scenario evaluated under this project adheres to both the **intent and the measurable standards** embedded in Vermont and federal programs.

Legislative and Policy Crosswalk

We will begin by developing a **legislative-policy alignment matrix** that explicitly maps the provisions of **Act 167 (2024)** and **Act 68 (2023)** to the analytical domains of this project, cost containment, quality improvement, access assurance, and sustainability modeling. The matrix will identify each legislative requirement, associated State agency responsibilities, and the specific data and modeling outputs that will demonstrate compliance. For example:

- Act 167 requirements related to “hospital sustainability, transparency, and accountability” will be mapped to Lewis & Ellis’s financial sustainability metrics, such as **operating margin forecasts, debt service coverage, and payer-mix stress testing**.
- Act 68 requirements to “support equitable access and affordability through regional planning” will be aligned with Laurel Health Advisors’ **geospatial access models**, which quantify **population-weighted travel times** and **service availability by region**.

Legislative and Policy Crosswalk			
Legislation	Analytical Domain	Requirements	Modeling Outputs
Act 167 (2024)	Sustainability Modeling	Hospital sustainability, transparency, and accountability	Operating margin forecasts, debt service coverage, payer-mix stress testing
Act 68 (2023)	Access Assurance	Support equitable access and affordability through regional planning	Population-weighted travel times, service availability
Federal CMS Rural Health Transformation Program	Quality Improvement	Integration of value-based payment structures and rural hospital technical support	Value-based readiness models policy impact analysis

Exhibit 3: Legislative and Policy Crosswalk

- Federal CMS Rural Health Transformation Program objectives, including the integration of **value-based payment structures** and **rural hospital technical support**, will be linked to Charm Economics’ **value-based readiness and policy impact models**, ensuring Vermont’s initiatives are eligible for continued federal support and technical assistance.

The alignment matrix will be developed collaboratively with

the **Agency of Human Services (AHS)**, the **Green Mountain Care Board (GMCB)**, and the **Vermont Office of Health Care Reform**, ensuring all measures and deliverables are verifiable and directly tied to statutory and programmatic obligations. An overview of this approach is shown in Exhibit 3.

Policy and Programmatic Modeling

To operationalize these alignments, we will deploy a **compliance and alignment modeling framework** that integrates financial, operational, and quality data with legislative and programmatic benchmarks. This will include three model components:



1. Policy Logic Model:

Using the **Harvard Logic Model and Results Framework methodology**, we will map the causal relationships among State reform inputs (legislation, funding streams, hospital transformation proposals), intermediate outputs (access and quality improvements), and long-term outcomes (cost control, sustainability, and population health gains). This model will identify measurable indicators, such as total cost of care per capita, hospital margin improvement, or CMS 5-Star quality gains, that will be monitored throughout the contract period to demonstrate progress toward Vermont's statutory goals.

2. Regulatory Compliance Simulation:

The team will employ **regression-based compliance modeling** using **R** and **Stata** to test how transformation proposals align with the quantitative and qualitative thresholds defined in Act 167 and Act 68. For example, models will simulate whether hospital-level service line adjustments meet statutory standards for maintaining network adequacy and equitable access. These models will be cross-referenced with the **CMS Rural Health Transformation Program's key performance indicators (KPIs)**—including cost trend moderation, rural hospital viability, and access maintenance—to confirm dual compliance with State and federal expectations.

3. Scenario Alignment Testing:

Charm Economics and Lewis & Ellis will develop a **scenario-testing framework** to evaluate proposed reforms for policy alignment before approval or implementation. This will include a **multi-criteria decision analysis (MCDA)** tool built in **Python and Tableau**, which will score each proposed initiative against defined Vermont and CMS criteria, such as:

- Alignment with Act 167 cost containment and transparency standards;
- Consistency with Act 68 access and regional equity principles;
- Compliance with CMS RHTP metrics for rural hospital transformation; and
- Demonstrated sustainability under projected reimbursement and workforce conditions.

Each reform scenario will receive a composite “alignment score,” which can be used by AHS and GMCB to prioritize or refine initiatives.

Use of Tools and Data

We will employ a robust suite of analytical and visualization tools to execute this work:

- **Python (Pandas, NumPy, and scikit-learn)** for data preparation, regression modeling, and policy alignment scoring.
- **R and Stata** for statistical modeling, legislative performance simulation, and multi-level compliance analysis.
- **Tableau and Power BI** for developing interactive dashboards that visualize Vermont's progress toward Act 167 and Act 68 targets.



- **ArcGIS** for geospatial policy alignment, linking regional performance outcomes (access, quality, and cost) to statutory thresholds for geographic equity.
- **Laurel Health Advisors' secure data ingestion and mapping system**, used in previous CMS and State Medicaid projects, to integrate hospital, claims, and financial data with policy variables for continuous compliance monitoring.

Alignment with Vermont's Needs

This structured, quantitative approach to alignment directly addresses Vermont's need for **transparency, accountability, and evidence-based policymaking** in hospital transformation. By embedding statutory and programmatic parameters into analytic models, the Laurel Health Team will ensure that every reform scenario can be evaluated for compliance and performance impact before implementation.

The outputs will include a **legislative alignment dashboard**, a **policy compliance matrix**, and a series of **data-driven policy briefs** that summarize how proposed initiatives advance the objectives of Acts 167 and 68 while maintaining full compliance with CMS Rural Health Transformation Program requirements. These deliverables will enable AHS and GMCB to make informed, defensible decisions that reinforce Vermont's leadership in equitable, sustainable, and transparent healthcare reform.

Task 2.2.1: Cost Savings – Enabling Data-Driven Reforms that Lower Statewide Healthcare Costs and Improve Affordability for Vermonters

We will implement a structured, data-driven, and outcome-oriented analytic approach to generate actionable insights that directly support Vermont's goal of achieving measurable **cost savings** and **improved affordability** across the healthcare system. This task will operationalize the integration of financial, clinical, and utilization data to identify where reforms can reduce system inefficiencies, optimize resource allocation, and guide payment and delivery redesign that maintains or improves quality and access. The analysis will translate complex datasets into concrete, decision-ready recommendations for the **Agency of Human Services (AHS)** and the **Green Mountain Care Board (GMCB)**, ensuring that Vermont's healthcare transformation efforts deliver real and sustained economic value for the State and its residents.

Analytical Framework and Data Architecture

The foundation of this task will be a **comprehensive cost and value analysis model**, built on harmonized data from multiple sources:

- Vermont hospital cost reports, discharge data, and budget submissions to GMCB;
- All-payer claims data (Medicare, Medicaid, and commercial payers);

Robert Murray was a principal architect of the **Maryland Total Cost of Care (TCOC) Model**, the nation's first statewide hospital global budgeting and cost control framework approved by CMS. Building on his earlier work designing Maryland's **All-Payer Rate Setting System** and **Global Budget Revenue (GBR) model**, Murray developed the methodologies and regulatory structures that allow Maryland to manage total healthcare expenditures while improving quality and maintaining access. His work included designing hospital global budgets, integrating cost, utilization, and quality metrics across payers, and creating actuarial and policy mechanisms for rate setting, case-mix adjustment, and performance-based incentives. Murray's framework aligned hospital incentives with population health goals, reduced unnecessary utilization, and ensured per capita spending growth remained below national trends—establishing Maryland as a **national model for sustainable, value-based health system reform**.

- Financial statements and audited balance sheets from hospitals and affiliated providers;
- Workforce and supply cost data from State and CMS sources; and
- Comparative benchmarks from the Laurel Health national data repository and the Charm Economics cost structure database.

These datasets will be securely integrated using **Laurel Health Advisors' Data Ingestion and Harmonization Framework**, which employs **Python, R, and SQL pipelines** with built-in data validation and de-duplication protocols. The harmonized analytic environment will serve as the basis for cost modeling, scenario testing, and benchmarking analyses.

Modeling Methods and Tools

To identify and quantify cost savings opportunities, the Laurel Health Team will use a combination of **actuarial, econometric, and systems-based models** designed to isolate cost drivers and simulate the impact of potential reforms:

1. Total Cost of Care (TCOC) Model:

Charm Economics will lead the development of a **TCOC model** to measure the aggregate cost per capita across inpatient, outpatient, post-acute, and community-based care. Using **risk-adjusted regression models** (generalized linear models and two-part models in **Stata** and **R**), the team will quantify how structural changes—such as regional service consolidation, expansion of advanced primary care, or telehealth substitution—affect total system spending and unit costs.

2. Service Line Cost and Margin Model:

Laurel Health and Lewis & Ellis will build **cost-allocation and service-line performance models** to identify high-cost, low-margin, or duplicative services across Vermont's hospitals. These models will use **activity-based costing (ABC)** and **multi-input cost decomposition** methods to reveal where consolidation, shared services, or modernization can yield efficiency gains without compromising access or quality. Tools such as **Python's NumPy, Pandas, and SciPy** libraries will support decomposition and optimization analyses.

3. Scenario-Based Savings Simulation:

Laurel Health Advisors will apply **system dynamics and predictive analytics models** using **Vensim** and **R's forecast and prophet packages** to simulate how reforms alter financial trajectories under multiple scenarios. Examples include:

- Shifting low-acuity inpatient care to outpatient or telehealth settings;
- Implementing shared administrative and procurement functions across regions;
- Reducing readmission and potentially avoidable utilization through targeted care coordination;
- Expanding preventive and chronic care management to offset acute care costs.

These simulations will produce quantifiable projections of **short-term (1–3 years)** and **long-term (5–10 years)** cost savings, segmented by payer type and service category.

4. Return on Investment (ROI) and Affordability Index Model:

To link financial outcomes to affordability for Vermonters, the team will develop an **Affordability Index** that measures how projected savings translate into reductions in commercial insurance premiums and out-of-pocket costs. The index will integrate data from the Vermont Health Care Uniform Rate Review and compare per capita costs to national and regional benchmarks.

Analytical Tools and Visualization

All models will be integrated into an **interactive cost-savings dashboard** built in **Power BI**, allowing State officials to visualize savings opportunities by region, service line, and policy lever. The dashboard will feature dynamic filters for scenario comparisons and time horizons, along with key performance indicators such as:

- Percentage change in total cost of care per capita;
- Hospital cost-to-charge ratio and operating margin improvement;
- Projected premium impact by payer type; and
- Estimated reinvestment potential for community and primary care infrastructure.

Pathway from Analysis to Business Value

The cost savings analysis is designed not merely to quantify financial outcomes, but to **drive decision-making and investment prioritization** that achieves the State's end goals of affordability, sustainability, and system efficiency. Each modeling output will be linked to **specific business value propositions**, including:

- **Operational Efficiency:** By identifying redundant service lines, administrative overlap, and sub-scale operations, Vermont will gain actionable pathways for reducing non-value-added expenditures.



- **Payment Reform Support:** The findings will directly inform GMCB's payment reform strategies by quantifying how bundled payments, global budgets, or value-based reimbursement models affect costs and margins.
- **Workforce Optimization:** The analysis will link staffing cost and productivity data to regional service demand, supporting workforce realignment that maintains service access while controlling labor costs.
- **Affordability Translation:** By connecting cost savings projections to premium rate modeling, the analysis will demonstrate how system efficiencies translate to affordability improvements for consumers and employers.

The result will be a **direct line of sight between data-driven findings and financial decision-making**, enabling AHS and GMCB to target reforms that deliver verifiable economic and public value.

Alignment with Vermont's Goals

This technical approach directly supports Vermont's statutory and policy objectives under Acts 167 and 68 by providing an analytical foundation for **sustainable, equitable, and data-validated cost reform**. By integrating advanced modeling, benchmarking, and scenario-based simulation, the Laurel Health Team will enable the State to:

- Identify and implement reforms that **measurably lower system costs**;
- Demonstrate **transparent financial stewardship** to stakeholders and residents; and
- Reinforce Vermont's leadership as a national model for **affordable, high-value healthcare transformation**.

Through this rigorous, replicable approach, the analysis under Task 2.2.1 will not only identify where savings can be realized but also **translate those savings into enduring business value**, supporting a sustainable and affordable healthcare future for all Vermonters.

Task 2.2.2: Customer Service Improvement – Enhancing Hospital and Regional Service Quality, Efficiency, and Access through Evidence-Based Modeling and Benchmarking

We will implement a comprehensive, data-informed, and performance-driven technical approach to improve hospital and regional service quality, operational efficiency, and patient access throughout Vermont. This task directly supports the State's goal of enhancing **customer service**, defined broadly as the experience, quality, and responsiveness of healthcare delivery to Vermonters, through the use of **evidence-based modeling, performance benchmarking, and process optimization**. Our methodology is designed to provide AHS and GMCB with measurable and actionable insights into how hospitals and regions can improve patient-centered performance, reduce avoidable utilization, and strengthen trust and satisfaction among patients, providers, and communities.

Integrated Performance and Quality Framework

The foundation of this work will be the development of an **Integrated Service Quality and Access Performance Framework (ISQAPF)**, which will combine quantitative modeling, qualitative benchmarking, and outcome measurement into a unified analytic structure. This framework will evaluate **efficiency, effectiveness, and experience of care** through four domains:

1. **Clinical Quality** – safety, effectiveness, readmission, and preventable event metrics.
2. **Operational Efficiency** – throughput, wait times, and resource utilization indicators.
3. **Access and Equity** – geographic and demographic accessibility, provider availability, and timeliness of care.
4. **Patient Experience and Responsiveness** – satisfaction, communication, and coordination of care.

The ISQAPF will integrate data from hospital reporting, all-payer claims, the Vermont Uniform Hospital Discharge Dataset, CMS Hospital Compare, and hospital-reported quality metrics.

Modeling Methodology and Analytic Tools

To systematically assess and improve customer service performance, the Laurel Health Team will use a multi-model analytic approach that blends **predictive analytics, benchmarking, and operational simulation**.

1. Predictive Service Quality Modeling:

Using **Python (scikit-learn, XGBoost)** and **R (caret and randomForest)**, we will build predictive models that forecast changes in service quality, patient satisfaction, and access outcomes under various reform scenarios. Inputs will include hospital staffing levels, service line adjustments, telehealth adoption rates, and patient flow data. Outputs will project potential changes in:

- Hospital star ratings and quality composite scores;
- Average emergency department (ED) and specialty wait times;
- Patient satisfaction (HCAHPS) indicators; and
- Avoidable utilization rates and clinical safety events.

Predictive models will be validated against historical hospital performance data and national comparators to ensure reliability and policy relevance.

2. Operational Efficiency and Flow Simulation:

We will lead **discrete-event simulation (DES)** and **process-optimization modeling** using **Simio** and **Arena Simulation Software** to analyze patient flow across hospitals and regions. These models will evaluate how proposed reforms (such as shared administrative functions,

centralized scheduling, or extended care networks) affect wait times, throughput, and resource utilization. Outputs will include process maps identifying bottlenecks and quantifying time and cost savings achieved through operational adjustments.

3. **Benchmarking and Performance Indexing:**

Lewis & Ellis will construct a **Hospital Performance and Service Index (HPSI)** to benchmark Vermont hospitals and regions against state, regional, and national standards. Using **CMS 5-Star, Leapfrog, PSI-90, and Joint Commission metrics**, the HPSI will score hospitals on cost efficiency, service quality, and responsiveness. The model will employ **multi-factor regression analysis** and **Data Envelopment Analysis (DEA)** to identify outliers and high-performing exemplars. Hospitals and regions will receive standardized performance profiles that identify specific improvement opportunities aligned with national best practices.

4. **Access and Equity Modeling:**

We will also conduct **geospatial access modeling** using **QGIS** to quantify differences in service accessibility across Vermont's rural and urban regions. The models will map population-weighted drive times, telehealth penetration, and provider-to-population ratios to reveal disparities and guide targeted interventions. These analyses will align directly with the equity goals of **Act 68**, ensuring improved access for underserved populations while maintaining efficiency.

Translating Analysis into Business Value and Customer Impact

Each analytic component is structured to produce measurable business and public value for Vermont. The outputs will directly inform operational, policy, and investment decisions, leading to demonstrable improvements in customer service outcomes.

- **Improved Operational Efficiency:** Simulation models will identify process inefficiencies and recommend streamlined workflows that reduce delays in admissions, discharges, and specialty referrals—enhancing both patient experience and staff productivity.
- **Enhanced Quality and Safety:** Predictive models will enable hospitals to proactively address areas of clinical risk and variation, improving safety and performance metrics while reducing costs associated with adverse events or readmissions.
- **Expanded Access:** GIS-based access models will help AHS and GMCB target investments—such as extended-hour clinics, telehealth expansion, and transportation coordination—to regions with identified access gaps.
- **Patient-Centered Care:** The integration of patient satisfaction and responsiveness metrics into predictive and performance models will create a data-driven basis for improving communication, care coordination, and trust in Vermont's healthcare system.

These improvements will translate to tangible system-level benefits: **shorter wait times, reduced preventable utilization, higher quality ratings, and greater patient satisfaction**. Over time, improved service quality and operational efficiency will contribute to **downward**



pressure on costs, reinforcing Vermont's goals under **Act 167** for sustainable and affordable healthcare delivery.

Alignment with Vermont's Goals

This approach fully aligns with Vermont's vision under **Acts 167 and 68** to strengthen hospital and regional service delivery by embedding **quality, access, and equity** into all aspects of reform. By combining **predictive analytics, simulation modeling, and benchmark-based performance evaluation**, we will deliver a robust framework for continuous improvement that empowers hospitals, policymakers, and communities to act on evidence rather than intuition.

The resulting deliverables, including the **Hospital Performance and Service Index, access improvement maps, and operational simulation dashboards**, will allow Vermont to make **transparent, data-informed decisions** that enhance customer service outcomes while supporting the broader goals of **affordability, sustainability, and health equity**. Through this work, we will help ensure that Vermonters experience a healthcare system that is more accessible, efficient, and responsive to their needs.

Task 2.2.3: Risk Reduction – Reducing Operational, Financial, and Quality-of-Care Risks through Predictive Analytics and Transformation Planning

We will employ a structured and data-intensive methodology to help the State of Vermont identify, quantify, and mitigate **operational, financial, and quality-of-care risks** associated with hospital and regional transformation. This work will enable the **AHS and GMCB** to make informed, evidence-based decisions that anticipate emerging vulnerabilities and promote long-term system stability. Using predictive analytics, scenario modeling, and continuous risk monitoring, the team will provide a forward-looking framework that strengthens resilience across Vermont's hospital and regional healthcare ecosystem.

Comprehensive Risk Modeling Framework

We will establish a **Comprehensive Risk Modeling Framework (CRMF)** that integrates three categories of risk —**operational, financial, and quality-related**—into a single analytic environment. This framework will align with Vermont's legislative mandates under **Act 167** (financial sustainability and system transparency) and **Act 68** (regional equity and access) as well as CMS's **Rural Health Transformation Program** objectives for sustainable care delivery in rural settings.

The CRMF will combine multiple analytic techniques to generate a real-time understanding of risk exposure and the effectiveness of proposed mitigation strategies:

1. **Operational Risk Modeling:** Evaluates hospital and regional system stability by examining utilization trends, workforce dynamics, patient flow, and capacity constraints.
2. **Financial Risk Modeling:** Quantifies fiscal exposure and sustainability using margin forecasts, revenue diversification, and payer-mix projections.

3. **Quality and Safety Risk Modeling:** Predicts how service line changes, workforce variations, and operational restructuring may impact patient safety, outcomes, and experience of care.

Each model will feed into an integrated **Risk Dashboard** that allows AHS and GMCB to assess cumulative and cross-domain risks in real time.

Modeling Methodology and Analytical Tools

We will use a blend of **predictive analytics, econometric simulation, and machine learning** to assess and mitigate system risks.

1. Predictive Risk Analytics:

We will apply **machine learning models**, including **logistic regression, gradient boosting, and Bayesian networks**, to predict hospital distress events such as declining operating margins, reductions in liquidity, or workforce shortages. These models will use multi-year datasets drawn from hospital financial statements, cost reports, and utilization data to identify early warning indicators. Tools used will include **Python (scikit-learn, TensorFlow)** and **R (glmnet, randomForest)**.

For example, a model may project the probability of hospital financial distress based on combinations of payer mix changes, readmission rates, and staffing turnover. The results will inform a **Risk Propensity Score** that allows policymakers to assess which hospitals or regions are most vulnerable to future shocks.

2. Financial Sustainability Simulation:

Lewis & Ellis will develop **financial and actuarial simulation models** to forecast the impact of proposed reforms on hospital solvency and statewide healthcare expenditures. Using **Monte Carlo simulations** and **sensitivity analysis** in **Stata and R**, the team will estimate the range and probability of financial outcomes under different policy and operational scenarios—such as service line realignments, changes in reimbursement models, or cross-border care shifts.

These models will produce **risk-adjusted sustainability metrics**, including operating margin probability distributions, days cash on hand, and debt service coverage ratios, benchmarked against national standards.

3. Operational and Workforce Resilience Modeling:

Charm Economics will lead the development of **systems dynamics models** to simulate how staffing, utilization, and capacity interrelate over time. These models will help Vermont predict and mitigate risks related to workforce shortages, seasonal capacity constraints, and care delivery bottlenecks. For example, simulations will evaluate the downstream effects of nursing shortages on emergency department throughput and patient satisfaction.

4. Quality and Safety Risk Prediction:

We will apply **quality-of-care forecasting models** using regression and classification techniques to estimate how proposed reforms affect performance on key safety and outcome

measures, including **PSI-90, readmissions, mortality, and adverse events**. Predictive models will integrate CMS and Leapfrog data with Vermont hospital-reported quality metrics to identify areas where service reconfiguration could pose quality or patient safety risks.

5. **Composite Risk Index (CRI):**

The team will synthesize results from all models into a **Composite Risk Index**, a multi-dimensional metric that quantifies overall system stability. The CRI will assign weighted scores to operational, financial, and clinical risk factors, enabling AHS and GMCB to rank hospitals or regions by vulnerability and resilience.

Application of Results to Vermont's Transformation Planning

The outputs of this analysis will not only identify risks but also inform **risk mitigation and transformation strategies** tailored to Vermont's hospital system. Each model will be designed to produce **actionable insights**, including:

- Identification of hospitals or regions at high risk of financial or operational instability;
- Quantification of the effects of workforce shortages or service consolidations on system resilience;
- Forecasting of potential access or quality degradation resulting from structural reforms;
- Evidence-based recommendations for financial interventions, technical assistance, or shared services that can stabilize at-risk providers.

The Laurel Health Team will deliver these findings through **interactive risk dashboards and scenario tools** developed in **Tableau** and **Power BI**, allowing AHS and GMCB to conduct “what-if” analyses, adjust model assumptions, and visualize projected outcomes under multiple transformation pathways.

Tools for Risk Reduction and Continuous Monitoring

To ensure ongoing utility, the Laurel Health Team will implement a **Risk Monitoring and Early Warning System (RMEWS)** using **Python-based predictive algorithms** and **Power BI dashboards**. This tool will automatically update risk scores as new data becomes available (e.g., quarterly hospital reporting or claims submissions) and will alert AHS and GMCB to changes in financial or operational conditions that may warrant attention.

Key outputs will include:

- **Early Warning Alerts:** Flags for hospitals showing deteriorating financial, operational, or quality metrics;
- **Scenario Comparison Views:** Visualizations showing how policy changes shift overall system risk profiles;
- **Trend Analysis Panels:** Longitudinal tracking of hospital and regional performance across risk categories; and



- **Mitigation Tracking Reports:** Documentation of interventions and their observed or projected effects.

Pathway from Analysis to Business Value

The Laurel Health Team's risk modeling and predictive analytics will produce **quantifiable business and policy value** that directly supports Vermont's strategic objectives:

- **Operational Stability:** Early detection of capacity and workforce challenges will enable AHS and GMCB to take targeted corrective actions before service disruptions occur.
- **Financial Resilience:** Predictive cost and margin modeling will identify hospitals requiring rate adjustments or transformation support, reducing the likelihood of insolvency.
- **Quality Assurance:** Real-time monitoring of safety and outcome metrics will minimize patient harm, reputational damage, and regulatory penalties.
- **Systemic Efficiency:** Integrated analysis across domains will help Vermont balance reform ambition with risk mitigation, ensuring transformation initiatives strengthen rather than destabilize the health system.

Alignment with Vermont's Goals

This risk-reduction approach directly supports Vermont's vision for a **sustainable, resilient, and equitable healthcare system**, as articulated in **Acts 167 and 68**. By using predictive analytics, simulation modeling, and continuous monitoring, the Laurel Health Team will enable the State to proactively manage risk while pursuing transformative reforms. The deliverable, including the **Composite Risk Index, Risk Dashboard**, and **RMEWS toolkit**, will provide Vermont with the analytical infrastructure necessary to sustain high-quality care, protect public investment, and maintain access for all Vermonters.

Through this structured, data-driven, and transparent approach, the Laurel Health Team will help Vermont move from reactive problem-solving to **anticipatory system management**, achieving measurable reductions in operational, financial, and quality-of-care risks while advancing the State's overarching goals of affordability, sustainability, and patient-centered excellence.

Task 2.2.4: Compliance – Ensuring Transformation Initiatives Meet State and Federal Standards and Align with CMS and Vermont Legislative Requirements

We will implement a rigorous, structured, and transparent approach to ensure that all transformation initiatives evaluated and modeled under this engagement comply fully with **State and Federal regulatory requirements**, including those established under **Vermont Acts 167 and 68** and the **Centers for Medicare & Medicaid Services (CMS) Rural Health Transformation Program**. This task will integrate policy, regulatory, and data compliance

processes into a unified analytical framework that ensures every proposed reform is **legally sound, technically defensible, and eligible for Federal financial participation (FFP)**.

The approach will focus on embedding compliance considerations throughout the analytic lifecycle, from data ingestion and modeling through to reporting and stakeholder communication, so that the resulting insights are both actionable and regulation-ready.

Compliance Integration Framework

The Laurel Health Team will establish a **Compliance Integration Framework (CIF)** designed to align all analytic and modeling activities with Vermont's legislative mandates and Federal program standards. The CIF will ensure that data collection, analytic methodologies, and deliverables adhere to all applicable **CMS, AHS, and GMCB** requirements for transparency, auditability, and reproducibility.

The framework will be structured around five interrelated compliance domains:

1. **Statutory Alignment:** Ensuring consistency with Vermont Acts 167 and 68 regarding affordability, access, transparency, and sustainability.
2. **Federal Program Compliance:** Aligning with CMS Rural Health Transformation Program, Medicaid Managed Care Rules (42 CFR §438), and Medicare Cost Report integrity standards.
3. **Data Governance and Security:** Meeting Federal HIPAA, 45 CFR §164, 42 CFR Part 2, and Vermont-specific data protection requirements for de-identified data handling and reporting.
4. **Methodological Integrity:** Ensuring models meet Federal standards for statistical transparency, replicability, and documentation.
5. **Reporting and Documentation Compliance:** Providing deliverables and reports that satisfy CMS, AHS, and GMCB audit, monitoring, and reporting requirements.

Technical Approach and Tools

To operationalize compliance throughout the project lifecycle, the Laurel Health Team will deploy a **Compliance Assurance Pipeline (CAP)**, a structured process that uses analytical and documentation tools to ensure that every data operation, model, and output conforms to relevant rules and guidance.

1. Regulatory Crosswalk Mapping:

We will develop a **compliance crosswalk matrix** that maps every element of Vermont's legislative and regulatory framework to corresponding modeling and reporting tasks. The matrix will include:

- Act 167 provisions on hospital financial oversight and affordability;
- Act 68 provisions on access, equity, and regional planning;



- CMS RHTP key performance indicators (KPI) on cost containment, sustainability, and access;
- HIPAA and 42 CFR Part 2 provisions on data security and PHI handling; and
- 2 CFR §200.321 requirements for socioeconomic contracting and Federal funding compliance.

Each task will have defined compliance checkpoints and responsible team leads to ensure traceability from policy requirement to analytic output.

2. **Automated Data Governance and Validation:**

Using **Python (pandas, cerberus)** and **R (validate, data.table)**, we will create automated scripts to validate datasets for completeness, accuracy, and de-identification prior to modeling. Each dataset will be reviewed against Vermont's data use agreements and CMS privacy standards, ensuring that no protected information is mishandled or disclosed during processing. Audit logs will document all data transformations for compliance verification.

3. **Model Validation and Auditability:**

Lewis & Ellis will ensure that all financial, quality, and access models meet Federal and State standards for methodological transparency and reproducibility. This will include:

- Cross-validation of cost and access projections using benchmark datasets (e.g., CMS Cost Reports, HRSA Access Data, AHRQ Quality Indicators);
- Peer review of analytic code and assumptions using **Git-based version control**; and
- Documentation of model structure, parameters, and versioning in compliance with **CMS Model Integrity Standards** and **AHS analytic documentation protocols**.

4. **Legislative and Federal Compliance Testing:**

We will employ a **Compliance Testing Engine (CTE)** built in **R Shiny** to test modeled outcomes against policy thresholds. For example:

- Ensuring service line reductions do not violate network adequacy thresholds established by Act 68;
- Verifying financial sustainability benchmarks under Act 167 are maintained;
- Confirming that reform initiatives meet CMS RHTP criteria for maintaining rural access and avoiding cost shifting; and
- Assessing adherence to State workforce and staffing adequacy mandates.

The CTE will allow AHS and GMCB staff to visualize compliance risks and identify specific adjustments needed to fully align transformation scenarios.



5. Compliance Reporting and Documentation:

All compliance documentation, including data lineage, analytic assumptions, model validations, and audit logs, will be compiled into a **Compliance and Assurance Report (CAR)** delivered with each major analytic output. Reports will follow CMS audit templates and Vermont AHS reporting standards, ensuring the State's ability to submit compliant documentation for Federal review and funding.

In each instance, the Laurel Health Team delivered **audit-ready, fully documented analytic products** that met the dual requirements of methodological rigor and regulatory compliance.

Deliverables and Analytical Tools

Deliverables for this task will include:

- **Compliance Crosswalk Matrix:** A detailed mapping of Vermont's legislative and CMS program requirements to project deliverables.
- **Automated Validation and Audit Scripts:** Tools that verify data accuracy, privacy compliance, and model integrity.
- **Compliance Testing Engine (CTE):** A user-friendly analytic tool allowing AHS and GMCB staff to test transformation scenarios for compliance with State and Federal standards.
- **Compliance and Assurance Report (CAR):** Comprehensive documentation of compliance status, gaps, and corrective actions for each project milestone.

All deliverables will be maintained in a secure, version-controlled repository and accessible through an audit trail consistent with **State and CMS data retention and verification requirements**.

Pathway from Analysis to Business Value

The integration of compliance processes into analytic workflows will provide Vermont with tangible and lasting business value:

- **Reduced Regulatory Risk:** Proactive identification of compliance gaps will prevent delays, denials, or penalties in CMS and State program reviews.
- **Enhanced Accountability:** Transparent documentation will strengthen public confidence in Vermont's transformation process.
- **Federal Funding Assurance:** Proper alignment with CMS RHTP and Medicaid rules will protect Vermont's eligibility for continued Federal financial participation and technical support.
- **Operational Efficiency:** Automated compliance verification tools will reduce administrative burden and accelerate the policy review cycle for proposed reforms.



- **Policy Credibility:** By ensuring all modeling and analysis adhere to legislative and Federal standards, Vermont will reinforce its reputation as a national leader in transparent, accountable, and data-driven healthcare transformation.

Alignment with Vermont's Goals

This compliance methodology is fully aligned with the objectives of **Acts 167 and 68**, as well as Vermont's participation in the **CMS Rural Health Transformation Program**. The Laurel Health Team's integrated compliance approach will enable the State to confidently advance transformative reforms, knowing that each initiative has been evaluated for legal sufficiency, methodological rigor, and Federal alignment.

By institutionalizing compliance throughout Vermont's hospital transformation process, this task ensures that **innovation and accountability progress hand in hand**, supporting the State's long-term goals of **sustainability, transparency, and equitable healthcare for all Vermonters**.

Task 2.2.5: Transparency and Accountability – Delivering Evidence-Based Reports and Visualizations to Support Open, Data-Driven Decision-Making

We will establish a robust and transparent analytic and reporting framework to ensure that all outputs produced under this engagement are **accessible, interpretable, and verifiable** by the State of Vermont, the **AHS and GMCB**, and the public. This task is central to Vermont's commitment under **Act 167** and **Act 68** to strengthen government and stakeholder trust through clear, evidence-based communication of the rationale and impact of healthcare transformation initiatives.

The team's approach will combine **data visualization, methodological transparency, and accountability reporting** into an integrated platform that documents every analytic step from data ingestion through to decision support. The outcome will be a set of reports and dashboards that enable policymakers, providers, and the public to trace how data-driven analyses translate into hospital- and regional-level transformation recommendations, ensuring accountability, reproducibility, and confidence in the State's reform process.

Transparency and Accountability Framework

The Laurel Health Team will implement a **Transparency and Accountability Framework (TAF)** that standardizes how data, models, and findings are shared and communicated. The framework will be grounded in three guiding principles:

1. **Traceability** – Every analytic process, dataset, and model will include documentation linking inputs, assumptions, and results to their original sources.
2. **Reproducibility** – All models and analyses will be fully version-controlled, allowing State analysts and auditors to replicate findings independently.

3. **Accessibility** – Reports, dashboards, and visualizations will be designed for diverse audiences, including policymakers, technical reviewers, hospital administrators, and the general public.

The TAF will operationalize these principles through a combination of structured documentation, interactive data visualization, and public-facing communication tools.

Technical Approach and Analytical Tools

1. Comprehensive Analytic Documentation and Data Lineage Tracking:

Laurel Health Advisors will use **Git-based version control** and **automated data lineage tools** (e.g., **OpenLineage**, **Great Expectations**, and **DataHub**) to track every transformation, cleaning, and modeling step performed on State and vendor-provided datasets. Each analytic product will include a corresponding **Data Lineage Report** that describes data sources, transformation logic, assumptions, and limitations.

All models will adhere to the **Transparency in Algorithms and Modeling Standards (TAMS)** framework developed by Laurel Health Advisors and Charm Economics, which specifies how input parameters, data sources, and code documentation must be presented for public accountability.

2. Interactive Reporting and Visualization:

The team will develop **interactive dashboards and reports** using **Tableau**, **Power BI**, and **R Shiny**, tailored for both internal (AHS, GMCB) and public audiences. These dashboards will display:

- Key metrics on cost, access, and quality outcomes under reform scenarios;
- Comparative benchmarks against national and regional peers;
- Forecasted impacts of transformation initiatives at hospital, regional, and statewide levels; and
- Indicators of system stability, such as hospital margin forecasts, bed occupancy, and workforce sustainability.

Visualizations will include drill-down capabilities, scenario toggles, and temporal views that allow users to see how reforms evolve over time.

3. Public Transparency Portal:

Laurel Health Advisors will develop a **Transparency Portal**—a web-based dashboard and data library for Vermont's hospital transformation initiative. Built with **Tableau Public** and **ArcGIS Online**, the portal will provide curated, de-identified summaries of:

- Regional access modeling and service distribution maps;
- Hospital and regional performance benchmarks;



- Progress toward Act 167 and Act 68 goals (affordability, access, and sustainability); and
- Metrics aligned with CMS's Rural Health Transformation Program.

The portal will enhance Vermont's capacity for stakeholder engagement and public accountability by providing clear, evidence-based visual narratives.

4. **Model Audit and Verification:**

Charm Economics and Lewis & Ellis will lead a **model verification and auditing process** to ensure that all analytic methods, formulas, and assumptions are documented and can be independently reviewed. Each analytic deliverable will include:

- **Methodology Annexes** detailing model specifications, data inputs, and sensitivity testing;
- **Validation Summaries** comparing model predictions with observed data; and
- **Reviewer Sign-off Logs** documenting quality assurance checks by subject matter experts.

5. **Performance and Accountability Reporting:**

The team will generate quarterly **Transformation Accountability Reports (TARs)** for AHS and GMCB, summarizing analytic findings, scenario outcomes, and progress toward legislative and CMS compliance goals. Each report will include key indicators of transparency and accountability, such as:

- Data quality and completeness scores;
- Documentation compliance rates;
- Variance analyses between modeled and observed outcomes; and
- Recommended corrective actions for identified data or performance gaps.
- **Washington State – Thriving Together NCW:** The team delivered an interactive analytics portal visualizing hospital access and performance data for rural communities, supporting community and regional decision-making.

Each of these projects combined transparency, stakeholder communication, and analytic rigor to build trust and accountability in the policymaking process.

Deliverables

The Laurel Health Team will produce a suite of deliverables designed to institutionalize transparency and accountability within Vermont's hospital transformation process:

- **Data Lineage and Methodology Reports** documenting all data and analytic operations;
- **Interactive Dashboards** for internal and public use;



- **Quarterly Transformation Accountability Reports (TARs)** summarizing analytic findings and performance;
- **Transparency Portal** providing curated, de-identified, and accessible information for public stakeholders; and
- **Audit-Ready Documentation Packages** for CMS and State review.

All deliverables will meet accessibility standards (Section 508 compliance) and will be designed for long-term use by AHS and GMCB beyond the life of this contract.

Pathway from Analysis to Business Value

This transparency and accountability framework produces both **governance value** and **operational efficiency** for Vermont:

- **Increased Public Trust:** Clear, accessible communication of data and findings will reinforce the integrity of the State's healthcare transformation process.
- **Policy Credibility and Defensibility:** Documentation and audit trails will enable AHS and GMCB to substantiate decisions with verifiable evidence, protecting against legal or stakeholder challenges.
- **Enhanced Decision Efficiency:** Interactive tools and dashboards will allow policymakers to quickly access key insights, reducing time spent interpreting or reconciling static reports.
- **Continuous Learning:** The transparent feedback loop between model results and observed outcomes will promote adaptive management and continuous improvement across transformation initiatives.
- **Replicability and Reuse:** Standardized methods and documentation will enable Vermont to reuse analytic infrastructure for future reform efforts, creating sustained return on investment.

Alignment with Vermont's Goals

This technical approach directly supports Vermont's legislative priorities and the **CMS Rural Health Transformation Program** requirement for transparent, evidence-based reporting. By embedding transparency and accountability into every phase of analysis, the Laurel Health Team will provide the State with a clear, defensible, and reproducible foundation for all transformation decisions.

Through this framework, Vermont will not only meet but exceed national best practices in healthcare transparency—demonstrating how **open, data-driven governance** can drive smarter, more equitable, and more sustainable healthcare reform for all Vermonters.

Past Performance

Our collective past performance demonstrates proven expertise in **hospital system modeling, Medicaid and Medicare program analytics, value-based care design, and policy alignment** across diverse healthcare environments. Over the past decade, the team has partnered with organizations such as the **Centers for Medicare & Medicaid Services (CMS)**, multiple State health departments, and leading hospital systems to develop **data-driven strategies that improve quality, reduce costs, and enhance access** to care. Our portfolio includes projects that align with the objectives of this engagement, developing **integrated analytical frameworks, predictive models, and benchmarking tools** to guide reform under complex legislative and regulatory requirements. These experiences have equipped our team with the technical capacity, methodological rigor, and practical understanding needed to deliver the **evidence-based insights and actionable recommendations** required by the State of Vermont. The following section highlights representative examples of past performance that demonstrate our ability to achieve measurable results and support sustainable healthcare transformation at both the State and system level

Experience with Vermont

Laurel Health Advisors has a longstanding and trusted relationship with the **State of Vermont**, supporting the development and advancement of the State's data infrastructure and health system transformation initiatives. Working in collaboration with the **Vermont Department of Health (VDH)** and the **Agency of Human Services (AHS)**, Laurel Health Advisors played a key role in the **design and implementation of a statewide data governance framework** that strengthened the Department's capacity to collect, manage, and use health data effectively across programs. This work involved establishing standardized data governance policies, defining stewardship roles and data-sharing protocols, and developing a unified framework for **data quality, accessibility, and interoperability** across the Department's public health, environmental health, and healthcare delivery programs. Laurel Health Advisors guided the operationalization of these governance policies through structured data management workflows, cross-agency coordination processes, and performance metrics that enabled VDH to align its data governance practices with federal expectations for transparency, privacy, and data-driven decision-making.

In parallel, Laurel Health Advisors supported Vermont in operationalizing **and managing the State's Health Information Exchange (HIE)**, assisting with integrating clinical, administrative, and public health data streams to create a cohesive data-sharing ecosystem. This work involved developing and refining HIE data ingestion and normalization protocols, improving data accuracy and completeness, and aligning the HIE's operations with the **Vermont Health Information Technology (HIT) Plan**. Laurel's work enabled the HIE to serve as a central infrastructure for interoperability, supporting initiatives in care coordination, quality improvement, and population health analytics. Through this effort, Laurel Health Advisors helped Vermont strengthen its ability to use real-time, multi-sector data to inform health system transformation and measure performance across hospitals, providers, and community-based organizations.

In concert with these efforts, **Robert Murray**, a recognized national expert in hospital rate regulation and payment reform, worked with Vermont to evaluate the feasibility and design parameters of a **global budgeting approach** tailored to the State's hospital system. Drawing from his experience as the principal architect of the **Maryland Total Cost of Care (TCOC) Model**, Murray provided technical and policy guidance to Vermont policymakers and the **Green Mountain Care Board (GMCB)** on how global budgets could stabilize hospital finances, control cost growth, and incentivize quality improvement. His work involved analyzing hospital cost structures, payer mix, and revenue trends to inform rate-setting strategies that balance fiscal sustainability with equitable access to care. Murray's expertise helped Vermont assess how Maryland's regulatory and payment methodologies could be adapted to Vermont's unique healthcare landscape, contributing to the State's ongoing exploration of **system-wide financial reform and all-payer rate regulation**.

Other Past Performance

Laurel Health Advisors has extensive and directly relevant experience working across multiple states to support **Medicaid program analysis, evaluation, and transformation**—experience that aligns closely with the scope and objectives of this proposal. Our team has provided analytic, policy, and technical assistance to state Medicaid agencies, health authorities, and legislative oversight bodies, building data-driven frameworks to assess cost, access, quality, and system performance. Our work integrates **claims analytics, financial modeling, data governance, and program evaluation** to help states design sustainable payment and delivery reforms that strengthen their Medicaid programs and improve population health outcomes.

In **New Mexico**, Laurel Health Advisors worked with the **Legislative Finance Committee (LFC)** to conduct a comprehensive review of Medicaid's financial performance and program impact. This work involved a detailed analysis of Medicaid spending trends, managed care organization (MCO) performance, and the relationship between Medicaid expansion and statewide healthcare utilization. The team evaluated hospital cost reports, uncompensated care patterns, and provider reimbursement structures to identify opportunities for improved fiscal management and policy alignment. Our findings informed the LFC's legislative recommendations on Medicaid rate-setting, value-based contracting, and health system accountability—helping New Mexico ensure that its Medicaid program delivered measurable results in cost efficiency and care quality.

In **Oklahoma**, Laurel Health Advisors supported the **Oklahoma Health Care Authority (OHCA)** in its Medicaid transformation and health system performance analytics. We led the integration of Medicaid claims, provider cost, and workforce data to evaluate the efficiency and sustainability of hospital operations under managed care transition scenarios. This work produced predictive models that identified financially vulnerable hospitals and quantified the impact of proposed Medicaid rate changes on rural access and total system cost. The analysis provided OHCA with actionable insights to guide reimbursement adjustments, provider stabilization strategies, and targeted rural health investments.

In **Berrien County, Michigan**, Laurel Health Advisors developed a **social determinants of health (SDOH) and Medicaid alignment framework** to help local policymakers and healthcare



organizations understand the intersection between Medicaid coverage, population health, and socioeconomic conditions. Our analysis assessed Medicaid enrollment, utilization, and service gaps among low-income and rural populations and proposed data governance structures to coordinate better local health and social service programs funded through Medicaid and community-based initiatives.

In **St. Louis, Missouri**, Laurel Health Advisors collaborated with regional healthcare providers and local government to analyze **Medicaid expansion impacts on urban hospitals**, focusing on access equity, uncompensated care reduction, and patient flow management. We provided strategic recommendations on aligning hospital systems and Medicaid managed care networks to improve capacity planning and resource allocation in high-need neighborhoods.

In **Washington State**, Laurel Health Advisors partnered with the State's Medicaid program and local coalitions under the **Thriving Together NCW** initiative to integrate Medicaid and public health data in support of rural health system transformation. We developed analytics and scenario models to assess Medicaid-funded service access, hospital sustainability, and telehealth adoption across four rural counties (Okanogan, Grant, Chelan, and Douglas). Our work demonstrated the impact of care coordination and virtual care services on Medicaid outcomes, supporting Washington's strategy to strengthen rural delivery networks under Medicaid managed care.

In **Oregon**, Laurel Health Advisors supported the State's Medicaid transformation initiatives by advising on **data integration, quality measurement, and program evaluation** for coordinated care organizations (CCOs). Our work focused on aligning CCO performance metrics with federal Medicaid standards, developing provider dashboards, and helping Oregon improve transparency in Medicaid financial reporting.

In **New Jersey**, Laurel Health Advisors partnered with the **Department of Health (DOH)** to evaluate **telehealth and Medicaid policy impacts** following the expansion of remote care delivery during and after the COVID-19 pandemic. We conducted a multi-dimensional analysis of Medicaid claims, hospital data, and patient outcomes to determine how telehealth utilization affected access, cost, and quality for Medicaid beneficiaries. Our findings guided the State's decisions on telehealth reimbursement parity and program integration within Medicaid.

In partnership with **AARP**, Laurel Health Advisors conducted a nationwide evaluation of the **impact of Medicaid expansion on hospital uncompensated care**. The analysis compared states that expanded Medicaid under the Affordable Care Act (ACA) with those that did not, using cost report and claims data to quantify reductions in uncompensated care costs, charity care, and bad debt. The results demonstrated significant improvements in hospital financial stability and access to care in expansion states, providing AARP and policymakers with evidence to support continued Medicaid expansion and improved payment equity.

Across these engagements, Laurel Health Advisors has consistently applied **advanced analytic modeling, cross-sector data integration, and evidence-based evaluation methods** to support Medicaid programs and their partners in achieving cost containment, quality improvement, and access equity. These experiences reflect our deep understanding of how Medicaid policy, finance, and operations intersect—and our ability to deliver actionable insights



that help states like Vermont build sustainable, transparent, and high-performing healthcare systems.

Lewis & Ellis has served as a long-term actuarial partner to the Texas Health and Human Services Commission (HHSC), developing and certifying **Medicaid managed care capitation rates** for multiple program lines, including STAR, STAR+PLUS, STAR Kids, and CHIP. The firm's work included establishing actuarially sound rates in compliance with **42 CFR §438**, incorporating medical trend analyses, utilization projections, and acuity adjustments across populations with complex needs. L&E also designed and implemented **risk adjustment methodologies** to ensure equitable rate structures for plans serving high-acuity enrollees, as well as financial monitoring frameworks to validate plan solvency and compliance. This work has directly supported the fiscal stability and operational efficiency of one of the nation's largest Medicaid managed care programs.

Lewis & Ellis provided actuarial and analytic support to the **Iowa Department of Health and Human Services** in the development and monitoring of Medicaid managed care capitation rates. L&E developed **risk adjustment methodologies** using demographic, diagnostic, and utilization-based models to ensure fair and equitable payments to managed care plans. The firm also conducted retrospective reviews to assess financial performance against budgeted projections, identifying drivers of cost and utilization variation. Their work enhanced the State's ability to manage financial risk and maintain compliance with CMS's Managed Care Final Rule requirements.

Lewis & Ellis assisted the **Arkansas Department of Human Services** in revising its **Medicaid provider reimbursement methodologies** for hospitals and nursing facilities. The firm performed comprehensive analyses of Medicaid cost reports, rebasing calculations, and inflationary trends to support updates to rate-setting policies under the State Plan Amendment (SPA). This included designing cost-based and case-mix adjusted payment models that balanced fiscal integrity with access to care. L&E's work improved transparency and predictability in provider payments and ensured compliance with CMS documentation and audit standards.

For the **Kansas Department of Health and Environment (KDHE)**, L&E developed long-range **Medicaid expenditure forecasts and fiscal impact analyses** to support policy decisions regarding program expansion and payment reform. The firm modeled multiple enrollment and expenditure scenarios, evaluated the financial effects of policy alternatives, and provided CMS-compliant actuarial certifications. This work equipped KDHE with robust projections to guide legislative and budgetary planning for its Medicaid program.

Personnel

The success of this engagement will be driven by the depth of the Laurel Health Team's expertise and proven track record. Together, our personnel bring decades of experience in **healthcare policy, Medicaid program management, actuarial analysis, data integration, and health system transformation** at the state and national levels. The team includes



nationally recognized leaders who have designed and implemented some of the country's most advanced health reform initiatives, including **statewide total-cost-of-care models, Medicaid transformation programs, and hospital sustainability frameworks**. Each team member has direct experience working with state agencies, hospital systems, and Medicaid programs, enabling them to deliver practical, actionable solutions that align with Vermont's legislative mandates under **Acts 167 and 68**, as well as the **CMS Rural Health Transformation Program**. Together, this multidisciplinary team will provide the analytical rigor, strategic insight, and operational expertise necessary to help Vermont achieve a transparent, sustainable, and high-performing healthcare system

Dr. Yael Harris, PhD, MHS

Dr. Yael Harris is a nationally recognized leader in health policy, data analytics, and health system transformation, with over 25 years of experience directing large-scale federal and state initiatives to improve the quality, accessibility, and efficiency of healthcare. As the **Chief Executive Officer of Laurel Health Advisors**, Dr. Harris leads multidisciplinary teams in designing and implementing evidence-based solutions that strengthen health system performance, advance Medicaid and Medicare innovation, and promote equitable care delivery.

Dr. Harris previously served as **Director of the Office of Health Information Technology and Quality** at the **Health Resources and Services Administration (HRSA)** within the U.S. Department of Health and Human Services (HHS), where she oversaw national programs on telehealth, health IT, and quality improvement in community health centers. Under her leadership, HRSA expanded health IT adoption, launched the national Telehealth Resource Center Program, and developed quality frameworks to support safety net providers in improving clinical outcomes.

She has led major research and technical assistance initiatives on behalf of the **Centers for Medicare & Medicaid Services (CMS)**, the **Office of the National Coordinator for Health IT (ONC)**, and the **Substance Abuse and Mental Health Services Administration (SAMHSA)**, focusing on interoperability, Medicaid program evaluation, and care coordination. Dr. Harris's recent work includes leading Laurel Health Advisors' contracts with **HRSA, New Jersey Department of Health, Washington State Medicaid, Connecticut Department of Social Services**, and the **Vermont Department of Health**, advancing telehealth, Medicaid transformation, and data governance strategies.

Dr. Harris holds a **PhD in Public Policy** from the **University of Maryland at Baltimore County** and a **Master of Health Science (MHS)** in Health Policy and Management from **Johns Hopkins University**. She is widely published in peer-reviewed journals and has served as a policy advisor to federal and state agencies on digital health, quality measurement, and data modernization.

Jason Goldwater, MPA, MA

Jason Goldwater is a senior health policy and data systems expert with more than 20 years of experience leading **Medicaid analytics, health information exchange (HIE) implementation, and hospital system transformation projects** for state and federal agencies. At **Laurel Health Advisors**, Mr. Goldwater leads initiatives focused on data integration, predictive analytics, and performance evaluation to inform health policy and system redesign.

Mr. Goldwater has served as a principal investigator and data strategist for projects funded by the **Centers for Medicare & Medicaid Services (CMS)**, the **Office of the National Coordinator for Health IT (ONC)**, and multiple state Medicaid agencies. He has developed and implemented **data ingestion and interoperability frameworks, Medicaid program dashboards, and population health analytics** that enable data-driven decision-making across large, multi-payer health systems. His work has supported **Oklahoma Medicaid (OHCA)**, **Connecticut Department of Social Services**, **New Jersey Department of Health**, **Washington State's Medicaid Transformation Project**, and **Vermont's Department of Health**.

Mr. Goldwater has deep experience in **hospital financial and clinical data integration**, including developing modeling tools for value-based contracting, risk adjustment, and total cost of care analyses. His recent work includes leading the design of Vermont's **Data Governance and Health Information Exchange (HIE) Operational Framework**, as well as supporting the **Minnesota Department of Human Services** in expanding the capacity of home and community-based service providers through data-driven performance systems.

Previously, Mr. Goldwater served as **Senior Director at NORC at the University of Chicago**, where he led national research and technical assistance programs for CMS and HHS, and as **Senior Policy Advisor at the Office of the National Coordinator for Health IT**, where he contributed to the development of ONC's **Nationwide Health Information Network (NHIN)** and **Meaningful Use** programs. He holds master's degrees in **Public Administration** and **Political Communication** and is a recognized national expert in **data interoperability, predictive modeling, and Medicaid program analytics**.

Dr. Adam Block, PhD

Dr. Adam Block is a nationally recognized **health economist** and the **President of Charm Economics**, specializing in hospital finance, health insurance markets, and healthcare policy modeling. With over 20 years of experience in healthcare economics and regulation, Dr. Block has developed and implemented complex models that assess the financial and operational impacts of hospital and health system reforms for state agencies, insurers, and provider networks.

Dr. Block's expertise includes **rate setting, cost modeling, market simulation, and value-based payment system design**. He has supported state and federal clients in developing **all-payer rate models, global budget frameworks, and Medicaid payment reform analyses**. A former economist for the **Centers for Medicare & Medicaid Services (CMS)**, Dr. Block contributed to the development of the **Affordable Care Act's rate review and cost containment policies**. He has also served as a faculty member at **New York Medical College**, where he teaches health economics, finance, and quantitative policy analysis. Dr. Block holds a

PhD in Health Economics from Harvard University and a BA in Economics from Tufts University.

Robert Murray, MA

Robert Murray is one of the nation's foremost experts in **hospital rate regulation, payment reform, and total-cost-of-care modeling**. As a **Senior Advisor** with Charm Economics, he provides strategic and technical guidance to states pursuing payment innovation and financial sustainability within their hospital systems. Mr. Murray served for nearly 20 years as the **Executive Director of the Maryland Health Services Cost Review Commission (HSCRC)**, where he was the principal architect of the **Maryland All-Payer Rate Setting System** and the **Maryland Total Cost of Care (TCOC) Model**, the first and only statewide hospital global budgeting system approved by CMS. Under his leadership, Maryland achieved national recognition for controlling hospital cost growth, improving quality outcomes, and maintaining financial viability across diverse hospital settings.

In Vermont, Mr. Murray advised the **Green Mountain Care Board (GMCB)** and the **Agency of Human Services (AHS)** on the application of global budgeting principles to hospital financial oversight and rate regulation. He provided technical and policy recommendations for adapting Maryland's TCOC methodologies to Vermont's rural healthcare landscape, emphasizing financial accountability, transparency, and protection of access.

Yidan Zhang, PhD

Yidan Zhang is a **Senior Health Economist** with extensive experience in **predictive modeling, statistical analysis, and healthcare data integration**. At Laurel Health Advisors, she leads the design and implementation of **data normalization, machine learning, and econometric models** that power state and federal health analytics projects.

Ms. Zhang has contributed to projects for **CMS, the Connecticut Department of Social Services, and New Mexico Medicaid**, focusing on data ingestion, quality control, and crosswalk creation for large administrative and clinical datasets. She has developed algorithms for **risk stratification, cost trend prediction, and network adequacy modeling**, supporting state Medicaid agencies in evidence-based policy planning. She holds a **PhD in Applied Economics from Emory University** and is proficient in **Python, R, and SQL**.

Dr. Fadi Martinos, PhD, MS

Dr. Fadi Martinos specializes in **health system simulation, econometric forecasting, and cost efficiency modeling**. Dr. Martinos has led the development of **dynamic simulation models** that assess the effects of policy and operational changes on healthcare system performance. He has contributed to projects for **Oregon Medicaid** and the **Washington State Health Authority**, applying advanced statistical and econometric methods to support hospital and Medicaid transformation initiatives. Dr. Martinos holds a **PhD in Applied Economics** and a **Master's degree in Econometrics from the University of Iowa**, and he has published extensively on healthcare market behavior and cost modeling methodologies.

Dr. Beth Linas, PhD, MHS

Dr. Beth Linas is an epidemiologist and population health scientist with more than 15 years of experience designing and implementing **data-driven public health, behavioral health, and healthcare evaluation studies**. She has led projects focused on **health outcomes measurement, Medicaid population analysis, and the integration of social determinants of health (SDOH)**. Dr. Linas's expertise lies in applying advanced epidemiologic methods and data integration techniques to inform Medicaid and public health policy. Her work has helped agencies align clinical, social, and economic data to assess program effectiveness.

Previously, Dr. Linas served as a **Research Scientist at Johns Hopkins University** and as a **Senior Epidemiologist** at the **National Institutes of Health (NIH)**, where she led multi-site studies on infectious disease surveillance, behavioral health, and social risk factors affecting vulnerable populations. She holds a **PhD in Epidemiology** and a **Master of Health Science (MHS)** from the **Johns Hopkins Bloomberg School of Public Health**. She is recognized nationally for her work at the intersection of data science, public health, and health equity.

Daniel Moore, FSA, MAAA, CERA

Daniel Moore is a **Vice President and Principal at Lewis & Ellis, Inc. (L&E)** with extensive experience in health insurance actuarial analysis, regulatory review, and financial modeling for both public and private sector clients. He has led actuarial teams supporting the **Centers for Medicare & Medicaid Services (CMS)** in **Medicare Advantage and Part D bid reviews**—personally reviewing over 1,000 bids—and has contributed to **Medicaid actuarial certification reviews**, ensuring compliance and accuracy in state-submitted rate structures. Mr. Moore's expertise spans **rate development, risk adjustment, reserve adequacy, and gain-sharing arrangements**, with significant experience advising insurers, state Departments of Insurance, and federal agencies. He has managed projects addressing **ACA rate filings, long-term care litigation support, and actuarial audits for the U.S. Department of Veterans Affairs**, as well as actuarial testimony and analysis for major regulatory cases. In addition to his government work, he has designed and priced a variety of accident and health products, including stop-loss, provider excess, and supplemental health coverage. A **Fellow of the Society of Actuaries, Chartered Enterprise Risk Analyst, and Member of the American Academy of Actuaries**, Mr. Moore brings a disciplined, data-driven approach to healthcare finance, regulation, and sustainability.

Dr. Matthew Kelsey, PhD, ASA, MAAA

Dr. Kelsey is a **Consulting Actuary and Principal at Lewis & Ellis, Inc. (L&E)** with specialized expertise in **Medicare, Medicaid, and commercial health insurance pricing, compliance, and regulatory review**. He has led multidisciplinary actuarial teams supporting the **Centers for Medicare & Medicaid Services (CMS)** in the review of thousands of **Medicare Advantage and Part D (MAPD)** bids, ensuring compliance with federal actuarial standards and program guidance. Dr. Kelsey has also contributed to **Medicaid waiver reviews**, including Home and Community-Based Services (HCBS) and PACE programs, in which he assessed financial soundness, cost-effectiveness, and compliance with CMS and state requirements. His experience extends to **Affordable Care Act (ACA) rate review** for multiple state Departments of Insurance, **mental health parity compliance analysis**, and **statistical modeling** to evaluate

adherence to the Mental Health Parity and Addiction Equity Act (MHPAEA). In addition to regulatory work, Dr. Kelsey has developed and implemented complex **pricing and competitive analysis models** for Medicare Advantage and Part D products, integrating SAS, Excel, and Tableau-based tools to project claims, model risk adjustment, and optimize bid strategies. A former university professor, he also leads training and mentorship programs for actuarial analysts and interns. Dr. Kelsey holds a **PhD and MA in Philosophy from Loyola University of Chicago** and a **BA from Baylor University** and is an **Associate of the Society of Actuaries** and a **Member of the American Academy of Actuaries**.

Allison Musso, ASA, MAAA

Ms. Musso is a **Vice President and Consulting Actuary at Lewis & Ellis, Inc. (L&E)**, with expertise in **financial modeling, regulatory compliance, and product development** across the health insurance industry. She has more than six years of experience supporting both public and private sector clients with **actuarial pricing, rate filings, and market analyses**. Ms. Musso has developed and priced a broad range of **Accident & Health (A&H)** products, including Accident, GAP, and Critical Illness insurance, as well as group and supplemental benefit plans. Her regulatory experience includes conducting **Affordable Care Act (ACA) rate and compliance reviews** for five state Departments of Insurance, assessing pricing methodologies, trend assumptions, and rate adequacy to ensure consistency with state and federal requirements. She has also performed detailed **Medicare Advantage and Part D bid desk reviews** for the **Centers for Medicare & Medicaid Services (CMS)**, analyzing actuarial assumptions, financial projections, and compliance with CMS bid submission guidelines. In addition, Ms. Musso has provided **actuarial audit support for the U.S. Department of Veterans Affairs**, contributed to **state financial examinations of health insurers**, and developed reserve models to ensure financial soundness and reporting accuracy. A **Member of the American Academy of Actuaries** and **Associate of the Society of Actuaries**, Ms. Musso holds a **Bachelor of Science in Mathematics** from the **University of Texas at Dallas** and has published industry insights on regulatory compliance and pharmacy benefit management.

Management Plan

We will implement a **comprehensive, disciplined management plan** to ensure seamless coordination, transparency, and accountability across all project activities. Our approach emphasizes **collaborative governance, clear lines of communication, proactive risk management, and rigorous quality control** across all phases of the engagement. The management structure integrates strategic oversight from Laurel Health Advisors with specialized analytical and actuarial expertise from Charm Economics and Lewis & Ellis, ensuring that each task is executed efficiently, on schedule, and in alignment with Vermont's legislative mandates under **Acts 167 and 68**, as well as the **CMS Rural Health Transformation Program**. The project will be managed through a structured framework that includes defined roles and responsibilities, milestone-driven progress tracking, and continuous engagement with the **AHS, GBCB, and the Office of Health Reform**. This management plan is designed to deliver a cohesive, data-informed, and fully compliant body of work that meets

Vermont's goals for hospital transformation, cost containment, quality improvement, and equitable access.

Organizational Capacity

The **Laurel Health Team** brings exceptional organizational depth, technical capacity, and subject-matter expertise to successfully execute this project. Together, the team unites nationally recognized leaders in **healthcare transformation, actuarial science, data analytics, and health policy** with decades of experience supporting state and federal agencies in implementing complex healthcare reform initiatives. The team's combined organizational infrastructure provides Vermont with a robust, multidisciplinary capability that integrates policy insight, economic and actuarial precision, and data-driven analytical excellence.

Laurel Health Advisors serves as the prime contractor and project integrator, providing overall strategic direction, project management, and analytic coordination. The firm has successfully led multi-year engagements for the **Centers for Medicare & Medicaid Services (CMS), Health Resources and Services Administration (HRSA)**, and multiple state agencies, including **Vermont, New Jersey, Washington, Connecticut, Oregon, and Oklahoma**. Laurel Health Advisors specializes in **data governance, interoperability, and health system analytics**, and has developed and managed complex data frameworks that inform payment reform, hospital transformation, and Medicaid program evaluation. The firm's structure supports rapid mobilization, secure data management, and agile project execution—backed by a team of experienced project managers, data scientists, and policy analysts who are adept at coordinating across multiple stakeholders and government entities.

Charm Economics contributes deep expertise in **healthcare market analysis, economic modeling, and financial simulation**. The organization has developed sophisticated econometric and system dynamics models that assess the financial and operational impacts of policy reforms on hospitals, payers, and state systems. Their experience includes constructing **rate-setting, cost containment, and global budget forecasting models** for state health authorities and health systems nationwide. Charm Economics brings a proven capacity to translate complex data into actionable financial and policy insights that support sustainable transformation—ensuring Vermont's modeling and analyses are rigorous, defensible, and policy-relevant.

Lewis & Ellis (L&E) provides nationally recognized **actuarial and financial modeling capabilities**, with decades of experience supporting **state Medicaid agencies, CMS, and Departments of Insurance**. L&E's team includes credentialed actuaries (FSA, ASA, MAAA) with specialized expertise in **Medicaid rate development, managed care financial analysis, and hospital reimbursement modeling**. Their organizational infrastructure includes a dedicated team of health actuaries supported by SAS, R, and Python-based analytic environments, enabling rapid validation of assumptions, replication of rate and cost methodologies, and comprehensive financial forecasting. L&E's experience in conducting **Medicare and Medicaid rate reviews, waiver analyses, and risk adjustment modeling** positions them to ensure all financial and actuarial outputs for Vermont meet CMS standards for transparency, accuracy, and audit readiness.

Together, these organizations have established **shared project management protocols, integrated analytical systems, and secure data-exchange processes** that ensure seamless collaboration across teams and deliverables. The Laurel Health Team employs a **joint governance model** supported by a cloud-based **collaboration and version-control environment**, enabling real-time coordination across modeling, data analysis, and policy development. Weekly leadership meetings, structured workstreams, and cross-organizational quality assurance reviews ensure consistent progress tracking and alignment with Vermont's objectives.

The Laurel Health Team's organizational capacity is anchored by its proven ability to manage large-scale, data-intensive projects that require both technical precision and policy insight. Collectively, the team offers Vermont an unmatched combination of **strategic leadership, actuarial rigor, analytic sophistication, and operational discipline**. This capacity ensures the State will receive comprehensive, evidence-based analyses and actionable recommendations that advance hospital sustainability, improve care access and quality, and achieve measurable cost and efficiency gains consistent with Vermont's transformation goals.

Roles and Responsibilities

The **Laurel Health Team** has assembled a multidisciplinary group of nationally recognized experts with the experience and technical capacity necessary to deliver on all aspects of this project. Each team member brings specialized expertise in **Medicaid policy, actuarial modeling, hospital finance, data analytics, and health system transformation**, ensuring that every component of the work—from strategic planning to technical execution—is managed by seasoned professionals. The following section outlines the **specific roles and responsibilities** of key personnel, highlighting how their individual contributions align with Vermont's goals for **cost containment, quality improvement, access assurance, and hospital sustainability**.

Dr. Yael Harris, PhD, MHS – Project Director (Laurel Health Advisors)

Dr. Harris will serve as the **Project Director**, providing overall strategic leadership, technical oversight, and quality assurance for all aspects of the project. She will be responsible for ensuring that all activities align with Vermont's legislative mandates and the strategic objectives of the **Agency of Human Services (AHS), Green Mountain Care Board (GMCB), and Office of Health Care Reform**. Dr. Harris will oversee project execution, stakeholder engagement, and deliverable review, ensuring that all analyses meet the highest standards of rigor, transparency, and compliance. Her extensive experience in health policy, Medicaid program evaluation, and federal and state healthcare transformation will guide the team in developing actionable, evidence-based recommendations.

Jason Goldwater, MPA, MA, PMP, LSSBB – Project Manager (Laurel Health Advisors)

Mr. Goldwater will serve as the **Project Manager**, responsible for the day-to-day management, coordination, and operational delivery of all project tasks. A **Project Management Professional**

(PMP) and certified **Lean Six Sigma Black Belt (LSSBB)**, Mr. Goldwater brings expertise in process optimization, data governance, and analytic project execution. He will lead the integration of technical workstreams across Laurel Health Advisors, Charm Economics, and Lewis & Ellis, ensuring adherence to schedule, budget, and quality objectives. Mr. Goldwater will also manage risk mitigation, oversee communication with state partners, and maintain the project's performance tracking and reporting systems to guarantee timely, accurate, and high-quality deliverables.

Dr. Adam Block, PhD – Health Economist (Charm Economics)

Dr. Block will lead the **economic and financial modeling** components of the project, providing expertise in cost modeling, market analysis, and value-based payment design. He will direct Charm Economics' work in developing econometric and system dynamics models that simulate the financial and operational effects of proposed reforms, ensuring Vermont's policy and fiscal decisions are supported by sound, data-driven analysis.

Robert Murray, MA – Senior Advisor, Hospital Finance and Rate Regulation (Laurel Health Advisors)

Mr. Murray will serve as the **Senior Advisor** on hospital rate regulation and payment reform, providing expert guidance on the development of Vermont's transformation and cost containment strategies. Drawing on his experience as the principal architect of the **Maryland Total Cost of Care Model**, he will advise the State on approaches to global budgeting, rate-setting, and all-payer financial modeling to ensure hospital sustainability and system-wide accountability.

Dr. Yidan Zhang, PhD – Senior Data Scientist (Laurel Health Advisors)

Ms. Zhang will oversee **data integration, normalization, and predictive analytics**, ensuring that all modeling and reporting are grounded in robust, validated data. She will design and manage analytic pipelines, apply machine learning and statistical methods to identify trends, and support scenario modeling and visualization for Vermont's hospital and regional analyses.

Dr. Fadi Martinos, PhD – Director of Quantitative Modeling (Laurel Health Advisors)

Dr. Martinos will lead the **quantitative and econometric modeling team**, develop simulation tools and forecasting models to evaluate hospital performance, cost trends, and reform impacts. He will ensure analytic consistency across economic and actuarial domains and provide validation of results to support data transparency and policy defensibility.

Dr. Beth Linas, PhD, MHS – Director of Epidemiology and Population Health (Laurel Health Advisors)

Dr. Linas will oversee all **population health and quality analytics**, integrating health outcomes and social determinants of health (SDOH) data into the evaluation framework. She will ensure that Vermont's analyses capture the impact of system changes on health equity, access, and outcomes across demographic and geographic populations.

Daniel Moore, FSA, MAAA, CERA – Vice President & Principal Actuary (Lewis & Ellis)



Mr. Moore will lead the **actuarial modeling and financial risk analysis** components of the project, overseeing Medicaid rate structure assessments, cost projections, and sustainability modeling. He will ensure all financial methodologies align with CMS actuarial standards and Vermont's fiscal reporting requirements.

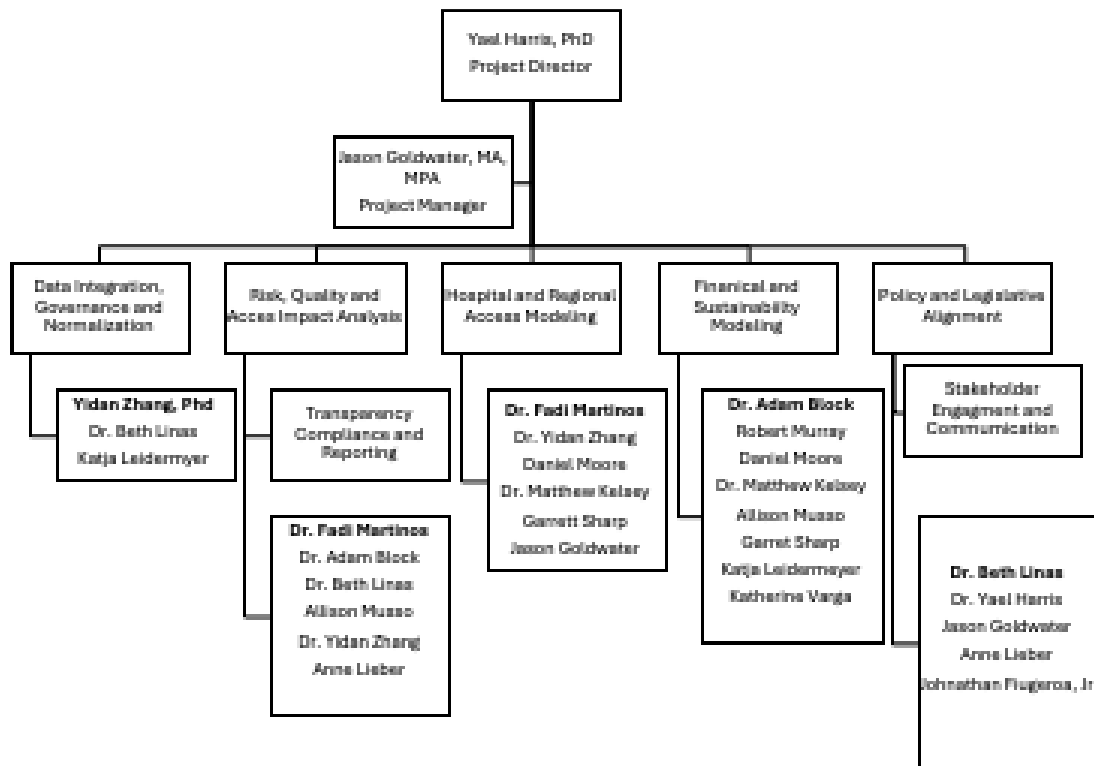
Dr. Matthew Kelsey, PhD, ASA, MAAA – Consulting Actuary & Principal (Lewis & Ellis)

Dr. Kelsey will provide **technical actuarial leadership** in developing rate, cost, and risk adjustment models. He will support financial forecasting, policy simulation, and regulatory compliance analysis for hospital and Medicaid transformation efforts, ensuring methodological accuracy and transparency.

Allison Musso, ASA, MAAA – Vice President & Consulting Actuary (Lewis & Ellis)

Ms. Musso will support **rate development, financial validation, and regulatory analysis** activities, focusing on compliance, reserve adequacy, and actuarial review. She will contribute to scenario modeling and ensure all actuarial deliverables meet CMS and state quality and documentation standards.

In addition, support for each of the required tasks will be provided by Anne Lieber, Katherine Varga, Johnathan Figueroa, Jr, Abby Sacks, Garrett Sharp, and Katja Leidermeyer. Our organizational chart is shown below.



Bold indicates Task Lead

Risk Management

The **Laurel Health Team** employs a comprehensive and proactive **Risk Management Framework** designed to identify, assess, mitigate, and monitor risks across all stages of the Vermont Hospital Transformation Analytical Support Project. This framework is grounded in **project management best practices** aligned with **PMI's PMBOK methodology** and enhanced by **Lean Six Sigma risk mitigation principles**, ensuring that potential risks are addressed early and that corrective measures are implemented swiftly to maintain project quality, timeliness, and compliance. Under the leadership of **Jason Goldwater, PMP, LSSBB**, the Project Manager, and with oversight from **Dr. Yael Harris**, Project Director, the risk management process will be integrated into daily operations and reinforced through structured reporting, communication, and performance review mechanisms.

The Laurel Health Team recognizes that this project operates within a complex environment that involves multiple stakeholders, diverse data systems, and interrelated technical and policy objectives. Accordingly, the team will implement a **four-tiered risk management strategy** encompassing:

1. Risk Identification:

Risks will be identified at the project's initiation and updated throughout its lifecycle through continuous collaboration among technical leads, subject-matter experts, and State partners. Key risk categories include:

- **Data and Technology Risks** – challenges with data access, integration, or quality; interoperability issues between hospital and State systems; or delays in data transfer.
- **Operational and Schedule Risks** – unanticipated delays in deliverables, dependencies between analytic and actuarial workstreams, or changes in scope or stakeholder priorities.
- **Policy and Compliance Risks** – evolving interpretations of Acts 167 and 68, CMS guidance updates, or misalignment between State and Federal compliance requirements.
- **Financial and Resource Risks** – potential budget constraints, staffing fluctuations, or unforeseen analytic complexities affecting labor allocation.
- **Stakeholder and Communication Risks** – differences in stakeholder expectations, coordination challenges across agencies, or miscommunication that may impact project execution or deliverable clarity.

2. Risk Assessment and Prioritization:

Identified risks will be evaluated using a standardized **Risk Probability and Impact Matrix** to determine their likelihood and potential severity. Each risk will be assigned a score and ranked as **low**, **moderate**, or **high** priority. The project leadership team will review and validate this assessment monthly, ensuring that high-impact or high-probability risks receive immediate attention and mitigation planning.

3. Risk Mitigation and Response Planning:

For each identified risk, a **Risk Mitigation Action Plan** will be developed that defines preventive measures, contingency actions, and responsible parties. Examples include:

- **Data Risks:** Establish redundant data transfer protocols, implement robust validation scripts in Python and R, and maintain real-time data integrity dashboards.
- **Operational Risks:** Employ proactive scheduling and dependency mapping using project management software (e.g., MS Project or Smartsheet), supported by biweekly progress reviews and adaptive resource reallocation.
- **Policy Risks:** Conduct quarterly crosswalk reviews between State legislation and CMS guidance to ensure analytic and financial models remain compliant with evolving requirements.
- **Stakeholder Risks:** Maintain structured communication pathways through a Stakeholder Engagement Plan that includes biweekly coordination calls, issue logs, and documentation of decisions.

4. Risk Monitoring and Reporting:

Mr. Goldwater will maintain a **Risk Register** as a living document throughout the engagement. This register will record identified risks, assigned owners, mitigation actions, and current status. A **Risk Review Committee**, led by Dr. Harris and Mr. Goldwater, will meet biweekly to monitor mitigation progress, assess emerging risks, and adjust action plans as needed. Significant risks will be escalated immediately to the Vermont Agency of Human Services (AHS) and Green Mountain Care Board (GMCB) for joint review and decision-making.

In addition to these structured processes, our approach to risk management emphasizes **early detection and transparency**. All project staff and partners—Laurel Health Advisors, Charm Economics, and Lewis & Ellis—will be empowered to flag emerging issues promptly through an open communication protocol. This collaborative, no-blame environment encourages proactive problem-solving and ensures that risks are addressed before they escalate.

Finally, as part of our quality assurance system, risk management activities will be fully integrated with the project's **Performance Monitoring Plan**. This linkage allows the team to track the effectiveness of mitigation strategies through key performance indicators (KPIs) such as deliverable timeliness, data integrity scores, and stakeholder satisfaction metrics.

Through this rigorous and transparent risk management approach, the Laurel Health Team will ensure that Vermont's hospital transformation project remains on schedule, within budget, and fully aligned with legislative and programmatic goals—delivering reliable, high-quality, and actionable results to guide the State's healthcare reform efforts.

Communication Management

We will implement a comprehensive **Communication Management Plan** designed to ensure clear, consistent, and transparent communication across all project stakeholders, partners, and technical teams. Effective communication is central to the success of the **Vermont Hospital Transformation Analytical Support Project**, as the work involves multiple agencies. Under the direction of **Dr. Yael Harris**, Project Director, and **Jason Goldwater, PMP, LSSBB**, Project Manager, the communication framework will facilitate timely information sharing, collaborative decision-making, and continuous alignment with Vermont's strategic goals.

The team's **Communication Management Framework** will be guided by four core principles: **clarity, consistency, accessibility, and accountability**. It integrates structured communication protocols, defined roles and responsibilities, and formal reporting tools to maintain stakeholder confidence and ensure the efficient flow of information throughout the project lifecycle.

1. Communication Governance and Structure

The Laurel Health Team will establish a clear governance structure for communication to ensure accountability and responsiveness:

- **Project Director (Dr. Yael Harris)** – Serves as the primary point of contact for AHS leadership, responsible for strategic communication, issue resolution, and high-level coordination across agencies.



- **Project Manager (Jason Goldwater, PMP, LSSBB)** – Manages daily communications with State partners, internal teams, and subcontractors, ensuring information flow between task leads, technical teams, and AHS/GMCB staff.
- **Task Leads (LHA, Charm Economics, Lewis & Ellis)** – Provide technical updates, summarize analytic progress, and elevate issues to the Project Manager.
- **AHS/GMCB/OHCR Representatives** – Participate in recurring meetings, review deliverables, and provide policy and operational feedback.

All communication channels will operate under a centralized **project communication matrix**, specifying audiences, purposes, formats, frequencies, and responsible parties.

2. Communication Channels and Frequency

The Laurel Health Team will maintain both **structured and flexible communication channels** to support collaboration, transparency, and accountability.

Formal Communication Mechanisms:

- **Weekly Project Team Meetings:** Led by the Project Manager to review progress, address emerging issues, and plan upcoming milestones.
- **Biweekly Leadership and Coordination Meetings:** Led by Dr. Harris with AHS, GMCB, and OHCR to discuss progress toward strategic goals, inter-agency coordination, and risk mitigation.
- **Monthly Technical Review Sessions:** Focused on data integration, modeling updates, and analytic findings; attended by subject-matter experts and technical leads from all partner organizations.
- **Quarterly Stakeholder Briefings:** Summarize key findings, trends, and recommendations for broader audiences including State policymakers, hospital representatives, and community stakeholders.
- **Ad Hoc Issue Resolution Meetings:** Convened by the Project Manager for rapid response to time-sensitive or high-priority issues.

Informal Communication Mechanisms:

- **Shared Collaboration Platforms:** Secure cloud-based project workspace (e.g., Microsoft Teams, SharePoint, or Smartsheet) for real-time document sharing, version control, and communication logs.
- **Instant Messaging Channels:** Secure, role-based channels for daily coordination between team members and technical partners.
- **Email Summaries and Bulletins:** Distributed for decision tracking, progress documentation, and follow-up actions.

3. Deliverable and Reporting Communication



All communication of deliverables, analytic findings, and progress reports will follow a structured process that ensures transparency, accuracy, and documentation.

- **Monthly Status Reports:** Submitted to AHS and GMCB summarizing accomplishments, challenges, risk status, and planned activities.
- **Technical Memoranda and Data Briefs:** Used to communicate intermediate findings and modeling results, ensuring stakeholders can review assumptions, methodologies, and implications.
- **Final Reports and Presentations:** Developed collaboratively and reviewed through internal quality control processes before submission to the State. Each report will include executive summaries, visualizations, and key recommendations designed for policymakers and the public.
- **Public Communication Materials:** Developed under AHS/GMCB guidance to ensure messaging consistency with Vermont's transparency and stakeholder engagement goals.

4. Stakeholder Engagement and Transparency

Given the project's significance to Vermont's health system and communities, stakeholder communication will be inclusive and transparent.

- **Stakeholder Engagement Plan:** The team will maintain a plan that defines audiences (e.g., hospitals, provider organizations, payers, regional planners, and community groups) and specifies communication modes, frequencies, and feedback mechanisms.
- **Collaborative Workshops and Briefings:** The team will host structured workshops to present findings, gather input, and refine modeling and recommendations based on stakeholder perspectives.
- **Communication Accessibility:** All public-facing materials will comply with **Section 508 accessibility standards**, ensuring usability for all audiences.

5. Communication Monitoring and Quality Assurance

Communication effectiveness will be monitored through regular feedback loops and quality reviews:

- The **Project Manager** will track the timeliness and responsiveness of communications and maintain an issue log summarizing questions, resolutions, and follow-up actions.
- The **Project Director** will review communication outputs for accuracy, clarity, and policy consistency before release to external audiences.
- A **Communication Effectiveness Review** will be conducted quarterly, assessing whether communication goals are being met and identifying areas for improvement.

6. Escalation and Decision Protocols

A defined escalation protocol will ensure that critical issues are resolved quickly and effectively:

- **Operational Issues:** Elevated from task leads to the Project Manager for immediate resolution.
- **Strategic or Policy Issues:** Escalated by the Project Director to AHS and GMCB leadership for guidance.
- **Data or Compliance Issues:** Jointly addressed by the Project Manager, Data Lead (Yidan Zhang), and the relevant State data governance staff.

Quality Assurance

the QA process will be governed by a structured system of checks and validations that ensure all analyses, models, and reports are accurate, replicable, and actionable. This system combines the **discipline of Lean Six Sigma methodologies** with industry-standard data and actuarial QA procedures to ensure a consistent, measurable, and transparent approach to quality management.

1. Quality Assurance Framework and Governance

The Laurel Health Team's QA Framework is built on four pillars: **Standards, Validation, Oversight, and Continuous Improvement.**

- **Project Director (Dr. Harris):** Provides executive oversight for all QA activities and ensures consistency with contractual, policy, and regulatory standards.
- **Project Manager (Mr. Goldwater):** Implements day-to-day QA processes, coordinates cross-team reviews, and ensures corrective actions are documented and resolved.
- **Technical Leads (Charm Economics, Lewis & Ellis, Laurel Health Advisors):** Conduct domain-specific quality reviews for modeling, actuarial, data, and policy outputs.
- **Quality Control (QC) Review Committee:** A standing group of senior analysts and subject-matter experts that perform periodic audits of analytical code, modeling assumptions, and deliverable content.

A **Quality Management Plan (QMP)** will be finalized during project initiation, outlining QA protocols, validation standards, and roles and responsibilities. This plan will be a living document, reviewed quarterly and adjusted as the project evolves.

2. Quality Control Across the Project Lifecycle

Quality assurance will be implemented at each phase of the project using structured checkpoints and standardized review processes:

a. Planning and Design Phase

- Conduct a **requirements validation workshop** with AHS and GMCB to confirm objectives, data sources, and deliverable expectations.



- Develop a **QA checklist** for each deliverable type, incorporating formatting, analytical, and documentation requirements.
- Establish a **baseline quality standard** for all data, models, and reports, consistent with CMS and AHS methodologies.

b. Data Integration and Processing Phase

- Implement **automated data validation scripts** in Python, R, and SQL to detect outliers, missing data, and ingestion inconsistencies.
- Conduct **crosswalk validation** to ensure consistency between hospital, payer, and State data sources.
- Maintain detailed **data provenance documentation** and transformation logs for auditability.

c. Analysis and Modeling Phase

- Require **dual independent model validation** for all predictive, actuarial, and econometric models—each model will be peer-reviewed by a separate team within the partner organization.
- Perform **parameter sensitivity testing** and **replication checks** to confirm model stability.
- Validate all financial and actuarial models against CMS and state-approved methods, ensuring compliance with **Actuarial Standards of Practice (ASOPs)** and CMS model integrity guidelines.

d. Deliverable and Reporting Phase

- Apply a **three-tier review process** for all reports and presentations:
 1. **Technical Review** by domain experts (actuarial, data science, economics).
 2. **Editorial Review** for clarity, accessibility, and compliance with Vermont's communication standards.
 3. **Executive Review** by the Project Director prior to submission to the State.
- All public-facing materials will be reviewed for **Section 508 accessibility compliance** and undergo a final QA sign-off checklist prior to release.

3. QA Tools, Systems, and Documentation

The team will employ secure, collaborative tools to manage QA activities and document review history:



- **Version Control Systems (GitHub or SharePoint)** to track revisions and maintain audit trails.
- **Automated QA Dashboards** for monitoring project quality indicators, error rates, and resolution timelines.
- **Standardized QA Templates** for deliverable reviews, including checklists for formatting, compliance, and analytic accuracy.
- **Data Validation Logs** documenting quality checks, test results, and corrective actions.

All QA documentation will be stored in a secure, shared environment accessible to the State for transparency and accountability.

4. Continuous Quality Improvement (CQI)

The Laurel Health Team will integrate **Continuous Quality Improvement** into all aspects of project execution through:

- **Quarterly Quality Review Meetings:** The QA Committee will evaluate process efficiency, identify patterns in errors or revisions, and implement improvements.
- **Root Cause Analysis:** When a quality issue arises, a structured RCA will be conducted to identify systemic causes and prevent recurrence.
- **Performance Metrics:** Key Quality Indicators (KQIs) such as accuracy rates, rework frequency, timeliness of deliverables, and stakeholder satisfaction will be tracked and reported to AHS.
- **Feedback Loops:** Regular feedback from AHS, GMCB, and other stakeholders will inform process refinements, ensuring the project remains responsive and adaptive.

5. Corrective Action and Escalation Procedures

If a quality issue is detected, the team will follow a defined escalation process to resolve it quickly and effectively:

1. **Issue Identification:** Any team member may report a potential issue in the QA tracking system.
2. **Assessment and Assignment:** The Project Manager evaluates the issue, assigns it to the responsible task lead, and categorizes its severity.
3. **Resolution:** The task lead develops and implements corrective actions, documented in the QA log.
4. **Verification:** The QA Committee validates that corrective measures are successful and sustainable.
5. **Reporting:** Major issues and resolutions are summarized in the monthly status report to AHS and GMCB.

6. Integration with Risk and Communication Management

QA will be tightly integrated with **Risk Management** and **Communication Management** processes. Quality risks—such as data integrity or analytic inconsistencies—will be tracked in the **Risk Register** and addressed collaboratively through the same escalation channels. QA progress and findings will be shared with AHS and GMCB through regular reporting, ensuring transparency and continuous improvement.

Timeline

The **Laurel Health Team** has developed a comprehensive **12-month project timeline** to guide the successful execution of the **Vermont Hospital Transformation Analytical Support Project**, beginning **December 1, 2025**, and concluding **November 30, 2026**. This timeline reflects a structured, milestone-driven approach designed to ensure timely delivery, methodological rigor, and alignment with Vermont’s legislative and programmatic goals. Each phase of the project builds on the preceding one—from project initiation and data integration through financial modeling, risk and access analysis, policy alignment, and final reporting—ensuring a logical progression that supports transparency, accuracy, and actionable results. Key milestones, deliverables, and responsible leads are clearly defined to promote accountability, facilitate coordination, and maintain momentum across all analytic and operational workstreams.

Proposed Project Timeline

Task # / Task Title	Timeline	Major Milestones	Key Deliverables	Task Lead(s)
1. Project Initiation and Mobilization	Dec 2025 – Jan 2026	- Kickoff meeting with AHS, GMCB, and OHCR - Finalization of Workplan and Schedule - Establishment of Data Use Agreements (DUAs) and Governance Protocols	- Final Project Workplan - Project Management Plan (PMP) - Communication and Risk Management Plans - Kickoff Summary Report	Dr. Yael Harris (Project Director) Jason Goldwater (Project Manager)
2. Data Acquisition, Integration, and Governance	Dec 2025 – Mar 2026	- Data inventory and mapping of sources - Secure data ingestion architecture operational - Completion of data normalization and validation protocols	- Data Governance Framework - Data Ingestion & Normalization Protocol Documentation - Integrated Analytic Dataset (Baseline)	Dr. Yidan Zhang (Data Lead) Jason Goldwater (PM) Dr. Beth Linas (Population Health Advisor)

Task # / Task Title	Timeline	Major Milestones	Key Deliverables	Task Lead(s)
			Monthly Data Quality Reports	
3. Hospital and Regional Access Modeling	Jan 2026 – Apr 2026	• GIS model design completed • Baseline access and utilization data validated • Completion of first scenario simulation (drive-time, seasonal variation, cross-border analysis)	• Access Modeling Technical Report • Regional Utilization Maps and Dashboards • Hospital Access Benchmarking Summary	Dr. Fadi Martinos (Quantitative Lead) Jason Goldwater (PM) Dr. Beth Linas (SDOH/Equity Advisor)
4. Financial Sustainability and Cost Modeling	Feb 2026 – Jun 2026	• Collection of hospital cost report and payer data • Preliminary Total Cost of Care (TCOC) modeling • Scenario modeling for service line realignment and cost containment	• Financial Sustainability Report • Hospital Fiscal Benchmark Models • TCOC Model Summary with Assumptions • Interactive Cost Projection Dashboard	Robert Murray (Hospital Finance Advisor) Dr. Adam Block (Health Economist) Daniel Moore (Actuarial Lead, L&E)
5. Legislative and Policy Alignment (Acts 167 & 68)	Mar 2026 – Jul 2026	• Draft of Legislative Crosswalk Matrix • Stakeholder alignment sessions with AHS and GMCB • Validation of CMS and State policy compliance	• Legislative-Policy Crosswalk Matrix • CMS and State Compliance Review Summary • Policy Recommendations Report	Dr. Yael Harris (Director) Robert Murray (Advisor) Jason Goldwater (PM)
6. Risk, Quality, and Access Impact Analysis	Apr 2026 – Aug 2026	• Completion of predictive modeling for financial and operational risk • Quality and outcome indicators integrated into analysis • Scenario testing of reform proposals	• Risk Analysis and Scenario Modeling Report • Quality Impact Dashboard • Population Health and Equity Summary	Dr. Fadi Martinos (Quantitative Lead) Dr. Beth Linas (Epidemiology Lead) Dr. Adam Block (Economics Lead)

Task # / Task Title	Timeline	Major Milestones	Key Deliverables	Task Lead(s)
7. Medicaid and Payment System Evaluation	May 2026 – Sep 2026	- Medicaid rate modeling integrated - Actuarial validation of rate adjustments and reimbursement models - Policy simulation for managed care and rural hospitals	- Medicaid Payment Analysis Report - Rate Modeling and Validation Documentation - Actuarial Certification Summary	Daniel Moore (Principal Actuary, L&E) Dr. Matthew Kelsey (Consulting Actuary, L&E) Allison Musso (VP, L&E)
8. Compliance and Transparency Framework	Jun 2026 – Sep 2026	- Compliance Integration Framework completed - Dashboard prototype for public-facing transparency - Audit-readiness review with AHS	- Compliance Integration Framework Report - Transparency and Accountability Dashboard - QA and Audit Checklist Documentation	Jason Goldwater (PM) Dr. Yael Harris (Director) Daniel Moore (L&E)
9. Stakeholder Engagement and Collaboration	Throughout (Dec 2025 – Nov 2026)	- Quarterly stakeholder briefings - Statewide coordination meetings - Presentation of analytic findings and models	- Stakeholder Communication Plan - Quarterly Engagement Briefs - Workshop Summaries and Feedback Reports	Dr. Yael Harris (Lead) Jason Goldwater (PM) Robert Murray (Policy Advisor)
10. Quality Assurance and Continuous Improvement	Throughout (Dec 2025 – Nov 2026)	- QA checkpoints at each deliverable phase - Mid-year QA review (June 2026) - Annual QA audit and final performance report	- Quality Management Plan (QMP) - QA Audit Reports (Mid-Year and Final) - Continuous Improvement Recommendations	Jason Goldwater (QA Lead) Dr. Yael Harris (Oversight) Daniel Moore (Technical QA, L&E)
11. Final Synthesis, Reporting, and Transition	Sep 2026 – Nov 2026	Draft final report review by AHS and GMCB	Final Analytic Report and Executive Summary	Dr. Yael Harris (Director) Jason Goldwater (PM) All Technical Leads

Task # / Task Title	Timeline	Major Milestones	Key Deliverables	Task Lead(s)
		- Final presentation and approval - Knowledge transfer and closeout session	- Policy Recommendations and Implementation Framework - Data Archive and Transition Plan - Project Closeout Report	

References

Laurel Health Advisors is pleased to provide the following three references for the State of Vermont.

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609 913-5008

Lisa.cummings@doh.nj.gov

CERTIFICATE OF COMPLIANCE

For your bid to be considered valid, this form must be completed in its entirety, executed by a duly authorized representative of the bidder, and submitted as part of the response to the proposal.

- A. **NON COLLUSION:** Bidder hereby certifies that the prices quoted have been arrived at without collusion and that no prior information concerning these prices has been received from or given to a competitive company. If there is sufficient evidence to warrant investigation of the bid/contract process by the Office of the Attorney General, bidder understands that this paragraph might be used as a basis for litigation.
- B. **CONTRACT TERMS:** Bidder hereby acknowledges that is has read, understands and agrees to the terms of this RFP, including Attachment C: Standard State Contract Provisions, and any other contract attachments included with this RFP.
- C. **WORKER CLASSIFICATION COMPLIANCE REQUIREMENT:** In accordance with Section 32 of The Vermont Recovery and Reinvestment Act of 2009 (Act No. 54), the following provisions and requirements apply to Bidder when the amount of its bid exceeds \$250,000.00.

Self-Reporting. Bidder hereby self-reports the following information relating to past violations, convictions, suspensions, and any other information related to past performance relative to coding and classification of workers, that occurred in the previous 12 months.

Summary of Detailed Information	Date of Notification	Outcome
N/A		

Subcontractor Reporting. Bidder hereby acknowledges and agrees that if it is a successful bidder, prior to execution of any contract resulting from this RFP, Bidder will provide to the State a list of all proposed subcontractors and subcontractors' subcontractors, together with the identity of those subcontractors' workers compensation insurance providers, and additional required or requested information, as applicable, in accordance with Section 32 of

The Vermont Recovery and Reinvestment Act of 2009 (Act No. 54), and Bidder will provide any

update of such list to the State as additional subcontractors are hired. Bidder further acknowledges and agrees that the failure to submit subcontractor reporting in accordance with Section 32 of The Vermont Recovery and Reinvestment Act of 2009 (Act No. 54) will constitute non-compliance and may result in cancellation of contract and/or restriction from bidding on future state contracts.

D. Executive Order 05 – 16: Climate Change Considerations in State Procurements Certification

Bidder certifies to the following (Bidder may attach any desired explanation or substantiation. Please also note that Bidder may be asked to provide documentation for any applicable claims):

1. Bidder owns, leases or utilizes, for business purposes, space that has received:

- ☒ Energy Star® Certification
- ☐ LEED®, Green Globes®, or Living Buildings ChallengeSM Certification
- ☐ Other internationally recognized building certification:

2. Bidder has received incentives or rebates from an Energy Efficiency Utility or Energy Efficiency Program in the last five years for energy efficient improvements made at bidder's place of business. Please explain:

N/A

3. Please Check all that apply:

- ☐ Bidder can claim on-site renewable power or anaerobic-digester power ("cow-power"). Or bidder consumes renewable electricity through voluntary purchase or offset, provided no such claimed power can be double-claimed by another party.
 - ☐ Bidder uses renewable biomass or bio-fuel for the purposes of thermal (heat) energy at its place of business.
 - ☒ Bidder's heating system has modern, high-efficiency units (boilers, furnaces, stoves, etc.), having reduced emissions of particulate matter and other air pollutants.
 - ☐ Bidder tracks its energy consumption and harmful greenhouse gas emissions. What tool is used to do this? _____
 - ☒ Bidder promotes the use of plug-in electric vehicles by providing electric vehicle charging, electric fleet vehicles, preferred parking, designated parking, purchase or lease incentives, etc..
 - ☐ Bidder offers employees an option for a fossil fuel divestment retirement account.
 - ☐ Bidder offers products or services that reduce waste, conserve water, or promote energy efficiency and conservation. Please explain:
-
-

4. Please list any additional practices that promote clean energy and take action to address climate change:

E. Executive Order 02 – 22: Solidarity with the Ukrainian People

- ☒ By checking this box, Bidder certifies that none of the goods, products, or materials offered in response to this solicitation are Russian-sourced goods or produced by Russian entities. If Bidder is unable to check the box, it shall indicate in the table below which of the applicable offerings are Russian-sourced goods and/or which are produced by Russian entities. An additional column is provided for any note or comment that you may have.

Provided Equipment or Product	Note or Comment

F. Certification Regarding Use of Contract Funds for Lobbying - Applicable to contracts over \$100,000.00 when federal monies are involved - this clause must be included in all subcontracts over \$100,000.00.

1. The prospective contractor certifies, to the best of his or her knowledge and belief, under the penalties of perjury under the laws of the State of Vermont and the United States that on behalf of the person, firm, association, or corporation he or she represents, that:

a. No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of any Federal agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.

b. If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any Federal agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative

agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure Form to Report Lobbying," in accordance with its instructions.

2. This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by 31 U.S.C. 1352. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

3. The prospective contractor also agrees that they shall require that the language of this certification be included in all lower tier subcontracts, which exceed \$100,000 and that all such recipients shall certify and disclose accordingly.

RFP/PROJECT:

DATE:
Page 4 of 4

Bidder Name: Laurel Health Advisors, LLC Contact Name: Jason Goldwater

Address: 8205 Sandy Stream Road Fax Number:

Laurel, MD 20723 Telephone: 301-529-4245

E-Mail: jgoldwater@lh-advisors.net

By: *Jason Goldwater* Name: Jason Goldwater
Signature of Bidder (or Representative) (Type or Print)

END OF CERTIFICATE OF COMPLIANCE

PRICE SCHEDULE

A. Fixed Price Deliverables:

Deliverable Description	Fixed Price
Data Integration & Analysis Plan	\$350,000
Hospital & Regional Access Modeling	\$750,000
Market & Financial Analysis	\$400,000
Evidence-Based Benchmarking Evaluation	\$400,000
Transformation Scenario Model	\$300,000
Final Regional and Hospital Transformation Analytics Report	\$200,000
Total Project Cost (Not to Exceed)	\$2,400,000.00

Name of Bidder: Laurel Health Advisors, LLC

Signature of Bidder: Jason Goldwater

Date: October 28, 2025

Appendix A:

Reporting Example



A COST STUDY ON PROVIDING REMOTE SPECIALITY CARE

Developed for: Vish Sankaran



Table of Contents

Executive Summary.....	2
Purpose of the Research Study	2
Summary of Findings	2
Introduction.....	4
Insurance Status, Prevalence, and Number of Providers.....	5
Capitation.....	12
Charlson Comorbidity Index	12
Medicaid Capitation	14
Medicare Fee-for-Service (FFS) Spending.....	22
Medicare Advantage	25
Cost Drivers	28
Area Deprivation Index.....	28
Readmissions and Other Factors	29
Value-Based Care	29
Cost-Benefit	31
Weight Management and Nutrition.....	34
Community Health Workers	34
Total Addressable Market.....	35
Conclusions and Recommendations	36
Endnotes.....	37

Executive Summary

Poor access to specialist care for patients with multimorbidity can be detrimental to their health. Excessive wait times and delays in care can cause anxiety in patients and their families. They can also reduce patients' ability to carry out day-to-day activities and lead to deterioration in their overall health as important diagnoses are delayed. In addition, many tests or services may be duplicated needlessly, which causes frustration for both patients and providers. Further, residents of rural areas are more likely to be hospitalized and die than those who live in cities primarily because they lack access to specialists. Medicare recipients with chronic conditions such as heart failure or diabetes who live in rural locations have higher death and hospitalization rates than those living in more urban settings. This is due to the lack of access to specialists such as cardiologists and endocrinologists.

Purpose of the Research Study

This research study sought to examine how to optimize remote interventions with specialist care amongst populations with multiple chronic conditions. These conditions of these individuals have exacerbated to a state in which their quality of life is compromised, the cost to the healthcare system is high, and the burden on patients, their caregivers, and the health system is significant. The focus of the study will be on counties within three specific states: Alabama, Mississippi, and Louisiana.

The study focused on multiple chronic conditions including the following metabolic diseases: cardiovascular disease (CVD), acute myocardial infarction (AMI), chronic kidney disease (CKD – 3a and 3b), and Type II Diabetes. The analysis will seek to answer the following questions:

1. What is the prevalence of these conditions among those with multiple chronic conditions and their associated risk factors among populations within each state?
2. What is the current monthly capitation rate for these populations and how does a lack of access to specialist care affect that amount?
3. What is the risk adjustment factor (RAF) for these populations with one or more of these conditions and how would that be affected by increased access to specialist care?
4. How does access to specialist care affect value-based payments?
5. What is the cost-benefit of remote specialist care interventions for these populations?

Summary of Findings

1. The overall PMPM rate for both Medicaid and Medicare Advantage is much higher than typical PMPM rates due to the severity of the conditions and the presence of multiple chronic conditions. The use of remote specialty care can potentially reduce overall healthcare utilization and costs, which drive down PMPM and save payers money.
2. Louisiana has the highest per capita Medicaid expenditure, followed closely by Mississippi. Mississippi has the fewest number of specialists in this study and has one of the lowest rates in the United States.
3. Most of the individuals with multiple chronic conditions have a 50% or lower 10-year survival rate and have conditions that would require at least one, if not two, specialists to provide the appropriate and needed care. In the examples provided for the Medicare population, those with multiple chronic conditions have a 0% survival rate after 10 years unless care is provided.
4. The total cost of care estimates does not factor in the cost of a referral, which could increase costs by 2% per person and are removed through remote specialty care.

5. There are significant cost-benefits for individuals using remote specialty care as it mitigates the risk of unnecessary referrals, eliminates patient travel, and limits the impact on patient productivity.
6. Research also shows cost-benefits to remote weight management and nutrition programs as well as using community workers.
7. The total addressable market for remote specialty care is approximately \$542,487,050 for Medicaid and \$1,240,206,960.40 for Medicare.

Introduction

The presence of multiple chronic conditions in a patient – also known as multimorbidity – has a profound impact on health, health care utilization, and associated costs.¹ The most recent literature on this issue focuses on a patient's number of comorbidities and the associated magnitude of required care processes and the potential influence of comorbidity characteristics on patient care and outcomes. When patients have multiple health problems, it affects their quality of life and their ability to work and is associated with increased rates of functional decline and mortality.² Furthermore, as a patient's number of chronic conditions increases, there is a rise in associated physician visits, prescriptions, hospitalizations, home health visits, out-of-pocket expenses, and total health care expenditures.³

Poor access to specialist care for patients with multimorbidity can be detrimental to their health. Excessive wait times and delays in care can cause anxiety in patients and their families. They can also reduce patients' ability to carry out day-to-day activities and lead to deterioration in their overall health as important diagnoses are delayed. In addition, many tests or services may be duplicated needlessly, which causes frustration for both patients and providers. Further, residents of rural areas are more likely to be hospitalized and die than those who live in cities primarily because they lack access to specialists. Medicare recipients with chronic conditions such as heart failure or diabetes who live in rural locations have higher death and hospitalization rates than those living in more urban settings.⁴ This is due to the lack of access to specialists such as cardiologists and endocrinologists.⁵

According to a report published by the Association of American Medical Colleges (AAMC), the United States could see an estimated shortage of between 37,800 and 124,000 physicians by 2034, including shortfalls in both primary and specialty care. Specifically, it is estimated that among medical specialties, the attrition rate could be between 3,800 and 13,400 providers. The study describes the magnitude of this loss by stating that if marginalized minority populations, people living in rural communities, and people without health insurance have the same health care use patterns as populations with fewer barriers to access, an additional 180,400 physicians would be needed.⁶

This research study seeks to examine how to optimize remote interventions with specialist care amongst populations with multiple chronic conditions. These conditions of these individuals have exacerbated to a state in which their quality of life is compromised, the cost to the healthcare system is high, and the burden on patients, their caregivers, and the health system is significant. The focus of the study will be on counties within three specific states: Alabama, Mississippi, and Louisiana.

- Alabama - Projected total cost of chronic disease from 2016-2030 in Alabama is \$671 billion. In 2015, 3.2 million people in Alabama had at least 1 chronic disease, and 1.4 million had 2 or more chronic diseases. Chronic disease could cost Alabama \$31.3 billion in medical costs and an extra \$13.4 billion annually in lost employee productivity (average per year 2016-2030).⁷
- Louisiana - Projected total cost of chronic disease from 2016-2030 in Louisiana is \$612 billion. In 2015, 2.9 million people in Louisiana had at least 1 chronic disease, and 1.2 million had 2 or more chronic diseases. Chronic disease could cost Louisiana \$28.8 billion in medical costs and an extra \$12 billion annually in lost employee productivity (average per year 2016-2030).⁸
- Mississippi - Projected total cost of chronic disease from 2016-2030 in Mississippi is \$405 billion. In 2015, 1.9 million people in Mississippi had at least 1 chronic disease, and 788 thousand had 2 or more chronic diseases. Chronic disease could cost Mississippi \$19.3 billion in medical costs and an extra \$7.7 billion annually in lost employee productivity (average per year 2016-2030).⁹

The study will focus on multiple chronic conditions including the following metabolic diseases: cardiovascular disease (CVD), acute myocardial infarction (AMI), chronic kidney disease (CKD – 3a and 3b), and Type II Diabetes. The analysis will seek to answer the following questions:

6. What is the prevalence of these conditions among those with multiple chronic conditions and their associated risk factors among populations within each state?
7. What is the current monthly capitation rate for these populations and how does a lack of access to specialist care affect that amount?
8. What is the risk adjustment factor (RAF) for these populations with one or more of these conditions and how would that be affected by increased access to specialist care?
9. How does access to specialist care affect value-based payments?
10. What is the cost-benefit of remote specialist care interventions for these populations?

Insurance Status, Prevalence, and Number of Providers

The analysis begins with the number of individuals within the identified states with various types on insurance including Employer-based insurance, Non-Group Insurance, Medicare, Medicaid, Military, or Uninsured, which is shown in Table 1

Table 1: Types of Health Insurance by Percentage of State Population

State	Employer	Non-Group	Medicaid	Medicare	Military	Uninsured
Alabama	47.2	5.5	19.5	21.0	2.1	9.7
Louisiana	41.8	4.9	29.3	19.0	1.4	8.9
Mississippi	42.2	4.7	24.2	21.0	1.8	12.9

Source: Kaiser Family Foundation, Health Insurance Coverage of the Total Population

A further breakdown, shown in Table 2, shows the number of residents within each state that are on Medicaid, Traditional Medicare (FFS), and Medicare Advantage.

Table 2 Medicaid, Medicare (FFS), and Medicare Advantage by Population

State	Medicaid	Medicare	Medicare Advantage
Alabama	1,223,250	572,444	487,638
Louisiana	1,400,000	592,680	305,320
Mississippi	720,826	512,468	104,963

Source: Kaiser Family Foundation, Health Insurance Coverage of Medicaid, Medicare and Medicare Advantage

To examine the prevalence of the conditions, a cohort of ten counties was selected within each state, which represents a mix of urban and rural environments. In Table 3, the prevalence of CVD within each county is shown. **The prevalence percentage in the United States of CVD for men is 8.3 and for women is 6.1 percent.**

Table 3: Prevalence Percentage of CVD in Alabama, Louisiana, and Mississippi

County	Prevalence Percentage of CVD (%)
--------	----------------------------------

Alabama	
Greene	12.5
Perry	11.8
Conecuh	11.7
Wilcox	11.2
Lamar	11.2
Monroe	11.1
Coosa	11.1
Choctaw	11.1
Winston	11.0
Lowndes	11.0
Louisiana	
Tensas	14.8
East Carroll	12.0
Bienville	11.5
Claiborne	10.9
Webster	10.8
Concordia	10.6
Washington	10.6
Franklin	10.6
Caldwell	10.5
Red River	10.5
Mississippi	
Sharkey	12.2
Jefferson	11.8
Humphreys	11.5
Issaquena	11.4
Quitman	11.4
Wilkinson	11.3
Amite	11.1
Jefferson Davis	11.0
Marion	10.8
Walthall	10.8

Source: CDC Interactive Atlas of Heart Disease and Stroke

The hospitalization rate for acute myocardial infarction (AMI) within the same counties (2017-2019) are shown in Table 4. **The national hospitalization rate for AMI is 6.3 per 1,000 beneficiaries**

Table 4: Hospitalization Rate for AMI in Alabama, Louisiana, and Mississippi

County	Hospitalization Rate Per 1,000 Beneficiaries
Alabama	
Greene	7.0
Perry	9.0
Conecuh	11.3
Wilcox	8.3

County	Hospitalization Rate Per 1,000 Beneficiaries
Lamar	8.0
Monroe	9.5
Coosa	7.2
Choctaw	8.9
Winston	9.5
Lowndes	7.7
Louisiana	
Tensas	11.0
East Carroll	12.5
Bienville	6.8
Claiborne	7.4
Webster	7.0
Concordia	8.1
Washington	10.6
Franklin	15.9
Caldwell	11.0
Red River	7.6
Mississippi	
Sharkey	8.1
Jefferson	8.7
Humphreys	10.4
Issaquena	8.7
Quitman	12.5
Wilkinson	6.3
Amite	8.2
Jefferson Davis	6.0
Marion	7.2
Walthall	5.1

Source: CDC Interactive Atlas of Heart Disease and Stroke

The prevalence rate of moderate Stage 3a and 3b of chronic kidney disease (CKD) within these counties in Alabama, Louisiana, and Mississippi are shown in Table 5. **The national prevalence count of moderate chronic kidney disease per 100 people is 19.4.**

Table 5: Prevalence Count of CKD in Alabama, Louisiana, and Mississippi

County	Prevalence Count per 100 (2019)
Alabama	
Greene	25.0
Perry	32.50
Conecuh	22.2
Wilcox	26.42
Lamar	21.10
Monroe	21.28
Coosa	34.69

County	Prevalence Count per 100 (2019)
Choctaw	16.3
Winston	21.99
Lowndes	40.0
Louisiana	
Tensas	13.04
East Carroll	25.00
Bienville	26.76
Claiborne	22.89
Webster	23.73
Concordia	23.81
Washington	27.75
Franklin	29.25
Caldwell	28.00
Red River	29.73
Mississippi	
Sharkey	18.75
Jefferson	23.68
Humphreys	21.82
Issaquena	14.52
Quitman	33.33
Wilkinson	22.22
Amite	25.88
Jefferson Davis	14.47
Marion	18.92
Walthall	16.44

Source: Centers for Disease Control and Prevention, Chronic Kidney Disease Surveillance System

Finally, the prevalence percentage of Type 2 diabetes within each of these counties for those 18 years of age or older from 2017 data is shown in Table 6. **The national prevalence percentage is 11.3%**

Table 6: Prevalence Count of Type II Diabetes in Alabama, Louisiana, and Mississippi

County	Overall Prevalence Percentage from 2021(%)
Alabama	
Greene	18.0
Perry	18.0
Conecuh	20.9
Wilcox	15.9
Lamar	19.3
Monroe	19.9
Coosa	23.1
Choctaw	13.9
Winston	20.0
Lowndes	14.2
Louisiana	

County	Overall Prevalence Percentage from 2021(%)
Tensas	10.5
East Carroll	21.3
Bienville	18.7
Claiborne	24.1
Webster	10.6
Concordia	16.9
Washington	16.2
Franklin	18.2
Caldwell	10.7
Red River	9.0
Mississippi	
Sharkey	11.8
Jefferson	14.7
Humphreys	13.6
Issaquena	11.0
Quitman	14.8
Wilkinson	10.5
Amite	14.0
Jefferson Davis	10.1
Marion	12.5
Walthall	19.4

Source: CDC Interactive Atlas of Heart Disease and Stroke

In addition to the prevalence and hospitalization rates of these metabolic chronic conditions, it is also critical to understand the prevalence of risk factors associated with these conditions. The specific focus was on obesity, hyperlipidemia, and high blood pressure.

The national obesity rate is 42.4%

The national hyperlipidemia rate is 38%

The national high blood pressure rate is 47%

The results are shown in Table 7

Table 7: First Set of Risk Factors in Alabama, Louisiana, and Mississippi

County	Obesity Rate (%)	Prevalence of Hyperlipidemia (%)	Prevalence of High Blood Pressure (%)
Alabama			
Greene	38.0	42.6	56.9
Perry	47.0	41.7	51.7
Conecuh	42.5	43.4	52.9
Wilcox	50.3	40.4	53.6
Lamar	34.5	43.2	46.4
Monroe	43.0	41.8	50.4
Coosa	44.6	42.9	49.5
Choctaw	46.7	42.3	51.5

County	Obesity Rate (%)	Prevalence of Hyperlipidemia (%)	Prevalence of High Blood Pressure (%)
Winston	35.7	44.0	45.3
Lowndes	43.4	41.2	54.6
Louisiana			
Tensas	45.0	45.9	56.9
East Carroll	48.0	40.4	51.9
Bienville	43.0	41.2	47.3
Claiborne	43.0	40.1	48.1
Webster	42.0	41.2	45.8
Concordia	42.0	40.1	46.6
Washington	39.0	41.0	45.8
Franklin	41.0	41.1	45.0
Caldwell	39.0	41.1	43.2
Red River	43.0	41.2	46.3
Mississippi			
Sharkey	49.0	42.5	53.1
Jefferson	51.0	41.2	55.4
Humphreys	50.0	41.8	53.4
Issaquena	48.0	41.1	50.6
Quitman	49.0	41.6	52.3
Wilkinson	50.0	41.4	51.2
Amite	42.0	42.5	48.5
Jefferson Davis	44.0	43.3	52.7
Marion	41.0	41.4	46.3
Walthall	44.0	41.8	49.0

Source: CDC Interactive Atlas of Heart Disease and Stroke

Additional risk factors that showed a significant effect on each of the chronic diseases included in this study were the percentage of individuals who are physically inactive, the percentage of individuals who are current smokers, and the percentage of individuals who are aware of their high cholesterol levels. The results are shown in Table 8.

The national prevalence percentage of physical inactivity is 25.3%

The national prevalence percentage of current smokers is 12.5%

The national prevalence percentage of high cholesterol awareness is 41.9%

Table 8: Second Set of Risk Factors in Alabama, Louisiana, Mississippi

County	Physical Inactivity (%)	Current Smoker Status (%)	High Cholesterol Awareness (%)
Alabama			
Greene	22.6	24.7	42.6
Perry	34.9	24.6	41.7
Conecuh	40.1	23.8	43.4
Wilcox	33.8	24.1	40.3

County	Physical Inactivity (%)	Current Smoker Status (%)	High Cholesterol Awareness (%)
Lamar	29.2	22.6	43.2
Monroe	34.7	23.8	41.8
Coosa	34.1	22.9	42.9
Choctaw	32.0	21.4	42.3
Winston	34.7	23.3	44.0
Lowndes	31.4	25.0	41.2
Louisiana			
Tensas	31.9	29.5	45.9
East Carroll	19.4	33.6	40.4
Bienville	25.3	26.3	41.2
Claiborne	32.7	27.1	40.1
Webster	30.6	25.5	41.2
Concordia	35.3	27.4	40.1
Washington	31.6	27.5	41.0
Franklin	36.2	26.5	41.1
Caldwell	27.7	27.5	41.1
Red River	32.9	25.3	41.2
Mississippi			
Sharkey	45.5	26.1	42.5
Jefferson	35.5	27.9	41.2
Humphreys	27.6	27.8	41.8
Issaquena	18.4	29.9	41.1
Quitman	38.0	27.3	41.6
Wilkinson	33.6	29.2	41.4
Amite	40.1	22.7	42.5
Jefferson Davis	31.5	23.0	43.3
Marion	28.4	25.7	41.4
Walthall	29.3	24.2	41.8

Source: America Health Rankings, 2021

A breakdown of each state's population with multiple chronic conditions by race is shown in Table 9

Table 9: Breakdown of Patients with Multiple Chronic Conditions by Race

State	Black	American Indian/Alaskan Native	Multiracial	White
Alabama	22.5	13.3	17.5	15.0
Louisiana	20.3	12.7	17.1	12.3
Mississippi	N/A	10.9	N/A	14.0

Source: CDC Behavioral Risk Factor and Surveillance System

Finally, in Table 10, the overall percentage of cardiologists, endocrinologists and metabolic specialists, nephrologists and gastroenterologists as compared to the United States is shown.

Table 10: Number of Specialists in Alabama, Louisiana, and Mississippi (per 100,000 people)

State	Cardiologists	Endocrinology, Diabetes, and Metabolism	Nephrologists	Gastroenterologists
Alabama	5.91	1.51	2.23	3.72
Louisiana	7.41	2.15	2.94	4.91
Mississippi	4.85	1.74	0.97	3.71

Source: Area Health Resource File. AMA Physician Master File, Kaiser Family Foundation, 2021

As a means of comparison, the analysis also looked at the ratio of specialists to the population in Massachusetts, Florida, and Illinois

Table 11: Number of Specialists in Massachusetts, Florida, and Illinois(per 100,000 people)

State	Cardiologists	Endocrinology, Diabetes, and Metabolism	Nephrologists	Gastroenterologists
Massachusetts	14.13	7.39	3.93	8.50
Florida	7.62	2.03	2.09	4.88
Illinois	7.25	3.16	2.33	4.62

Source: Area Health Resource File. AMA Physician Master File, Kaiser Family Foundation, 2021

Capitation

To understand the costs associated with patients with multiple chronic conditions, it is important to first identify categories of individuals with multimorbidity. In a review of the literature, information was gathered from aggregate spending reports from State Medicaid Agencies in Alabama, Louisiana, and Mississippi, peer-reviewed articles on Medicaid and Medicare spending on chronic disease, and payment information from the Centers for Medicare & Medicaid Services (CMS), the following were identified as the most common and frequent patients utilizing care.^{10,11,12,13,14,15}

- A male aged <50 years with myocardial infarction (MI), congestive heart failure (CHF), mild liver disease, and uncomplicated diabetes.
- A male or female <50 years with myocardial infarction (MI) and uncomplicated diabetes
- A female 50-59 with myocardial infarction (MI), congestive heart failure (CHF), mild liver disease, and uncomplicated diabetes.
- A male or female aged 60-69 Years with congestive heart failure, transient ischemic attack (TIA), mild liver disease, and uncomplicated diabetes
- A male 65-69 years with moderate to severe chronic kidney disease and diabetes with end-organ damage
- A male 70 years or older with congestive heart failure and transient ischemic attack.

Charlson Comorbidity Index

The Charlson Comorbidity Index was first developed in 1987 by Mary Charlson and colleagues as a weighted index to predict the risk of death within 1 year of hospitalization for patients with specific comorbid conditions.¹⁶ Nineteen conditions were included in the index. Each condition was assigned a weight from 1

to 6, based on the estimated 1-year mortality hazard ratio from a Cox proportional hazards model. These weights were summed to produce the Charlson comorbidity score. The score offers a 10-year survival prediction in patients with multiple comorbidities. In Table 12, the survival percentage is provided alongside the identified patients.

Table 12: Estimated 10-Year Survival Rates for Select Patients with Multiple Chronic Conditions

Conditions	Estimated 10-Year Survival
<u>Patient #1</u> Male, 45 to 64 Cardiovascular disease (High) Type II Diabetes CKD (3a) Metabolic disease Mediations for CVD, diabetes, liver disease, CKD	2%
<u>Patient #2</u> Female 25-44 Cardiovascular disease (High) Metabolic disorder Type II Diabetes Pulmonary disease Medications for diabetes, CVD, hyperlipidemia and asthma	53%
<u>Patient #3</u> Male 25-44 Cardiovascular disease (High) Type II diabetes CKD (3b) Metabolic disorder Pulmonary disease Medications for diabetes, CVD, hyperlipidemia and asthma	21%
<u>Patient #4</u> Female, 45-64 Cardiovascular (Medium) Type II Diabetes CKD (3b) Metabolic disorder Pulmonary disease Medications for diabetes, CVD, hyperlipidemia and asthma	0%
<u>Patient #5</u> Male, Age 60-69 Years CHF Transient Ischemic Attack (TIA) Mild Liver Disease Type II Diabetes	0%

Conditions	Estimated 10-Year Survival
Metabolic disorder	
<u>Patient #6</u> Female, Age 60-69 Years Diabetes with End-Organ Damage Chronic Kidney Disease (3b) TIA CHF Metabolic disorder	0%
<u>Patient #7</u> Male Age 70-79 Years Diabetes with End-Organ Damage Chronic Kidney Disease (3b) TIA CHF Metabolic disorder Pulmonary disease	0%
<u>Patient #8</u> Female, Age 70-79 Years Type II Diabetes TIA CHF Metabolic disorder Pulmonary disease	0%

By understanding what the survival rate is for these patients, it is possible to identify the costs associated with Medicaid, Traditional Medicare, and Medicare Advantage in providing care

Medicaid Capitation

Under the capitated model for Medicaid, the Centers for Medicare & Medicaid Services (CMS), a state, and a health plan enter into a three-way contract to provide comprehensive, coordinated care, with CMS and the state each paying a health plan a prospective capitation payment. These capitation payments are typically made on a per member per month (PMPM) basis. Managed care organizations negotiate with providers to provide services to their enrollees, either on a fee-for-service (FFS) basis or through arrangements under which they pay providers a fixed periodic amount to provide services. A brief description of each state's Medicaid payment program is provided below.

Alabama

In July 2011, almost two-thirds of Alabama Medicaid beneficiaries were enrolled in one of two statewide mandatory managed care programs. Alabama's Maternity Care Program, which began in 1988, provides maternity care for most pregnant women in Alabama Medicaid. The state administers the program through ten contractors, which are paid a capitated rate to cover all pregnancy-related care, except for those provided by a teaching hospital. Alabama operates a statewide PCCM program called Patient 1st. The patient first program began in 1997 operating under 1915(b) authority; however, in 2013 the program transitioned out of the 1915(b) waiver and is currently operating under 1932(a) authority. The Patient 1st program serves children, low-income families, and individuals who are aged, blind or disabled. Patient 1st

members are assigned to contracting primary care providers who are paid a small case management fee to cover the cost of coordinating care and referrals to specialty care.¹⁷

In August 2011, Alabama added an enhanced PCCM program to its Patient 1st program, called the Patient Care Network of Alabama (PCNA), in which primary medical providers (PMPs) in four geographic regions are given the option to contract with a primary care network (PCN) and serve as a medical home for high-risk individuals. Similar to North Carolina's Community Care Networks, Alabama's PCNs support the PMPs by providing intensive case management, reviewing service utilization data, monitoring capacity and referrals, facilitating quality improvement and the use of evidence-based care, and conducting education and outreach. In 2012, Alabama began operating a Health Home program that serves people with two chronic conditions, or one chronic condition and the risk of developing another. In 2013, Alabama passed legislation to develop Regional Care Organizations (RCOs) that build on the existing PCNAs to create risk-based managed care entities that will be paid a capitated rate to cover a variety of services for almost all Medicaid members in Alabama. The RCOs began operations in 2016.¹⁸

Louisiana

On July 1, 2016, per Executive Order JBE 16-01, Louisiana expanded Medicaid coverage under the Affordable Care Act to adults ages 19 through 64 under 138% of the Federal Poverty Level. The majority of the expansion group receives full Medicaid coverage through enrollment in one of five contracted Medicaid MCOs. Each month less than 1% of expansion enrollees are enrolled in cases such as incarceration or enrollment in Medicare that may require a period of coverage for some services under fee for service (FFS). During the current reporting quarter, October 1, 2021 through December 31, 2021, the unduplicated count of expansion enrollees enrolled in an MCO was 712,274.¹⁹

Mississippi

Mississippi's Medicaid program is administered by the Division of Medicaid, housed within the Office of the Governor. The federal government's matching contribution—the Federal Medical Assistance Percentage, or FMAP—is “designed so that the federal government pays a larger portion of Medicaid costs in states with lower per capita incomes relative to the national average” (Congressional Research Service, 2020). As of FY 2021, Mississippi's FMAP of 77.76% is the highest in the nation. For every dollar spent in the state's Medicaid program, the federal government pays \$0.7776, while the state of Mississippi pays the remaining \$0.2224.

Mississippi launched its managed care program, MSCAN, in 2011. Currently, the Mississippi Division of Medicaid contracts with three payers as coordinated care organizations (CCOs): Molina Health, Magnolia Health (owned by Centene Corporation), and United Healthcare. Participation in MSCAN is mandatory for all eligible coverage groups except children aged 0 to 19 who have disabilities or are in foster care. Dually eligible participants are not eligible for MSCAN, nor are individuals in any waiver program or institutional setting. As of December 2019, 438,029 participants were enrolled in MSCAN, or 65.4% of the total Medicaid population. Under Medicaid managed care, CCOs receive monthly capitated payments from the state for Medicaid participants. The CCOs, in turn, are then responsible for building networks of providers and reimbursing providers at contractually agreed-upon amounts.²⁰

In Table 13, the Medicaid population by county is identified along with the total number of benefit payments made at the end of 2021. An assumption was made that each Medicaid beneficiary remained in the program for all of 2021 and the calculation did not account for churn.

Table 13: Medicaid Population by County and Average Annual Payment Per Eligible

County	Medicaid Population by County	Total Number of Benefit Payments	Average Annual Payment Per Eligible
Greene	3,277	\$16,406,554	\$5,006
Perry	3,816	\$26,271,548	\$3,816
Conecuh	3,881	\$22,667,642	\$5,841
Wilcox	4,887	\$27,449,228	\$5,617
Lamar	3,544	\$25,661,355	\$7,240
Monroe	5,622	\$34,467,815	\$6,131
Coosa	2,203	\$10,202,815	\$4,631
Choctaw	3,604	\$20,389,400	\$5,657
Winston	5,357	\$35,603,871	\$6,646
Lowndes	3,873	\$20,171,553	\$5,208
Louisiana			
Tensas	2,666	\$16,119,064	\$6,046
East Carroll	4,501	\$32,028,710	\$7,116
Bienville	7,440	\$49,399,434	\$6,640
Claiborne	6,523	\$41,770,494	\$6,404
Webster	19,485	\$119,938,526	\$6,135
Concordia	10,903	\$64,576,436	\$5,923
Washington	25,595	\$164,176,980	\$6,414
Franklin	12,135	\$78,173,225	\$6,442
Caldwell	5,644	\$33,800,654	\$5,989
Red River	4,599	\$27,738,324	\$6,031
Mississippi			
Sharkey	3,092	\$16,407,104	\$5,306
Jefferson	21,714	\$120,677,879	\$5,558
Humphreys	5,618	\$42,333,413	\$7,535
Issaquena	2,219	\$18,608,800	\$8,386
Quitman	5,665	\$52,219,900	\$9,218
Wilkinson	4,971	\$32,865,221	\$6,611
Amite	2,311	\$19,812,311	\$8,573
Jefferson Davis	1,812	\$13,224,120	\$7,298
Marion	3,978	\$27,664,871	\$6,954
Walthall	3,451	\$23,235,134	\$6,733

Source: Medicaid Annual State Reports from Alabama, Louisiana, and Mississippi, 2021

In Table 14, the cost of multiple chronic conditions is shown in 2020 dollars. The dollar figures represent an average amount and do not stratify by inpatient, outpatient, or emergency room visits.

Table 14: Average Cost of Multiple Chronic Conditions for Medicare and Medicaid

Number of Chronic Conditions	Total Amount (\$)
Medicare	

Number of Chronic Conditions	Total Amount (\$)
2-3	\$ 6,683.59
4-5	\$ 11,241.16
6+	\$ 22,363.15
Medicaid	
2-3	\$ 5,410.02
4-5	\$ 12,993.53
6+	\$ 27,350.43

In Table 15, the annual average payment for Medicaid is compared to the annual average costs for multiple chronic conditions among the Medicaid population within each county in Alabama, Louisiana, and Mississippi to determine if that amount covers the total cost of care.

Table 15: Comparison of Annual Average Payment by Medicaid Beneficiary, by County, against Annual Average Cost of Multiple Chronic Conditions

County	Average Payment Per Eligible	2 to 3 Conditions	Profit/Loss	4-5 Conditions	Profit/Loss	6+ Conditions	Profit/Loss
Alabama							
Greene	\$5,006	\$ 5,410.00	\$ (404.00)	\$ 12,993.53	\$ (7,987.53)	\$ 22,363.15	\$ (17,357.15)
Perry	\$3,816	\$ 5,410.00	\$ (1,594.00)	\$ 12,993.53	\$ (9,177.53)	\$ 22,363.15	\$ (18,547.15)
Conecuh	\$5,841	\$ 5,410.00	\$ 431.00	\$ 12,993.53	\$ (7,152.53)	\$ 22,363.15	\$ (16,522.15)
Wilcox	\$5,617	\$ 5,410.00	\$ 207.00	\$ 12,993.53	\$ (7,376.53)	\$ 22,363.15	\$ (16,746.15)
Lamar	\$7,240	\$ 5,410.00	\$ 1,830.00	\$ 12,993.53	\$ (5,753.53)	\$ 22,363.15	\$ (15,123.15)
Monroe	\$6,131	\$ 5,410.00	\$ 721.00	\$ 12,993.53	\$ (6,862.53)	\$ 22,363.15	\$ (16,232.15)
Coosa	\$4,631	\$ 5,410.00	\$ (779.00)	\$ 12,993.53	\$ (8,362.53)	\$ 22,363.15	\$ (17,732.15)
Choctaw	\$5,657	\$ 5,410.00	\$ 247.00	\$ 12,993.53	\$ (7,336.53)	\$ 22,363.15	\$ (16,706.15)
Winston	\$6,646	\$ 5,410.00	\$ 1,236.00	\$ 12,993.53	\$ (6,347.53)	\$ 22,363.15	\$ (15,717.15)
Lowndes	\$5,208	\$ 5,410.00	\$ (202.00)	\$ 12,993.53	\$ (7,785.53)	\$ 22,363.15	\$ (17,155.15)
Louisiana							
Tensas	\$6,046	\$ 5,410.00	\$ 636.00	\$ 12,993.53	\$ (6,947.53)	\$ 22,363.15	\$ (16,317.15)
East Carroll	\$7,116	\$ 5,410.00	\$ 1,706.00	\$ 12,993.53	\$ (5,877.53)	\$ 22,363.15	\$ (15,247.15)
Bienville	\$6,640	\$ 5,410.00	\$ 1,230.00	\$ 12,993.53	\$ (6,353.53)	\$ 22,363.15	\$ (15,723.15)
Claiborne	\$6,404	\$ 5,410.00	\$ 994.00	\$ 12,993.53	\$ (6,589.53)	\$ 22,363.15	\$ (15,959.15)
Webster	\$6,135	\$ 5,410.00	\$ 725.00	\$ 12,993.53	\$ (6,858.53)	\$ 22,363.15	\$ (16,228.15)
Concordia	\$5,923	\$ 5,410.00	\$ 513.00	\$ 12,993.53	\$ (7,070.53)	\$ 22,363.15	\$ (16,440.15)
Washington	\$6,414	\$ 5,410.00	\$ 1,004.00	\$ 12,993.53	\$ (6,579.53)	\$ 22,363.15	\$ (15,949.15)
Franklin	\$6,442	\$ 5,410.00	\$ 1,032.00	\$ 12,993.53	\$ (6,551.53)	\$ 22,363.15	\$ (15,921.15)
Caldwell	\$5,989	\$ 5,410.00	\$ 579.00	\$ 12,993.53	\$ (7,004.53)	\$ 22,363.15	\$ (16,374.15)
Red River	\$6,031	\$ 5,410.00	\$ 621.00	\$ 12,993.53	\$ (6,962.53)	\$ 22,363.15	\$ (16,332.15)
Mississippi							
Sharkey	\$5,306	\$ 5,410.00	\$ (104.00)	\$ 12,993.53	\$ (7,687.53)	\$ 22,363.15	\$ (17,057.15)
Jefferson	\$5,558	\$ 5,410.00	\$ 148.00	\$ 12,993.53	\$ (7,435.53)	\$ 22,363.15	\$ (16,805.15)
Humphreys	\$7,535	\$ 5,410.00	\$ 2,125.00	\$ 12,993.53	\$ (5,458.53)	\$ 22,363.15	\$ (14,828.15)
Issaquena	\$8,386	\$ 5,410.00	\$ 2,976.00	\$ 12,993.53	\$ (4,607.53)	\$ 22,363.15	\$ (13,977.15)
Quitman	\$9,218	\$ 5,410.00	\$ 3,808.00	\$ 12,993.53	\$ (3,775.53)	\$ 22,363.15	\$ (13,145.15)
Wilkinson	\$6,611	\$ 5,410.00	\$ 1,201.00	\$ 12,993.53	\$ (6,382.53)	\$ 22,363.15	\$ (15,752.15)
Amite	\$8,573	\$ 5,410.00	\$ 3,163.00	\$ 12,993.53	\$ (4,420.53)	\$ 22,363.15	\$ (13,790.15)
Jefferson Davis	\$7,298	\$ 5,410.00	\$ 1,888.00	\$ 12,993.53	\$ (5,695.53)	\$ 22,363.15	\$ (15,065.15)
Marion	\$6,954	\$ 5,410.00	\$ 1,544.00	\$ 12,993.53	\$ (6,039.53)	\$ 22,363.15	\$ (15,409.15)
Walthall	\$6,733	\$ 5,410.00	\$ 1,323.00	\$ 12,993.53	\$ (6,260.53)	\$ 22,363.15	\$ (15,630.15)

Per member-per-month (PMPM) is an important metric to calculate premiums for group health insurance or capitated payments to providers. The metric applies to revenue or cost for each enrolled member each month. It is calculated by dividing the units of service or product by member months.

Interpreting the PMPM is identifying the fixed amount of money per potential payment per unit of time paid to a healthcare organization in advance to perform needed services. By lowering the total cost of care and thereby reducing the overall PMPM, the payer saves money while potentially improving healthcare outcomes.

In the analysis that follows, we attempt to address the following questions.

1. What are the PMPM rates that Medicaid must set to cover medical costs for patients with multiple chronic conditions plus administrative expenses?
2. What would the rate be if utilization of services were reduced for each patient by 10% or 20%?
3. What rates would be set if the average cost of services for patients with multiple chronic conditions were reduced by 10%?
4. What would the premium be assuming that both utilization and cost reductions were made?

In Table 16, the number of member months was calculated by the number of beneficiaries enrolled in Medicaid in AL, LA and MS multiplied by 12. This analysis did not account for Medicaid churn.

Table 16: Number of Member Months for Medicaid Programs in AL, LA, and MS

State	Number of Medicaid Beneficiaries	Member Months
AL	1,000,000	12,000,000
LA	1,000,000	12,000,000
MS	860,000	10,080,000

From this analysis, we calculated the number of inpatient days per 1,000 enrollees by identifying the average length and cost of a hospital stay for a patient with multiple chronic conditions:

The average length of stay: 5 days

The average cost for 5 days: \$27,530.43

The average cost per day: \$5,470.86

In Table 17, the following analysis is shown:

1. Inpatient days per 1,000 enrollees
2. Average cost per day
3. The annual cost for 1,000 enrollees
4. The annual cost for 1,000 enrollees with 5% administrative cost
5. The annual cost for 50,000 enrollees with 5% administrative cost
6. PMPM Rate

Table 17: Baseline PMPM Rate for Medicaid Programs in AL, LA, and MS

Service Category	Inpatient Days Per 1,000	Average Cost Per Day	Annual Cost for 1,000 enrollees	Annual Cost for 1,000 enrollees with administration cost	Annual Cost for 50,000 enrollees with administration cost	PMPM rate
Alabama						
General	5000	\$5,470.86	\$27,354,300	\$28,794,000	\$1,439,700,000	\$119.98
Louisiana						
General	5000	\$5,470.86	\$27,354,300	\$28,794,000	\$1,439,700,000	\$119.98
Mississippi						
General	5000	\$5,470.86	\$27,354,300	\$28,794,000	\$1,439,700,000	\$142.83

Medicaid uses the Chronic Illness and Disability Payment (CDPS) for risk adjustment weights for Medicaid beneficiaries. The CDPS is a diagnostic-based risk adjustment model that is widely used to adjust capitated payments for health plans. CDPS uses International Classification of Disease (ICD) codes to assign CDPS Categories that indicate illness burden related to major body systems (e.g. cardiovascular) or types of chronic disease (e.g. diabetes). Within each major category is a hierarchy reflecting both the clinical severity of the condition and its expected effect on future costs. Each of the hierarchical CDPS Categories is assigned a CDPS weight. CDPS weights are additive across major categories. Medicaid Rx (MRX) is a pharmacy-based system that uses National Drug Classification (NDC) codes to assign MRX categories based on types of pharmacotherapy. Each category is assigned an MRX weight.

CDPS risk scores are calculated at the individual level but are applied at the group level. An individual risk score includes the sum of the intercept (or baseline), demographic (i.e. age and gender) weights, and weights for all indicated CDPS (or MRX or CDPS+Rx) categories. As described above, CDPS weights are hierarchical within major categories but are additive across major categories. In practice, this means, for example, that an individual will receive higher scores for multiple chronic conditions across different major categories, such as Cardiovascular and Diabetes, but will only receive the highest score for multiple conditions within a major category (e.g. for heart failure but not also for hypertension).

Using patients one through four as shown in Table 18, the cumulative CDPS scores are shown in Table 15

Table 18: Cumulative CDPS Scores for Medicaid Beneficiaries with Multiple Chronic Conditions

Patient #1	Patient #2	Patient #3	Patient #4
6.0094	4.30214	4.97456	2.86713

From this data, the adjusted PMPM rates are shown in Table 19.

Table 19: Adjusted PMPM Rates for AL, LA, and MS

Baseline PMPM	Adjusted PMPM #1	Adjusted PMPM #2	Adjusted PMPM #3	Adjusted PMPM #4
Alabama				
\$119.98	\$720.98	\$516.15	\$596.82	\$343.98
Louisiana				
\$119.98	\$720.98	\$516.15	\$596.82	\$343.98
Mississippi				

Baseline PMPM	Adjusted PMPM #1	Adjusted PMPM #2	Adjusted PMPM #3	Adjusted PMPM #4
\$142.83	\$858.31	\$614.46	\$710.50	\$409.50

In Table 20, we simulate the following scenarios to modify the PMPM in addition to the risk adjustment.

1. A reduction in utilization of 10%
2. A reduction in utilization of 20%
3. A reduction of cost by 10% and utilization by 10%

Table 20: Adjusted PMPM by Utilization and Cost Reductions

Category	Baseline	Adjusted #1	Adjusted #2	Adjusted #3	Adjusted #4
Alabama					
10% reduction in utilization	\$107.98	\$648.88	\$464.53	\$537.14	\$309.59
20% reduction in utilization	\$95.98	\$576.78	\$412.92	\$477.46	\$275.19
10% cost reduction and 10% utilization reduction	\$97.16	\$583.90	\$418.02	\$483.35	\$278.58
Louisiana					
10% reduction in utilization	\$107.98	\$648.88	\$464.53	\$537.14	\$309.59
20% reduction in utilization	\$95.98	\$576.78	\$412.92	\$477.46	\$275.19
10% cost reduction and 10% utilization reduction	\$97.16	\$583.90	\$418.02	\$483.35	\$278.58
Mississippi					
10% reduction in utilization	\$128.54	\$772.48	\$553.02	\$639.45	\$368.55
20% reduction in utilization	\$114.26	\$686.65	\$491.57	\$568.40	\$327.60
10% cost reduction and 10%	\$115.67	\$695.12	\$497.64	\$575.42	\$331.65

Category	Baseline	Adjusted #1	Adjusted #2	Adjusted #3	Adjusted #4
utilization reduction					

This analysis demonstrates that by reducing the utilization and cost of care, the overall PMPM will decrease, which produces cost-savings for payers. In Table 20, if that

Medicare Fee-for-Service (FFS) Spending

The actual per capita spending for all Medicare Fee-For-Service Beneficiaries by state and county. for multiple chronic conditions is shown in Table 21. The number of chronic conditions was segmented in 2-3, 4-5, and 6+

Table 21: Medicare FFS Spending on Patients with Multiple Chronic Conditions

County	Number of Chronic Conditions 2-3	Number of Chronic Conditions 4-5	Number of Chronic Conditions 6+
Alabama			
Greene	\$4,906.1	\$9,924.4	\$26,502.8
Perry	\$4,742.0	\$9,420.0	\$26,903.7
Conecuh	\$4,800.3	\$8,386.1	\$25,782.2
Wilcox	\$5,933.4	\$10,401.7	\$29,461.3
Lamar	\$4,095.4	\$9,981.3	\$25,862.1
Monroe	\$5,426.0	\$12,521.2	\$29,336.7
Coosa	\$5,168.0	\$10,638.5	\$26,744.8
Choctaw	\$5,995.6	\$9,951.4	\$28,649.2
Winston	\$4,828.6	\$8,435.6	\$23,495.5
Lowndes	\$5,018.1	\$12,343.8	\$30,612.4
Louisiana			
Tensas	\$6,053.8	\$12,927.8	\$35,434.6
East Carroll	\$4,805.7	\$11,153.9	\$35,552.2
Bienville	\$7,423.5	\$13,132.8	\$41,663.9
Claiborne	\$5,977.6	\$11,696.6	\$31,294.0
Webster	\$5,677.3	\$12,936.4	\$33,525.0
Concordia	\$5,572.2	\$11,204.9	\$35,728.1
Washington	\$5,788.1	\$12,141.6	\$35,735.4
Franklin	\$7,006.4	\$13,002.1	\$33,035.5
Caldwell	\$6,546.7	\$11,946.4	\$30,610.8
Red River	\$5,967.5	\$12,554.9	\$43,059.6
Mississippi			
Sharkey	\$5,132.7	\$11,608.9	\$31,594.6
Jefferson	\$5,367.1	\$10,627.9	\$36,125.3
Humphreys	\$5,905.8	\$12,788.4	\$38,643.8
Issaquena	\$7,301.5	\$13,848.5	\$41,509.7
Quitman	\$6,107.9	\$11,761.5	\$33,552.4
Wilkinson	\$7,538.0	\$9,630.7	\$29,883.5

Amite	\$5,129.1	\$11,936.4	\$31,142.1
Jefferson Davis	\$6,022.8	\$9,622.8	\$28,169.8
Marion	\$5,279.1	\$10,301.5	\$26,775.5
Walthall	\$5,052.5	\$10,485.5	\$27,810.1

Source: Centers for Medicare & Medicaid Services

In Table 22, a comparison of the annual spending rate for Medicare FFS against those beneficiaries with multiple chronic conditions is shown. The number of chronic conditions is stratified by 2-3, 4-5, and 6+. A comparison was made between the annual cost per beneficiary and the amount each program spends per beneficiary to identify any shortfalls.

Table 22: Comparison of Annual Average Payment by Medicare FFS Beneficiary, by County, against Annual Average Cost of Multiple Chronic Conditions

County	Average Payment Per Eligible	2 to 3 Conditions	Profit/Loss	Average Payment Per Eligible (4-5)	4-5 Conditions	Profit/Loss	Average Payment Per Eligible (6+)	6+ Conditions	Profit/Loss
Alabama									
Greene	\$4,906	\$ 6,683.59	\$ (1,777.49)	\$ 9,924.40	\$ 11,241.16	\$ (1,316.76)	\$ 26,502.80	\$ 22,363.15	\$ 4,139.65
Perry	\$4,742	\$ 6,683.59	\$ (1,941.59)	\$ 9,420.00	\$ 11,241.16	\$ (1,821.16)	\$ 26,903.70	\$ 22,363.15	\$ 4,540.55
Conecuh	\$4,800	\$ 6,683.59	\$ (1,883.29)	\$ 8,386.10	\$ 11,241.16	\$ (2,855.06)	\$ 25,782.20	\$ 22,363.15	\$ 3,419.05
Wilcox	\$5,933	\$ 6,683.59	\$ (750.19)	\$ 10,401.70	\$ 11,241.16	\$ (839.46)	\$ 29,461.30	\$ 22,363.15	\$ 7,098.15
Lamar	\$4,095	\$ 6,683.59	\$ (2,588.19)	\$ 9,981.30	\$ 11,241.16	\$ (1,259.86)	\$ 25,862.10	\$ 22,363.15	\$ 3,498.95
Monroe	\$5,426	\$ 6,683.59	\$ (1,257.59)	\$ 12,521.20	\$ 11,241.16	\$ 1,280.04	\$ 29,336.70	\$ 22,363.15	\$ 6,973.55
Coosa	\$5,168	\$ 6,683.59	\$ (1,515.59)	\$ 10,638.50	\$ 11,241.16	\$ (602.66)	\$ 26,744.80	\$ 22,363.15	\$ 4,381.65
Choctaw	\$5,996	\$ 6,683.59	\$ (687.99)	\$ 9,951.40	\$ 11,241.16	\$ (1,289.76)	\$ 28,649.20	\$ 22,363.15	\$ 6,286.05
Winston	\$4,829	\$ 6,683.59	\$ (1,854.99)	\$ 8,435.60	\$ 11,241.16	\$ (2,805.56)	\$ 23,495.50	\$ 22,363.15	\$ 1,132.35
Lowndes	\$5,018	\$ 6,683.59	\$ (1,665.49)	\$ 12,343.80	\$ 11,241.16	\$ 1,102.64	\$ 30,612.40	\$ 22,363.15	\$ 8,249.25
Louisiana									
Tensas	\$6,054	\$ 6,683.59	\$ (629.79)	\$ 12,927.80	\$ 11,241.16	\$ 1,686.64	\$ 35,434.60	\$ 22,363.15	\$ 13,071.45
East Carroll	\$4,806	\$ 6,683.59	\$ (1,877.89)	\$ 11,153.90	\$ 11,241.16	\$ (87.26)	\$ 35,552.20	\$ 22,363.15	\$ 13,189.05
Bienville	\$7,424	\$ 6,683.59	\$ 739.91	\$ 13,132.80	\$ 11,241.16	\$ 1,891.64	\$ 41,663.90	\$ 22,363.15	\$ 19,300.75
Claiborne	\$5,978	\$ 6,683.59	\$ (705.99)	\$ 11,696.60	\$ 11,241.16	\$ 455.44	\$ 31,294.00	\$ 22,363.15	\$ 8,930.85
Webster	\$5,677	\$ 6,683.59	\$ (1,006.29)	\$ 12,936.40	\$ 11,241.16	\$ 1,695.24	\$ 33,525.00	\$ 22,363.15	\$ 11,161.85
Concordia	\$5,572	\$ 6,683.59	\$ (1,111.39)	\$ 11,204.90	\$ 11,241.16	\$ (36.26)	\$ 35,728.10	\$ 22,363.15	\$ 13,364.95
Washington	\$5,788	\$ 6,683.59	\$ (895.49)	\$ 12,141.60	\$ 11,241.16	\$ 900.44	\$ 35,735.40	\$ 22,363.15	\$ 13,372.25
Franklin	\$7,006	\$ 6,683.59	\$ 322.81	\$ 13,002.10	\$ 11,241.16	\$ 1,760.94	\$ 33,035.50	\$ 22,363.15	\$ 10,672.35
Caldwell	\$6,547	\$ 6,683.59	\$ (136.89)	\$ 11,946.40	\$ 11,241.16	\$ 705.24	\$ 30,610.80	\$ 22,363.15	\$ 8,247.65
Red River	\$5,968	\$ 6,683.59	\$ (716.09)	\$ 12,554.90	\$ 11,241.16	\$ 1,313.74	\$ 43,059.60	\$ 22,363.15	\$ 20,696.45
Mississippi									
Sharkey	\$5,133	\$ 6,683.59	\$ (1,550.89)	\$ 11,608.90	\$ 11,241.16	\$ 367.74	\$ 31,594.60	\$ 22,363.15	\$ 9,231.45
Jefferson	\$5,367	\$ 6,683.59	\$ (1,316.49)	\$ 10,627.90	\$ 11,241.16	\$ (613.26)	\$ 36,125.30	\$ 22,363.15	\$ 13,762.15
Humphreys	\$5,906	\$ 6,683.59	\$ (777.79)	\$ 12,788.40	\$ 11,241.16	\$ 1,547.24	\$ 38,643.80	\$ 22,363.15	\$ 16,280.65
Issaquena	\$7,302	\$ 6,683.59	\$ 617.91	\$ 13,848.50	\$ 11,241.16	\$ 2,607.34	\$ 41,509.70	\$ 22,363.15	\$ 19,146.55
Quitman	\$6,108	\$ 6,683.59	\$ (575.69)	\$ 11,761.50	\$ 11,241.16	\$ 520.34	\$ 33,552.40	\$ 22,363.15	\$ 11,189.25
Wilkinson	\$7,538	\$ 6,683.59	\$ 854.41	\$ 9,630.70	\$ 11,241.16	\$ (1,610.46)	\$ 29,883.50	\$ 22,363.15	\$ 7,520.35
Amite	\$5,129	\$ 6,683.59	\$ (1,554.49)	\$ 11,936.40	\$ 11,241.16	\$ 695.24	\$ 31,142.10	\$ 22,363.15	\$ 8,778.95
Jefferson Davis	\$6,023	\$ 6,683.59	\$ (660.79)	\$ 9,622.80	\$ 11,241.16	\$ (1,618.36)	\$ 28,169.80	\$ 22,363.15	\$ 5,806.65
Marion	\$5,279	\$ 6,683.59	\$ (1,404.49)	\$ 10,301.50	\$ 11,241.16	\$ (939.66)	\$ 26,775.50	\$ 22,363.15	\$ 4,412.35
Walthall	\$5,053	\$ 6,683.59	\$ (1,631.09)	\$ 10,485.50	\$ 11,241.16	\$ (755.66)	\$ 27,810.10	\$ 22,363.15	\$ 5,446.95

Medicare Advantage

The rates for Medicare Advantage (MA) are calculated by taking the 2022 Rate Score issued by CMS (which is also inclusive of the 5% Quality Bonus Payment) and multiplying it by the risk adjustment factor for the identified categories of patients that qualify for MA. The use of CMS Geographic Base Index and Hierarchical Condition Categories (HCCs) were used to derive the average risk adjustment factor (RAF) and the PMPM was calculated by multiplying the monthly capitation rate by the RAF. The RAFs for Patients five through eight (see Table 11) are shown in Table 23.

Table 23: Risk Adjustment Factors (RAFTs) for Medicare Advantage Patients

Patient #5	Patient #6	Patient #7	Patient #8
1.84	2.11	2.24	1.90

Using this data, we can calculate the adjusted PMPM scores across counties in AL, LA, and MS as shown in Table 24. Given the diversity of the patient populations within each of the counties, each one was randomly assigned an RAF, which represented the most frequent patients with multiple chronic conditions that were 65 or over.

Table 24: Adjusted PMPM Rate for Medicare Advantage in AL, LA, and MS

County	MA Monthly Capitation Rate	Average Risk Score (RAF)	Per Member Per Month (PMPM) for MA
Alabama			
Greene	1034.11	1.84	\$1,902.76
Perry	955.79	1.84	\$1,758.65
Conecuh	1093.46	2.11	\$2,307.20
Wilcox	1035.81	2.11	\$2,185.56
Lamar	1019.04	2.24	\$2,282.65
Monroe	1081.42	2.24	\$2,422.38
Coosa	1034.15	1.90	\$1,964.89
Choctaw	1065.51	1.90	\$2,024.47
Winston	1037.99	1.84	\$1,909.90
Lowndes	1100.91	2.11	\$2,322.92
Louisiana			
Tensas	1145.45	1.84	\$2,107.63
East Carroll	1144.81	1.84	\$2,106.45
Bienville	1265.63	2.11	\$2,670.48
Claiborne	1101.29	2.11	\$2,323.72
Webster	1092.84	2.24	\$2,447.96
Concordia	1132.89	2.24	\$2,537.67
Washington	1165.37	1.90	\$2,214.20
Franklin	1207.34	1.90	\$2,293.95
Caldwell	1210.56	1.84	\$2,227.43
Red River	1364.56	2.11	\$2,879.22
Mississippi			
Sharkey	988.16	1.84	\$1,818.21
Jefferson	1107.54	1.84	\$2,037.87

County	MA Monthly Capitation Rate	Average Risk Score (RAF)	Per Member Per Month (PMPM) for MA
Humphreys	1056.54	2.11	\$2,229.30
Issaquena	1050.02	2.11	\$2,215.54
Quitman	1121.42	2.24	\$2,511.98
Wilkinson	1077.42	2.24	\$2,413.42
Amite	1073.15	1.90	\$2,038.99
Jefferson Davis	1078.48	1.90	\$2,049.11
Marion	1058.66	1.84	\$1,947.93
Walthall	1086.03	2.11	\$2,291.52

In Table 25, the comparison of the MA PMPM rates against the cost of multiple chronic conditions is shown.

Table 25: Comparison of Annual Average Payment by MA Beneficiary, by County, against Annual Average Cost of Multiple Chronic Conditions

County	Average Annual Payment Per Eligible	2-3 Conditions	Profit/Loss	4-5 Conditions	Profit/Loss	6+ Conditions	Profit/Loss
Alabama							
Greene	\$ 22,833.12	\$ 6,683.59	\$ 16,149.53	\$ 11,241.16	\$ 11,591.96	\$ 22,363.15	\$ 469.97
Perry	\$ 21,103.80	\$ 6,683.59	\$ 14,420.21	\$ 11,241.16	\$ 9,862.64	\$ 22,363.15	\$ (1,259.35)
Conecuh	\$ 27,686.40	\$ 6,683.59	\$ 21,002.81	\$ 11,241.16	\$ 16,445.24	\$ 22,363.15	\$ 5,323.25
Wilcox	\$ 26,226.72	\$ 6,683.59	\$ 19,543.13	\$ 11,241.16	\$ 14,985.56	\$ 22,363.15	\$ 3,863.57
Lamar	\$ 27,391.80	\$ 6,683.59	\$ 20,708.21	\$ 11,241.16	\$ 16,150.64	\$ 22,363.15	\$ 5,028.65
Monroe	\$ 29,068.56	\$ 6,683.59	\$ 22,384.97	\$ 11,241.16	\$ 17,827.40	\$ 22,363.15	\$ 6,705.41
Coosa	\$ 23,578.68	\$ 6,683.59	\$ 16,895.09	\$ 11,241.16	\$ 12,337.52	\$ 22,363.15	\$ 1,215.53
Choctaw	\$ 24,293.64	\$ 6,683.59	\$ 17,610.05	\$ 11,241.16	\$ 13,052.48	\$ 22,363.15	\$ 1,930.49
Winston	\$ 22,918.80	\$ 6,683.59	\$ 16,235.21	\$ 11,241.16	\$ 11,677.64	\$ 22,363.15	\$ 555.65
Lowndes	\$ 27,875.04	\$ 6,683.59	\$ 21,191.45	\$ 11,241.16	\$ 16,633.88	\$ 22,363.15	\$ 5,511.89
Louisiana							
Tensas	\$ 13,745.40	\$ 6,683.59	\$ 7,061.81	\$ 11,241.16	\$ 2,504.24	\$ 22,363.15	\$ (8,617.75)
East Carroll	\$ 13,737.72	\$ 6,683.59	\$ 7,054.13	\$ 11,241.16	\$ 2,496.56	\$ 22,363.15	\$ (8,625.43)
Bienville	\$ 15,187.56	\$ 6,683.59	\$ 8,503.97	\$ 11,241.16	\$ 3,946.40	\$ 22,363.15	\$ (7,175.59)
Claiborne	\$ 13,215.48	\$ 6,683.59	\$ 6,531.89	\$ 11,241.16	\$ 1,974.32	\$ 22,363.15	\$ (9,147.67)
Webster	\$ 13,114.08	\$ 6,683.59	\$ 6,430.49	\$ 11,241.16	\$ 1,872.92	\$ 22,363.15	\$ (9,249.07)
Concordia	\$ 13,594.68	\$ 6,683.59	\$ 6,911.09	\$ 11,241.16	\$ 2,353.52	\$ 22,363.15	\$ (8,768.47)
Washington	\$ 13,984.44	\$ 6,683.59	\$ 7,300.85	\$ 11,241.16	\$ 2,743.28	\$ 22,363.15	\$ (8,378.71)
Franklin	\$ 14,488.08	\$ 6,683.59	\$ 7,804.49	\$ 11,241.16	\$ 3,246.92	\$ 22,363.15	\$ (7,875.07)
Caldwell	\$ 14,526.72	\$ 6,683.59	\$ 7,843.13	\$ 11,241.16	\$ 3,285.56	\$ 22,363.15	\$ (7,836.43)
Red River	\$ 16,374.72	\$ 6,683.59	\$ 9,691.13	\$ 11,241.16	\$ 5,133.56	\$ 22,363.15	\$ (5,988.43)
Mississippi							
Sharkey	\$ 11,857.92	\$ 6,683.59	\$ 5,174.33	\$ 11,241.16	\$ 616.76	\$ 22,363.15	\$ (10,505.23)
Jefferson	\$ 13,290.48	\$ 6,683.59	\$ 6,606.89	\$ 11,241.16	\$ 2,049.32	\$ 22,363.15	\$ (9,072.67)
Humphreys	\$ 12,678.48	\$ 6,683.59	\$ 5,994.89	\$ 11,241.16	\$ 1,437.32	\$ 22,363.15	\$ (9,684.67)
Issaquena	\$ 12,600.24	\$ 6,683.59	\$ 5,916.65	\$ 11,241.16	\$ 1,359.08	\$ 22,363.15	\$ (9,762.91)
Quitman	\$ 13,457.04	\$ 6,683.59	\$ 6,773.45	\$ 11,241.16	\$ 2,215.88	\$ 22,363.15	\$ (8,906.11)
Wilkinson	\$ 12,929.04	\$ 6,683.59	\$ 6,245.45	\$ 11,241.16	\$ 1,687.88	\$ 22,363.15	\$ (9,434.11)
Amite	\$ 12,877.80	\$ 6,683.59	\$ 6,194.21	\$ 11,241.16	\$ 1,636.64	\$ 22,363.15	\$ (9,485.35)
Jefferson Davis	\$ 12,941.76	\$ 6,683.59	\$ 6,258.17	\$ 11,241.16	\$ 1,700.60	\$ 22,363.15	\$ (9,421.39)
Marion	\$ 12,703.92	\$ 6,683.59	\$ 6,020.33	\$ 11,241.16	\$ 1,462.76	\$ 22,363.15	\$ (9,659.23)
Walthall	\$ 13,032.36	\$ 6,683.59	\$ 6,348.77	\$ 11,241.16	\$ 1,791.20	\$ 22,363.15	\$ (9,330.79)

Cost Drivers

There are also additional drivers that increase the overall cost of care for patients with multiple chronic conditions. These include the Area Deprivation Index score for each county, the rate of hospital readmissions for patients with multiple chronic conditions, and the early death rate. Each of these raises the overall health care costs due to barriers to access to appropriate care, limited or non-existent follow-up after discharge from a hospital, and no care management for individuals.

Area Deprivation Index

The Area Deprivation Index (ADI) is based on a measure created by the Health Resources & Services Administration (HRSA) over three decades ago, and has since been refined, adapted, and validated to the Census Block Group neighborhood level by the research team at the University of Wisconsin-Madison. It allows for rankings of neighborhoods by a socioeconomic disadvantage in a region of interest (e.g., at the state or national level).²¹ It includes factors for the theoretical domains of income, education, employment, and housing quality. It can be used to inform health delivery and policy, especially for the most disadvantaged neighborhood groups. The scores are from 1-10 within state counties and the higher the score, the more disadvantaged it is (low income, high unemployment, travel, and housing barriers, food insecurity, etc.). The state score for each county identified in the study is shown in Table 26.

Table 26: Average Area Deprivation (ADI) Scores by County

County	ADI Score
Alabama	
Greene	9
Perry	7
Conecuh	6
Wilcox	8
Lamar	9
Monroe	6
Coosa	2
Choctaw	8
Winston	9
Lowndes	6
Louisiana	
Tensas	9
East Carroll	6
Bienville	8
Claiborne	9
Webster	10
Concordia	5
Washington	7
Franklin	7
Caldwell	10
Red River	7
Mississippi	
Sharkey	9

County	ADI Score
Jefferson	8
Humphreys	8
Issaquena	7
Quitman	7
Wilkinson	7
Amite	5
Jefferson Davis	7
Marion	6
Walthall	6

Source: Health Resources and Services Administration

Readmissions and Other Factors

Additionally, information from the County Health Rankings and CMS, provide information about the percentage of hospital readmissions, preventable hospitalizations, and early death, as shown in Table 27.

Table 27: Readmissions, Preventable Hospitalizations, and Early Data by State

State	Hospital Readmissions (65-74) (%)	Preventable Hospitalizations – Ages 65-74 (Number)	Early Death – Ages 65-74 (Number)
Alabama	16	2,332	2,767
Louisiana	16	2,276	2,691
Mississippi	16	2,321	3,024

Source: America's Health Rankings and the Centers for Medicare & Medicaid Services

Health care systems originally designed to treat single conditions may not have well-developed processes to support MCC care delivery. Individuals with MCC often require multiple providers, and MCC populations are more likely than lower morbidity populations to experience financial constraints and associated social needs.²² Impaired communication between providers can result in contradictory advice for patients, insufficient chronic care coordination, and missed opportunities to leverage patient self-management. Interactions between patients' social and medical needs can further complicate care delivery and worsen health outcomes for MCC individuals.²³ Further, current reimbursement for care delivery is insufficient to support the longer visit times needed for patient-centered approaches to complex care needs.

Current reimbursement structures (e.g., fee-for-service or FFS, risk-based or capitated payments) may impede the system's ability to demonstrate any net cost savings or a return on investment (ROI) for targeted MCC services. Effective approaches to MCC care lay at the intersection of meeting patients' needs, addressing system needs, available community resources, and aligning financial incentives. Several studies identified that meeting patients' social needs and increasing access to necessary care decreases the likelihood of unnecessary emergency or hospital utilization.²⁴

Value-Based Care

Certain value-based care programs, such as Accountable Care Organizations, drive population-based approaches and work to align financial incentives to support best practices. Calculating a PMPM amount for a value-based approach includes using the current PMPM for Medicare Advantage and multiplying that by the current percentage of cost-of-care trends (published by CMS) to derive an adjusted cost-of-care PMPM. That figure is multiplied by the aggregated adjustment factor (also published by CMS) to derive a

program's performance year target PMPM. The results of this analysis for the state countries included in this study are shown in Table 28.

Table 28: Performance Target PMPM for Value-Based Care Models

County	PMPM	Cost-of-Care Trends	Adjusted Cost-of-Care PMPM	Aggregated Adjustment Factor	Performance Year Target PMPM
Alabama					
Greene	\$1,902.76	7%	\$2,035.95	1.00	\$2,035.95
Perry	\$1,758.65	7%	\$1,881.76	1.00	\$1,881.76
Conecuh	\$2,307.20	7%	\$2,468.70	1.00	\$2,468.70
Wilcox	\$2,185.56	7%	\$2,338.55	1.00	\$2,338.55
Lamar	\$2,282.65	7%	\$2,442.44	1.00	\$2,442.44
Monroe	\$2,422.38	7%	\$2,591.95	1.00	\$2,591.95
Coosa	\$1,964.89	7%	\$2,102.43	1.00	\$2,102.43
Choctaw	\$2,024.47	7%	\$2,166.18	1.00	\$2,166.18
Winston	\$1,909.90	7%	\$2,043.59	1.00	\$2,043.59
Lowndes	\$2,322.92	7%	\$2,485.52	1.00	\$2,485.52
Louisiana					
Tensas	\$2,107.63	7%	\$2,255.16	1.00	\$2,255.16
East Carroll	\$2,106.45	7%	\$2,253.90	1.00	\$2,253.90
Bienville	\$2,670.48	7%	\$2,857.41	1.00	\$2,857.41
Claiborne	\$2,323.72	7%	\$2,486.38	1.00	\$2,486.38
Webster	\$2,447.96	7%	\$2,619.32	1.00	\$2,619.32
Concordia	\$2,537.67	7%	\$2,715.31	1.00	\$2,715.31
Washington	\$2,214.20	7%	\$2,369.19	1.00	\$2,369.19
Franklin	\$2,293.95	7%	\$2,454.53	1.00	\$2,454.53
Caldwell	\$2,227.43	7%	\$2,383.35	1.00	\$2,383.35
Red River	\$2,879.22	7%	\$3,080.77	1.00	\$3,080.77
Mississippi					
Sharkey	\$1,818.21	7%	\$1,945.48	1.00	\$1,945.48
Jefferson	\$2,037.87	7%	\$2,180.52	1.00	\$2,180.52
Humphreys	\$2,229.30	7%	\$2,385.35	1.00	\$2,385.35
Issaquena	\$2,215.54	7%	\$2,370.63	1.00	\$2,370.63
Quitman	\$2,511.98	7%	\$2,687.82	1.00	\$2,687.82
Wilkinson	\$2,413.42	7%	\$2,582.36	1.00	\$2,582.36

County	PMPM	Cost-of-Care Trends	Adjusted Cost-of-Care PMPM	Aggregated Adjustment Factor	Performance Year Target PMPM
Amite	\$2,038.99	7%	\$2,181.72	1.00	\$2,181.72
Jefferson Davis	\$2,049.11	7%	\$2,192.55	1.00	\$2,192.55
Marion	\$1,947.93	7%	\$2,084.29	1.00	\$2,084.29
Walthall	\$2,291.52	7%	\$2,451.93	1.00	\$2,451.93

Source: Centers for Medicare & Medicaid Services

- The initial PMPM includes the 5% bonus payment, which when included with the cost-of-care trends and the aggregated adjustment factor led to a performance year target PMPM that meets the cost of patients with multiple chronic conditions
- To meet the target and achieve the 5% bonus payment, the emphasis is on quality outcomes and reduced costs of care. Given the lack of specialists within Alabama, Louisiana, and Mississippi, the need for remote specialty care may be critical to reaching these thresholds.

Cost-Benefit

Given the severity of illness of those patients with multiple chronic conditions, particularly in Medicaid managed care where the cost is extremely high, it is critical to examine the cost-benefit of providing specialty and targeted services for these high acuity patients. Rather than staffing remote locations with specialty providers, telehealth allows specialists to connect with patients in rural and underserved areas and providers virtually.²⁵ This expands patients' access to specialty care and enables primary care providers to engage and connect with specialty providers, allowing them to better serve their patients. These types of remote consults could potentially reduce health care costs by facilitating efficient access to specialty care services and reducing unnecessary face-to-face visits. However, remote specialty consultants may increase health care costs if they lead to unnecessary diagnostic testing, provide lower-quality care, or delay (not supplant) the need for face-to-face visits. A 2018 publication examined the association between remote consultation by cardiologists and health care costs for 369 Medicaid recipients cared for at a federally qualified health care center. They reported an approximate \$466 cost savings per patient receiving a cardiology consult.²⁶

A 2021 study published in *The American Journal of Managed Care* found that the use of remote specialty consultations led to cost savings after three- and six months in both outpatient and inpatient settings. In Table 29, the percentage difference in the remote consultant in an outpatient setting is shown. In almost all of the findings, the three- and six-month outpatient costs were less as compared to a face-to-face encounter. The only exception was cardiology, which showed an increase after three months.²⁷

Table 29: Cost Savings of Outpatient Remote Specialty Consultations

Specialty	Three Months	Six Months
Cardiology	6.24%	-2.35%
Diabetes	-23.37%	-25.94%
Endocrinology	-6.86%	-8.33%
Pulmonary	-14.85%	-16.50%
Nephrology	-11.44%	-16.05%

Source: American Journal of Managed Care

In Table 30, the three- and six-month inpatient cost outcomes are shown.

Table 30: Cost Savings of Inpatient Remote Specialty Consultations

Specialty	Three Months	Six Months
Cardiology	-5.24%	20.64%
Diabetes	-11.23%	-1.78%
Endocrinology	-4.00%	2.51%
Pulmonary	-12.99%	-10.12%
Nephrology	-9.34%	-2.66%

Source: American Journal of Managed Care

The overall cost of remote specialty consultation would include direct (costs to the payer) and indirect costs (cost to the patients). Direct costs consist of three elements: delivery costs, consultation-specific costs, and referral costs. Delivery costs include administrative costs, support costs, and the costs associated with the registration and orientation of new patients who are using this service for the first time. Administrative costs consist of time spent on preparation prior to the patient encounter and dissemination of information to primary care providers and other personnel managing the care of the patients. Support costs consist of time following up with primary care providers/specialists on cases and outstanding requests.

Consultation-specific costs include staff time to assign each new patient to the appropriate specialist and the costs associated with remunerations. Added referral costs are based on those consultations where the primary care providers were not contemplating a referral, but as a result of a consultation with a patient, one was initiated. The total potential savings as a result of remote specialty consultation are calculated based on direct (savings for the payer) and indirect savings (savings for the patient).

Potential savings could come from the following:

- All avoided referrals (patient was able to communicate directly with a specialist and that information was relayed to a primary care provider)
- Travel costs – the patient does not have to travel to see a specialist
- Costs associated with lost wages/productivity

For this cost-benefit analysis, a standard rate of \$780 per referral was used.

Patient Travel. To determine costs associated with patient travel, the analysis assumed that, in the absence of direct specialty care, patients would use their vehicle (or public transportation) to make one round-trip visit to a provider closest to their residence. Travel distance was calculated by identifying a provider address using a State's Medicaid site or Medicare.gov (dependent upon the payer source for the encounter) and randomly choosing an address within the same county. Google Maps provided mileage and time for the fastest route assuming 55 miles per hour (MPH) on non-freeway lanes and 65 MPH on highways. The Internal Revenue Service (IRS) rate of \$0.68 per mile was used to determine the travel cost.

Patient Productivity. Another source of savings is the increased productivity of patients when they engage in a telehealth visit and do not have to travel to the providers' office and take leave from work. This analysis assumes that, on average, wages are lost each time a patient travels to a location for an in-person visit.

The productivity analysis also shows that wages are an adequate proxy for a patient's benefits when actively working. Given that many Medicare beneficiaries are retirees, The patient productivity analysis focused solely on Medicaid populations and used information from State Medicaid programs to determine the salary thresholds to qualify for the program. Assuming a four-person household, the annual minimum income rate to qualify for Medicaid was calculated and divided by 12 to derive a monthly income. That figure was then divided by 160 to calculate an hourly rate. The following formula to determine the total wages lost:

Time in hour increments to travel to an in-person location (calculated using the Google Maps fastest route estimate divided by 60 minutes)

Multiplied by calculated *hourly wage* (based on a threshold for Medicaid eligibility)

Multiplied by two (to account for *round trip* travel).

This calculation provides the wages lost for a round-trip visit to a provider per patient enrolled in Medicaid.

The analysis relied on data gathered from March through September of 2020 that showed the patients using telehealth services for health care. The data was segmented by state, with Alabama, Louisiana, and Mississippi used for the analysis. Those states were further segmented by county, and a cost-benefit analysis showing the potential savings per person is found in Table 31 for both the Medicaid and Medicare programs

Table 31: Estimated Cost Benefit of Remote Specialty Consultation for Medicaid and Medicare

County	Medicaid Cost Savings	Medicare Cost Savings (both FFS and MA)
Alabama		
Greene	\$1,674	\$1,124
Perry	\$1,231	\$1,132
Conecuh	\$1,876	\$1,298
Wilcox	\$1,987	\$1,076
Lamar	\$1,844	\$1,334
Monroe	\$1,533	\$1,001
Coosa	\$1,673	\$1,341
Choctaw	\$1,981	\$1,109
Winston	\$1,762	\$1,192
Lowndes	\$1,926	\$1,246
Louisiana		
Tensas	\$1,997	\$1,151
East Carroll	\$1,814	\$1,716
Bienville	\$1,478	\$1,443
Claiborne	\$1,598	\$1,053
Webster	\$1,625	\$1,523
Concordia	\$1,973	\$1,687
Washington	\$1,376	\$1,109
Franklin	\$1,321	\$1,312
Caldwell	\$1,869	\$1,101

County	Medicaid Cost Savings	Medicare Cost Savings (both FFS and MA)
Red River	\$1,978	\$1,452
Mississippi		
Sharkey	\$1,155	\$1,121
Jefferson	\$1,415	\$1,098
Humphreys	\$1,900	\$1,514
Issaquena	\$1,835	\$1,786
Quitman	\$1,921	\$1,526
Wilkinson	\$1,935	\$1,513
Amite	\$1,748	\$1,433
Jefferson Davis	\$1,876	\$1,064
Marion	\$1,933	\$1,321
Walthall	\$1,451	\$1,578

Weight Management and Nutrition

Guidelines issued by both the CDC and the National Institutes of Health outline an evidence-based process to guide comprehensive weight management of individuals by medical and allied health professionals.²⁸ A key recommendation in the guidelines for adults is regular follow-up with ongoing, long-term support provided by a clinician. However, weight management interventions often have high attrition rates, with logistical aspects of attendance reported as common reasons to drop out. The use of a remote weight management and nutrition program could include the following²⁹:

- One-on-one consultation – An initial consultation followed by review appointments. Consultations would consist of patient assessment, education, and counseling tailored to the individual.
- Assessment of health behaviors, including diet and physical activity, comorbidities, medications, smoking, weight history, and readiness to change.
- Provision of advice such as the health benefits of weight loss, meal plans, and dietary and physical activity guidelines

Literature on the cost-benefit of remote weight management programs included direct costs such as materials (e.g., video platform, web cameras, etc.), personnel, and other resources (recurring costs of treatment). The estimated costs of a remote program were approximately \$389.78 with the costs decreasing the longer the patient stayed in the program (presumably because patients would not have to travel and with scheduled follow-ups would be less likely to quit).³⁰

Community Health Workers

A growing number of health organizations have hired community health workers to provide social support, care coordination, and advocacy for high-risk patients. Several research studies have described socio-behavioral interventions delivered by community health workers that are effective in improving clinical outcomes for those with chronic diseases. A cost-benefit analysis of community health workers conducted in Pennsylvania factored direct costs, such as personnel, equipment and services, and office space rent. The analysis also included inpatient and outpatient costs for patients. The analysis showed a cost reduction in the use of community health workers within the state Medicaid program of 37% for both inpatient and outpatient use.³¹

Total Addressable Market

Using a bottom-up approach that looks at the subset of the specialty care market for Alabama, Louisiana, and Mississippi to ascertain the total market size. The analysis included the number of individuals on Medicaid and Medicare within each county and the amount of Medicaid and Medicaid dollars spent on those with multiple chronic conditions. Some assumptions were made in the creation of this table.

- It cannot be assumed that everyone currently on Medicaid or Medicare has multiple chronic conditions. For this analysis, the total number of beneficiaries was multiplied by 25% to arrive at a number for the analysis.
- It was also difficult to identify the number of Medicare (either FFS or MA) beneficiaries by county since that data resides with CMS. The number of Medicare beneficiaries by state was identified and that was divided by the number of counties in the state, and then that number was multiplied by 25% to arrive at a figure for the analysis.

The results are found in Table 32

Table 32: Estimated Total Addressable Market Size for Medicaid and Medicare, by State and County, for People with Multiple Chronic Conditions.

County	Total Number of Medicaid Beneficiaries with MCC	Cost of those with 2-3 Chronic Diseases	Total Medicaid Addressable Market	Total Number of Medicare Beneficiaries with MCC	Cost of those with 2-3 Chronic Diseases	Total Medicare Addressable Market
Alabama						
Greene	4,320	\$ 5,410.02	\$ 23,371,286	7930	\$ 6,683.59	\$ 53,000,869
Perry	2,250	\$ 5,410.02	\$ 12,169,840	7930	\$ 6,683.59	\$ 53,000,869
Conecuh	7,280	\$ 5,410.02	\$ 39,382,241	7930	\$ 6,683.59	\$ 53,000,869
Wilcox	2,805	\$ 5,410.02	\$ 15,175,106	7930	\$ 6,683.59	\$ 53,000,869
Lamar	2,102	\$ 5,410.02	\$ 11,371,862	7930	\$ 6,683.59	\$ 53,000,869
Monroe	3,418	\$ 5,410.02	\$ 18,491,448	7930	\$ 6,683.59	\$ 53,000,869
Coosa	1,376	\$ 5,410.02	\$ 7,441,483	7930	\$ 6,683.59	\$ 53,000,869
Choctaw	2,108	\$ 5,410.02	\$ 11,401,617	7930	\$ 6,683.59	\$ 53,000,869
Winston	3,304	\$ 5,410.02	\$ 17,872,001	7930	\$ 6,683.59	\$ 53,000,869
Lowndes	2,330	\$ 5,410.02	\$ 12,605,347	7930	\$ 6,683.59	\$ 53,000,869
Louisiana						
Tensas	1,160	\$ 5,410.02	\$ 6,275,623	6908	\$ 6,683.59	\$ 46,170,240
East Carroll	2,000	\$ 5,410.02	\$ 10,820,040	6908	\$ 6,683.59	\$ 46,170,240
Bienville	3,060	\$ 5,410.02	\$ 16,554,661	6908	\$ 6,683.59	\$ 46,170,240
Claiborne	2,730	\$ 5,410.02	\$ 14,766,650	6908	\$ 6,683.59	\$ 46,170,240
Webster	7,978	\$ 5,410.02	\$ 43,161,140	6908	\$ 6,683.59	\$ 46,170,240
Concordia	4,567	\$ 5,410.02	\$ 24,704,856	6908	\$ 6,683.59	\$ 46,170,240
Washington	10,812	\$ 5,410.02	\$ 58,490,431	6908	\$ 6,683.59	\$ 46,170,240
Franklin	5,061	\$ 5,410.02	\$ 27,377,406	6908	\$ 6,683.59	\$ 46,170,240
Caldwell	2,292	\$ 5,410.02	\$ 12,399,766	6908	\$ 6,683.59	\$ 46,170,240
Red River	1,911	\$ 5,410.02	\$ 10,335,843	6908	\$ 6,683.59	\$ 46,170,240
Mississippi						
Sharkey	1,546	\$ 5,410.02	\$ 8,363,891	3718	\$ 6,683.59	\$ 24,849,588
Jefferson	10,857	\$ 5,410.02	\$ 58,736,587	3718	\$ 6,683.59	\$ 24,849,588
Humphreys	2,809	\$ 5,410.02	\$ 15,196,746	3718	\$ 6,683.59	\$ 24,849,588
Issaquena	1,110	\$ 5,410.02	\$ 6,002,417	3718	\$ 6,683.59	\$ 24,849,588
Quitman	2,833	\$ 5,410.02	\$ 15,323,882	3718	\$ 6,683.59	\$ 24,849,588
Wilkinson	2,486	\$ 5,410.02	\$ 13,446,605	3718	\$ 6,683.59	\$ 24,849,588
Amite	1,156	\$ 5,410.02	\$ 6,251,278	3718	\$ 6,683.59	\$ 24,849,588
Jefferson Davis	906	\$ 5,410.02	\$ 4,901,478	3718	\$ 6,683.59	\$ 24,849,588
Marion	1,989	\$ 5,410.02	\$ 10,760,530	3718	\$ 6,683.59	\$ 24,849,588
Walthall	1,726	\$ 5,410.02	\$ 9,334,990	3718	\$ 6,683.59	\$ 24,849,588
			\$ 542,487,050			\$ 1,240,206,960.40

Conclusions and Recommendations

Based on this analysis, the following conclusions and recommendations can be reached:

1. The overall PMPM rate for both Medicaid and Medicare Advantage is much higher than typical PMPM rates due to the severity of the conditions and the presence of multiple chronic conditions. The use of remote specialty care can potentially reduce overall healthcare utilization and costs, which drive down PMPM and save payers money.
2. Louisiana has the highest per capita Medicaid expenditure, followed closely by Mississippi. Mississippi has the fewest number of specialists in this study and has one of the lowest rates in the United States.
3. Most of the individuals with multiple chronic conditions have a 50% or lower 10-year survival rate and have conditions that would require at least one, if not two, specialists to provide the appropriate and needed care. In the examples provided for the Medicare population, those with multiple chronic conditions have a 0% survival rate after 10 years unless care is provided.
4. The total cost of care estimates does not factor in the cost of a referral, which could increase costs by 2% per person and are removed through remote specialty care.
5. There are significant cost-benefits for individuals using remote specialty care as it mitigates the risk of unnecessary referrals, eliminates patient travel, and limits the impact on patient productivity.
6. Research also shows cost-benefits to remote weight management and nutrition programs as well as using community workers.

7. The total addressable market for remote specialty care is approximately \$542,487,050 for Medicaid and \$1,240,206,960.40 for Medicare.

Endnotes

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- ¹ Sambamoorthi, Usha et al. "Multiple chronic conditions and healthcare costs among adults." Expert review of pharmacoeconomics & outcomes research vol. 15,5 (2015): 823-32. doi:10.1586/14737167.2015.1091730
 - ² Ryan, Aine et al. "Multimorbidity and functional decline in community-dwelling adults: a systematic review." Health and quality of life outcomes vol. 13 168. 15 Oct. 2015, doi:10.1186/s12955-015-0355-9
 - ³ McPhail, Steven M. "Multimorbidity in chronic disease: impact on health care resources and costs." Risk management and healthcare policy vol. 9 143-56. 5 Jul. 2016, doi:10.2147/RMHP.S97248
 - ⁴ Manemann, Sheila M et al. "Rurality, Death, and Healthcare Utilization in Heart Failure in the Community." Journal of the American Heart Association vol. 10,4 (2021): e018026. doi:10.1161/JAHA.120.018026
 - ⁵ Lagasse, Jeff. "Shortage of Specialists Negatively Impacting Patients in Rural Areas." Healthcare Finance News, <https://www.healthcarefinancenews.com/news/shortage-specialists-negatively-impacting-patients-rural-areas>.
 - ⁶ "AAMC Report Reinforces Mounting Physician Shortage." AAMC, 11 June 2021, <https://www.aamc.org/news-insights/press-releases/aamc-report-reinforces-mounting-physician-shortage>.
 - ⁷ "Alabama: Keeping Education Active: Partnership to Fight Chronic Disease." Alabama | Keeping Education ACTIVE | Partnership to Fight Chronic Disease, <https://www.fightchronicdisease.org/states/alabama#:~:text=Chronic%20disease%20could%20cost%20Alabama,per%20year%202016%2D2030>).
 - ⁸ "Louisiana: Keeping Education Active: Partnership to Fight Chronic Disease." Louisiana | Keeping Education ACTIVE | Partnership to Fight Chronic Disease, <https://www.fightchronicdisease.org/states/louisiana>.
 - ⁹ "Mississippi: Keeping Education Active: Partnership to Fight Chronic Disease." Mississippi | Keeping Education ACTIVE | Partnership to Fight Chronic Disease, <https://www.fightchronicdisease.org/states/mississippi#:~:text=Chronic%20disease%20could%20cost%20Mississippi,per%20year%202016%2D2030>).
 - ¹⁰ Alabama Medicaid, <https://medicaid.alabama.gov/>.
 - ¹¹ Louisiana Medicaid 2020 Annual Report - Louisiana Department of Health. <https://www.ldh.la.gov/assets/medicaid/AnnualReports/MedicaidAnnualReport2020.pdf>.
 - ¹² 2021 - Mississippi Division of Medicaid. <https://medicaid.ms.gov/wp-content/uploads/2021/12/2021-Annual-Report.pdf>.
 - ¹³ Vogeli, C., Shields, A.E., Lee, T.A. et al. Multiple Chronic Conditions: Prevalence, Health Consequences, and Implications for Quality, Care Management, and Costs. J GEN INTERN MED 22 (Suppl 3), 391–395 (2007). <https://doi.org/10.1007/s11606-007-0322-1>
 - ¹⁴ Lehnert, Thomas, et al. "Health care utilization and costs of elderly persons with multiple chronic conditions." Medical Care Research and Review 68.4 (2011): 387-420.
 - ¹⁵ "Ratebooks & Supporting Data." CMS, <https://www.cms.gov/Medicare/Health-Plans/MedicareAdvtgSpecRateStats/Ratebooks-and-Supporting-Data>.
 - ¹⁶ "Charlson Comorbidity Index (CCI)." MDCalc, <https://www.mdcalc.com/calc/3917/charlson-comorbidity-index-cci>.
 - ¹⁷ Managed Care in Alabama - Medicaid.gov. <https://www.medicaid.gov/Medicaid/downloads/alabama-mcp.pdf>.
 - ¹⁸ Managed Care in Alabama - Medicaid.gov. <https://www.medicaid.gov/Medicaid/downloads/alabama-mcp.pdf>.
 - ¹⁹ "Medicaid Expansion - Adult Group." Medicaid Expansion - Adult Group | La Dept. of Health, <https://ldh.la.gov/page/2327>.
 - ²⁰ "Managed Care - Mississippi Division of Medicaid." Mississippi Division of Medicaid -, 13 July 2022, <https://medicaid.ms.gov/programs/managed-care/>.
 - ²¹ Zuelsdorff M;Larson JL;Hunt JFV;Kim AJ;Koscik RL;Buckingham WR;Gleason CE;Johnson SC;Asthana S;Rissman RA;Bendlin BB;Kind AJH; "The Area Deprivation Index: A Novel Tool for Harmonizable Risk Assessment in Alzheimer's Disease Research." Alzheimer's & Dementia (New York, N. Y.), U.S. National Library of Medicine, <https://pubmed.ncbi.nlm.nih.gov/32548238/>.
 - ²² Zuelsdorff M;Larson JL;Hunt JFV;Kim AJ;Koscik RL;Buckingham WR;Gleason CE;Johnson SC;Asthana S;Rissman RA;Bendlin BB;Kind AJH; "The Area Deprivation Index: A Novel Tool for Harmonizable Risk Assessment in Alzheimer's Disease Research." Alzheimer's & Dementia (New York, N. Y.), U.S. National Library of Medicine, <https://pubmed.ncbi.nlm.nih.gov/32548238/>.
 - ²³ Clarke, Janice L et al. "An Innovative Approach to Health Care Delivery for Patients with Chronic Conditions." Population health management vol. 20,1 (2017): 23-30. doi:10.1089/pop.2016.0076
 - ²⁴ Bhatt, Jay, and Priya Bathija. "Ensuring Access to Quality Health Care in Vulnerable Communities." Academic medicine : journal of the Association of American Medical Colleges vol. 93,9 (2018): 1271-1275. doi:10.1097/ACM.0000000000002254
 - ²⁵ "Telehealth Models for Increasing Access to Specialty Care - RHIHUB Toolkit." Telehealth Models for Increasing Access to Specialty Care - RHIhub Toolkit, <https://www.ruralhealthinfo.org/toolkits/telehealth/2/care-delivery/specialty-care>.

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- ²⁶ Anderson, Daren et al. "A cost-effectiveness analysis of cardiology eConsults for Medicaid patients." *The American journal of managed care* vol. 24,1 e9-e16. 1 Jan. 2018
- ²⁷ Anderson, Daren, et al. "Reduced Cost of Specialty Care Using Electronic Consultations for Medicaid Patients." *Health Affairs*, vol. 37, no. 12, 2018, pp. 2031–2036., <https://doi.org/10.1377/hlthaff.2018.05124>.
- ²⁸ Rollo, Megan E., et al. "Cost Evaluation of Providing Evidence-Based Dietetic Services for Weight Management in Adults: In-Person versus Ehealth Delivery." *Nutrition & Dietetics*, vol. 75, no. 1, 2017, pp. 35–43., <https://doi.org/10.1111/1747-0080.12335>.
- ²⁹ Moran, Lisa, et al. "Predictors of Lifestyle Intervention Attrition or Weight Loss Success in Women with Polycystic Ovary Syndrome Who Are Overweight or Obese." *Nutrients*, vol. 11, no. 3, 2019, p. 492., <https://doi.org/10.3390/nu11030492>.
- ³⁰ Rollo, Megan E., et al. "Cost Evaluation of Providing Evidence-Based Dietetic Services for Weight Management in Adults: In-Person versus Ehealth Delivery." *Nutrition & Dietetics*, vol. 75, no. 1, 2017, pp. 35–43., <https://doi.org/10.1111/1747-0080.12335>.
- ³¹ Kangovi, Shreya, et al. "Evidence-Based Community Health Worker Program Addresses Unmet Social Needs and Generates Positive Return on Investment." *Health Affairs*, vol. 39, no. 2, 2020, pp. 207–213., <https://doi.org/10.1377/hlthaff.2019.00981>.